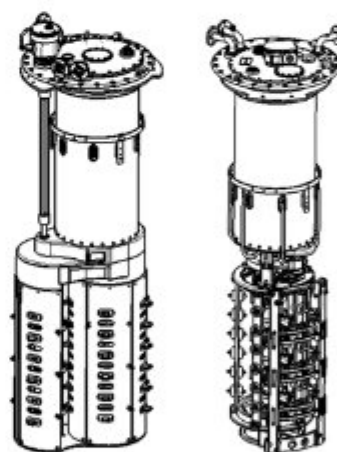




TECHNICAL DATA

TYPE SHZV VACUUM ON-LOAD TAP CHANGER FOR OIL-IMMERSED TRANSFORMER

HM0.154.3901-02.06/2018



SHZV(cylinder tap selector) SHZV(cage tap selector)

SHANGHAI HUAMING POWER EQUIPMENT CO., LTD.

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1. General

Type SHZV vacuum on-load tap changer (herein referred to as tap changer) is of combined structure, applicable to oil-immersed voltage regulating transformers. The tap changer is composed of a diverter switch and a tap selector, immersed in transformer oil. Diverter switch is in a separate oil compartment. Tap selector is in the same transformer oil tank as the windings. Tap changer is to be mounted on transformer tank top by means of bell type mounting through a top flange. Tap changer is operated by a motor drive unit. Tap changer and motor drive unit are connected by an upper gearbox, driving shaft and a bevel gearbox. Tap changer provides both local and remote operation modes.

Tap changer connection includes neutral point of star connection for three-phase and any connection for single phase. Three units of single phase tap changers can be used for any selectable winding connections for a three phase transformer. Its basic connection diagram is shown in Fig. 1 below.

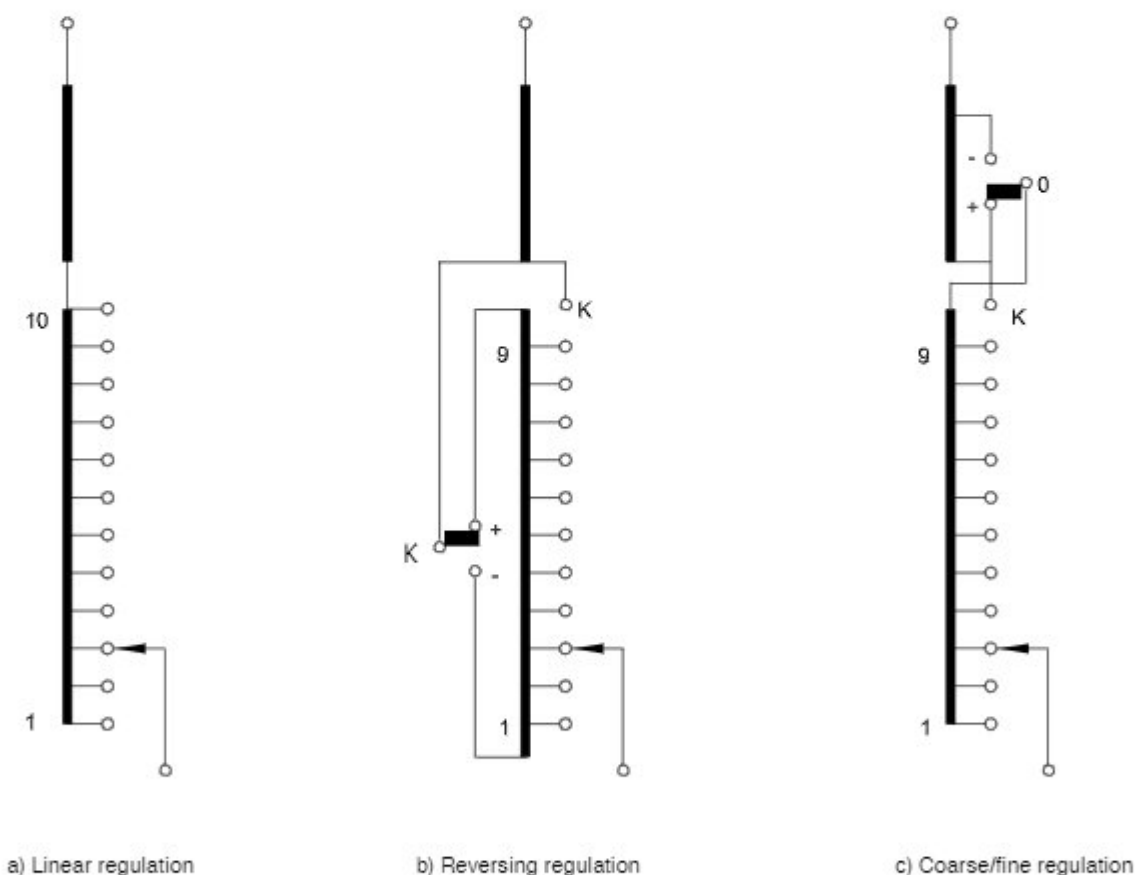


Fig.1 Basic Connection Diagram of Tap Winding

2. Technical specifications

Type SHZV OLTC complies with IEC 60214-1:2014 standard. Tap changer technical data is listed in Table 1-1 and Table 1-2 below.

Table 1-1 Type SHZV (Cylinder Tap Selector) Series of On-Load Tap Changer Technical Data

Item	Description		SHZVIII				SHZVI					
1	Max. rated through current (A)		400	600	1000	1300	400	600	1000	1600	2400	3000
2	Rated frequency (Hz)		50 or 60									
3	Connection		3-phase Y-connection for neutral point only				Single-phase for any selectable winding connection					
4	Max. rated step voltage (V)		4000									
5	Rated step capacity		1500	1600	3000	3300	1500	1600	3000	4400	5600	6000
6	Short-circuit current test (kA)	Thermal (3s)	6	8	12	15	6	8	12	24	24	30
		Dynamic (peak)	15	20	30	37.5	15	20	30	60	60	75
7	Max. operating positions		35 without change-over selector; 107 with change-over selector									
8	Insulation to ground	The highest voltage for equipment (kV)	72.5			126		170			252	
		Rated separate source AC withstand voltage (kV/50Hz, 1min)	140			230		350			460	
		Rated lightning impulse withstand voltage (kV, 1.2/50µs)	350			550		750			1050	
9	Tap selector		Categorized into B, C, D & DE four sizes									
10	Mechanical life		Not less than 1,500,000 operations									
11	Electrical life		Not less than 600,000 operations									
12	Oil compartment of diverter switch	Service pressure	0.03 MPa									
		Leakage test	No leakage under 0.08 MPa for 24 hours									
		Over pressure protection	Rupture disc bursts at 300 ±20% KPa									
		Protective relay	Set oil flow speed at 1.0m/s ± 10%									
13	Motor drive unit		SHM-D or CMA7									

Note: SHZV with cylinder tap selector is applicable for the contact pitches of 10, 12 and 14 in linear, reversing and coarse and fine regulating.

**Table 1-2 Type SHZV (Cage Tap Selector) Series of
On-Load Tap Changer Technical Data**

Item	Description		SHZVIII		SHZVI		
1	Max. rated through current (A)		400	600	600	1200	1500
2	Rated frequency (Hz)		50 or 60				
3	Connection		3-phase Y-connection for neutral point only		Single-phase for any selectable winding connection		
4	Max. rated step voltage (V)		4000				
5	Rated step capacity		1500	1600	1600	3000	4000
6	Short-circuit current test (kA)	Thermal (3s)	6	8	8	24	24
		Dynamic (peak)	15	20	20	60	60
7	Max. operating positions		18 without change-over selector; 35 with change-over selector				
8	Insulation to ground	The highest voltage for equipment (kV)	72.5	126	170	252	
		Rated separate source AC withstand voltage(kV/50Hz,1min)	140	230	350	460	
		Rated lightning impulse withstand voltage (kV,1.2/50μs)	350	550	750	1050	
9	Tap selector		Categorized into B, C, D, DE four sizes				
10	Mechanical life		Not less than 1,500,000 operations				
11	Electrical life		Not less than 600,000 operations				
12	Oil compartment of diverter switch	Service pressure	0.03 MPa				
		Leakage test	No leakage under 0.08 MPa for 24 hours				
		Over pressure protection	Rupture disc bursts at 300 ±20% KPa				
		Protective relay	Set oil flow speed at 1.0m/s ±10%				
13	Motor drive unit		SHM-D or CMA7				

3. Type designation

3.1. Type designation

Due to the different combinations of number of phases, maximum rated through current, the highest voltage for equipment, selector size and connections, type SHZV comes with various models. Hence, the type designation shall provide all the above technical parameter and below is its detailed explanation in Fig. 2.

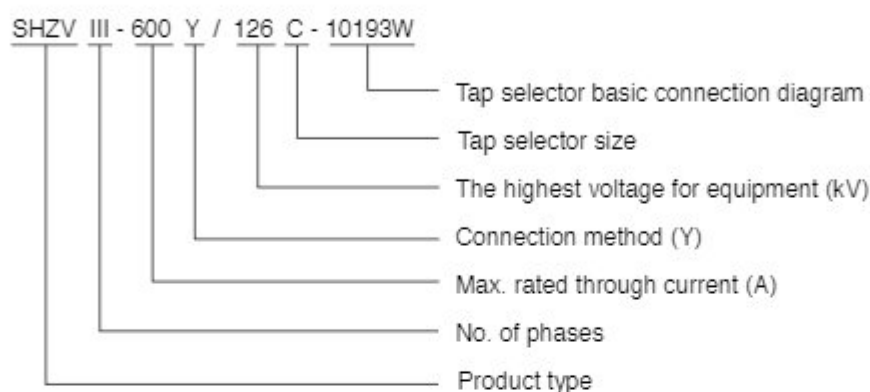


Fig. 2 Tap Changer Type Explanation

3.2. Tap selector basic connection method

Because of voltage regulation range difference and winding connection variations, tap selector has a number of different specifications. Tap selector specification is decided by number of inherent contacts, number of operating positions, number of mid positions and type of change-over selector. Please refer to Fig. 3. for indications of different tap selector parameters.

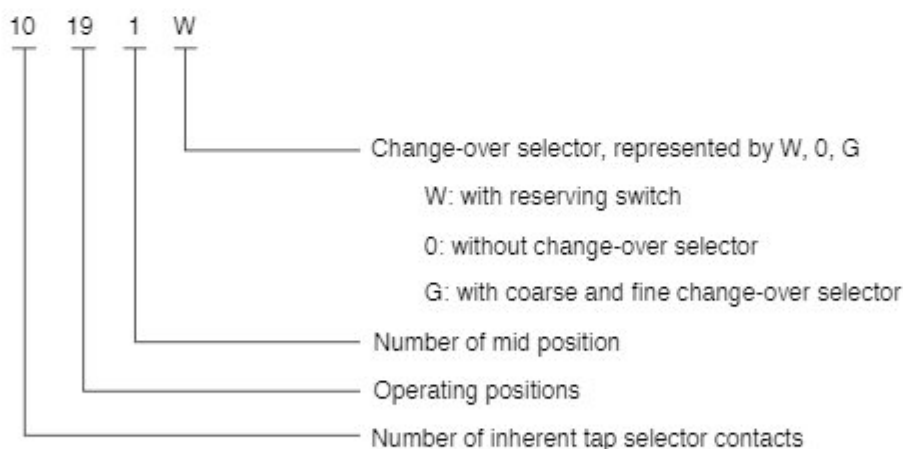


Fig. 3. Tap Selector Basic Connection Method Explanation

3.3. Tap selector basic connection diagram

Different transformer tapping corresponds to different tap selector basic connection diagrams. Fig. 4 shows common basic connection diagrams. Special requirement can also be specially designed.

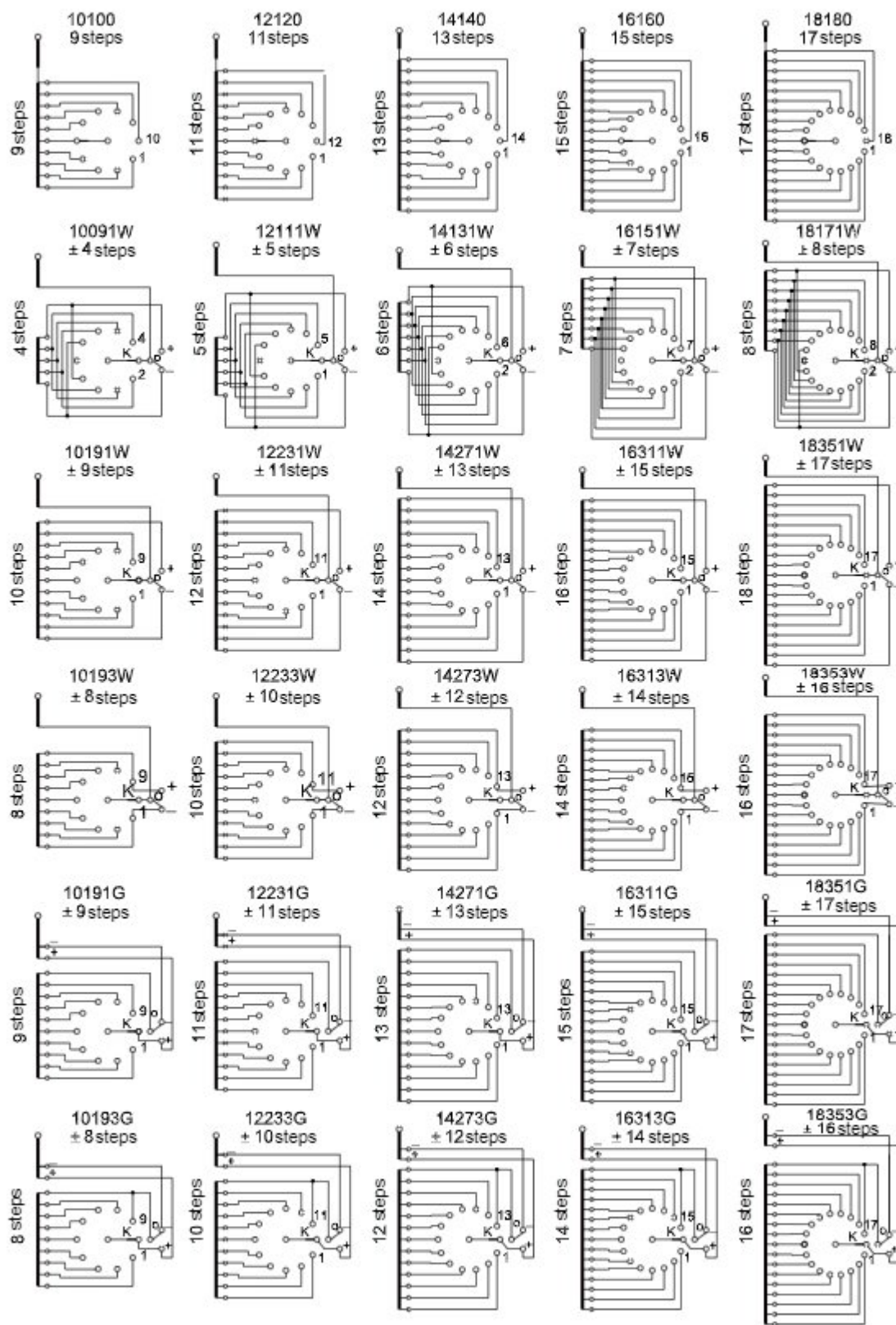


Fig. 4 Tap Selector Basic Connection Diagram

4. Terms and definitions

4.1. Rated through-current

Rated through current I_u : The current flowing through an on-load tap changer toward the external circuit, which can be transferring from one tap to the other at the relevant rated step voltage and which can be carried continuously while meeting the requirement.

Maximum rated through-current I_{um} : The highest rated through-current for which the tap changer is designed for and which forms the basis for all current related tests.

4.2. Step voltage

Rated step voltage U_i : For each value of rated through current, the highest permissible voltage between terminals which are intended to be connected to successive taps of the transformer.

Maximum rated step voltage U_{im} : The highest value of the rated step voltage for which the tap changer is designed. The maximum rated step voltage for type SHZV OLTC is 4000V.

4.3. Breaking capacity

Please refer to Fig.5, Fig.6 for curvy of the rated step capacity.

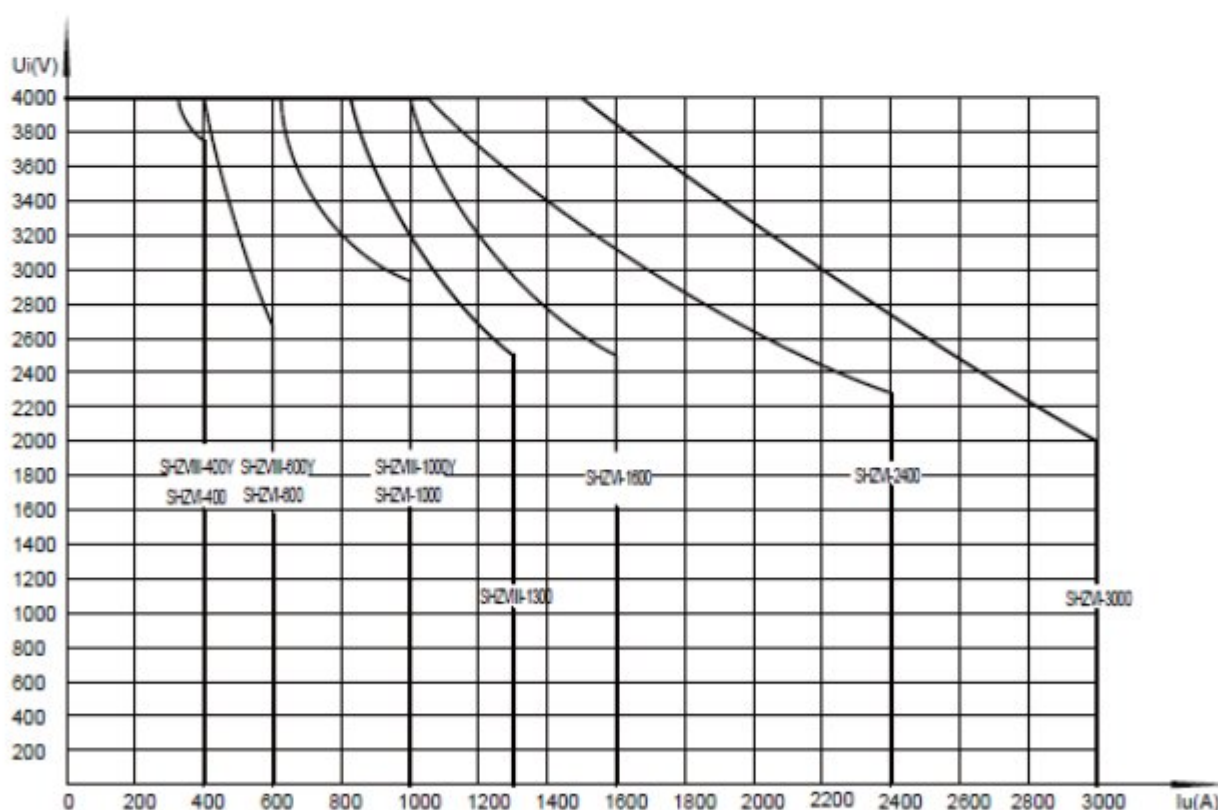


Fig. 5 Curve of SHZV Rated Step Capacity
(Take Cylinder Tap Selector Structure as Example)

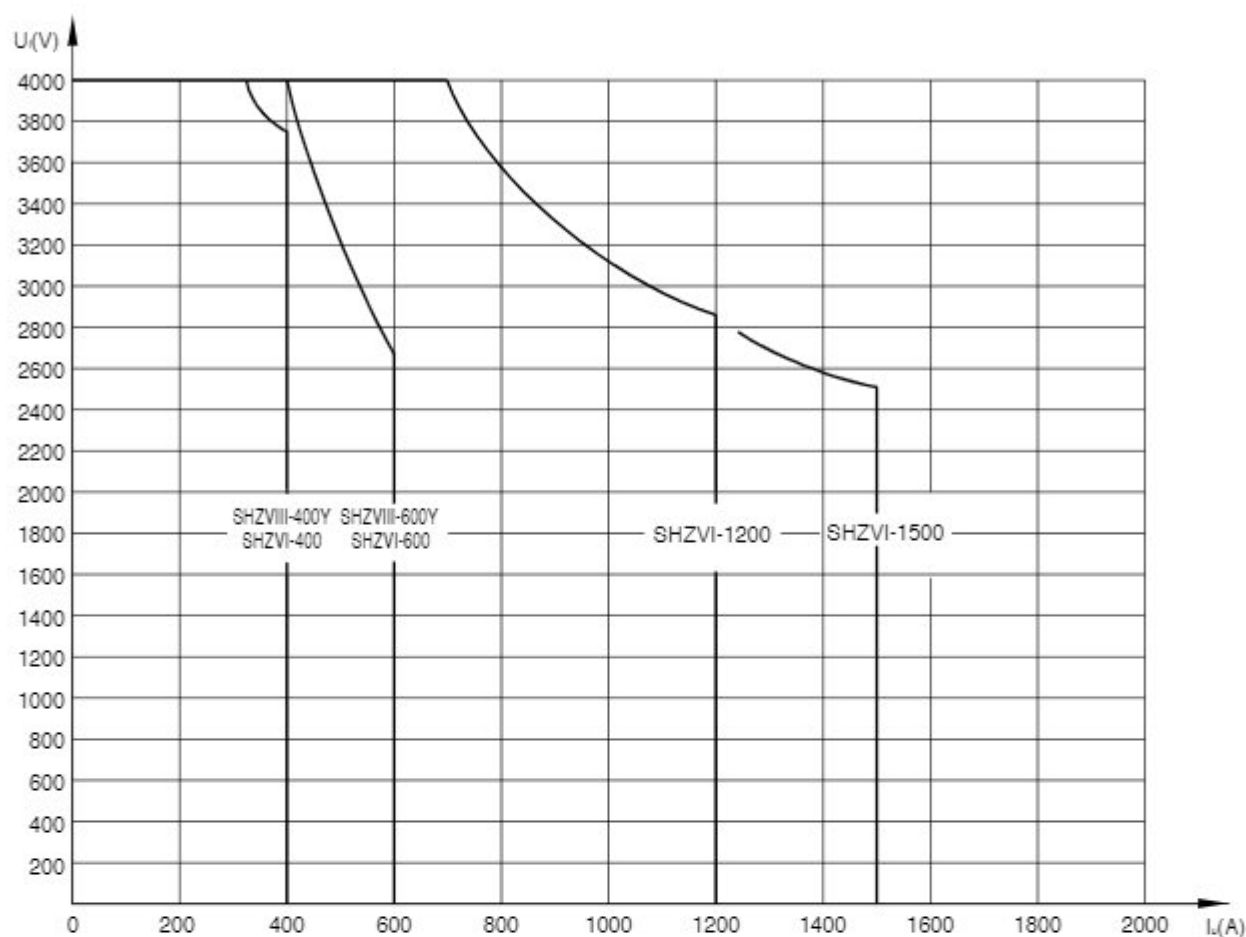


Fig. 6 Curve of SHZV Rated Step Capacity
(Take Cage Tape Selector Structure as Example)

According to stipulations of IEC60214-1: 2014, tap changer shall be able to break two times of maximum rated through-current and its relevant step voltage for 40 operations. Breaking capacity of type SHZV tap changer is

$$P_{st,max} = 2P_{stV} \approx 2I_{um} \times U_{stV} \text{ Where,}$$

P_{stV} : rated step capacity

I_{um} : maximum rated through-current

U_{stV} : relevant rated step voltage

4.4. Electrical life of vacuum interrupter

The electrical life of vacuum interrupt is not less than 600,000 operations.

4.5. Short-circuit current test

According to IEC 60214-1: 2014, all contacts continuously carrying the current shall be able to withstand 2s ($\pm 10\%$) short circuit test current without melting, deformation or mechanical damage. Meanwhile the starting peak current value shall be 2.5 ($\pm 5\%$) times of the root means square value of rated short circuit test current. Refer the short circuit test current values to Table 1-1 and Table 1-2 Type SHZV Series of On-Load Tap Changer Technical Data.

4.6. Service condition of tap changer

4.6.1. Service temperature range of tap changer in oil is $-25^{\circ}\text{C} \sim +105^{\circ}\text{C}$

4.6.2. Service ambient air temperature range of tap changer is $-25^{\circ}\text{C} \sim +40^{\circ}\text{C}$ and humidity $\leq 85\%$.

If the service ambient temperature is $>40^{\circ}\text{C}$ or $<-25^{\circ}\text{C}$, it should be indicated in the order specifications.

The material and accessories will be specially treated to suit the needs.

4.6.3. Perpendicular deflection between ground and tap changer after being mounting on transformer shall be less than 2%

4.6.4. There shall be no serious dust, explosive gas or corrosive gas on service site

Remark: Please contact us if special application required.

4.7. Tap changer insulation level to earth

Tap changer insulation level to earth is the insulation between tap changer live parts and grounding part. It is determined by dielectric tests according to IEC-60214-1-2014. (Please refer to Table 2) The requirement of which correlates to the transformer tap winding location, regulation range & regulation method, winding connection & arrangement and rated voltage of transformer winding. It's decided by the insulation to earth of transformer tap winding.

Table 2 Tap Changer Insulation Level to ground

(unit: kV)

The highest voltage for equipment U_m (kV)	Rated separate source AC withstand voltage (kV/50Hz, 1min)	Rated lightning impulse withstand voltage (1.2/50 μ s)
72.5	140	350
126	230	550
170	325	750
252	460	1050

4.8. Internal insulation level of tap changer

The internal insulation level of type SHZV tap changer is categorized into B, C, D, DE four sizes. Refer the internal insulation level to table 3. Basic connection diagram and insulation distance mark is shown in Fig. 7. Internal insulation must be checked when selecting the proper tap changer whether it is qualified for the voltage withstanding requirement.

Table 3 Tap Changer Internal Insulation Level

(unit: kV)

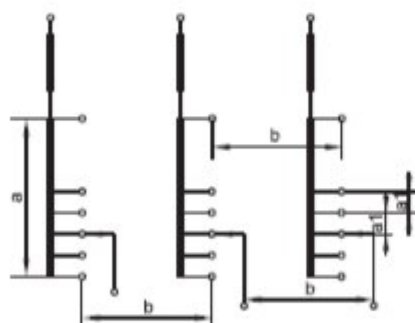
Designation code	Tap selector size B		Tap selector size C		Tap selector size D		Tap selector size DE	
	1.2/50 μ s	50Hz 1min	1.2/50 μ s	50Hz 1min	1.2/50 μ s	50Hz 1min	1.2/50 μ s	50Hz 1min
a	265	50	365	82	490	105	550	120
b	265	50	350	82	490	146	550	160
a ₀₁	90	20	90	20	90	20	90	20
a ₀₂	70	20	70	20	70	20	70	20
a1	150	30	150	30	150	30	150	30
c1	500	145	550	180	590	225	660	230
c2	500	145	550	195	590	225	660	250

Note: When a₀₁ represents Insulation of spark protection gap, its Insulation is 1.2/50 μ s, 90~130kV, 130kV 100% responsive.

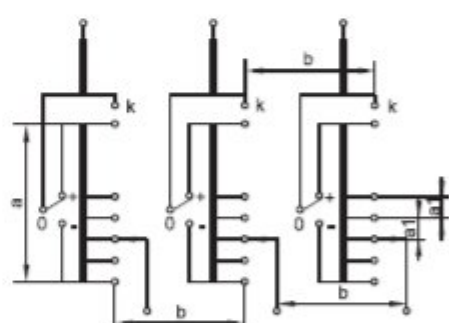
When a₀₂ represents Insulation of ZnO protection, its Insulation is 1.2/50 μ s, 50~70kV, 70kV 100% responsive.

4.9. Tap changer mounting method

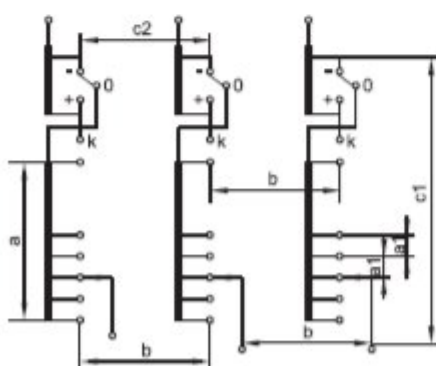
Type SHZV tap changer is mounted to transformer tank top by a head flange. Hence, a mounting flange shall be provided by transformer producer, the dimension of which shall refer to the drawing of Appendix 52. Type SHZV tap changer is only applicable to bell type mounting. The supporting flange of the tap changer is only for temporary support during the transformer conductor connection. After putting the bell tank, tap changer shall be fixed to the mounting flange of the transformer. The connection pipes on head flange of type SHZV tap changer is of spread around arrangement (Refer details to appendix). Only SHZV with cage tap selector is applicable to standard tank mounting transformer (Please refer to Appendix 71 for its supporting flange).



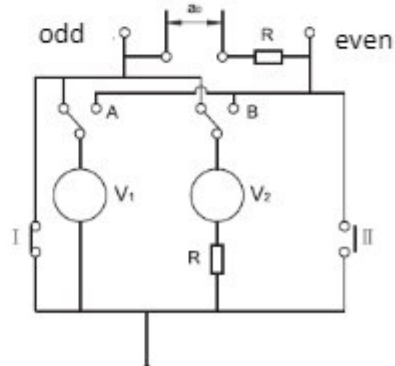
Linear Regulation



Reversing Regulation



Coarse/Fine Regulation



Diverter switch internal insulation

- a: between start and end of a fine tap winding; also between start and end of coarse tap winding;
- b: between any tapping of different fine tap windings, or between ends of different coarse windings;
- a0: between selected and preset diverter switch tapping;
- a1: between any selected and preselected taps of the tap selector;
- c1: between the start of coarse tap winding and the current take-off terminal for the same phase;
- c2: between start contacts (-) of coarse winding for different phases;
- A,B: Isolating contact;
- V₁: Main breaking vacuum interrupter;
- V₂: Transition vacuum interrupter;
- R: Resistor;
- I: Odd number main contact;
- II: Even number main contact.

Fig. 7 Basic Connection Diagram and Insulation Distance Mark

5. Special designs

5.1. Potential connection of the tap winding

For transformers with high voltage rating and big regulation range, during the operation of the change-over selector, the tap winding is disconnected momentarily from the main winding and in a so-called "suspension" status. At that moment, the tap winding takes a new potential which is determined together by the coupling capacitance to ground C_e and coupling capacitance to the adjacent winding C_w (refer details to Fig.9). Usually this potential is different from the previous potential of the tap winding before the operation. The difference between the two is called bias voltage. This bias voltage turns out to be the recovery voltage on the gap of the change-over selector. When the recovery voltage exceeds a certain critical value, the change-over selector would discharge electricity and produce considerable amount of gas. This could be a serious problem. Therefore, potential connection of the tap winding must be considered when this recovery voltage exceeds a certain value, in order to avoid the discharge during the operation of the change-over selector.

The permissible recovery voltage to type SHZV tap changer is 35kV. In case the recovery voltage of the change-over selector exceeds this value, a tie-in resistor with fixed value shall be permanently connected into the tap winding (refer to Fig. 8). The mounting location and dimension of tie-in resistor for SHZV can be found in Appendix.

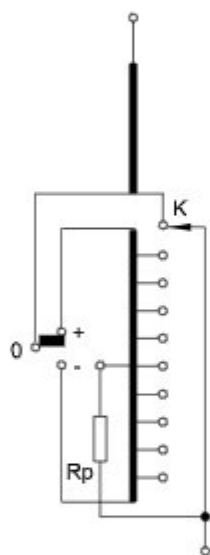


Fig. 8 Potential Connection by Tie-In Resistor

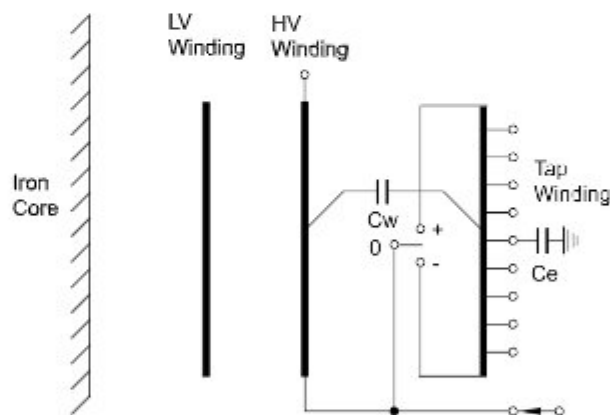


Fig.9 Arrangement of Winding and Coupling Capacitance

For the convenience of Huaming to define the load of the change-over selector and size & quantity of the tie-in resistor, please provide the following transformer data when selecting tap changer:

- Complete transformer data: rated capacity, rated voltage, regulation range, connection of winding and insulation level, etc.;
- Arrangement of winding: relative location between tap winding and adjacent winding or winding part;
- Operating A.C. voltage across windings or layers of windings adjacent to the tap windings;
- Capacitance between tap winding and adjacent winding (C_w);
- Capacitance of the tap winding to ground or grounded adjacent windings (if exist) (C_e);
- Voltage stress across half the tap winding at lightning impulse voltage test;
- A.C. voltage across half the tap winding under operation and test conditions. (is normally derived from order specification sheet for tap changer) .

5.2 Two phase and single phase of type SHZV

Type SHZV tap changer can be designed as one motor drive unit (or three motor drive unit) driving three single phase tap changers or one two phases plus one single phase tap changer, for regulation of delta connection or other regulation locations other than neutral point.

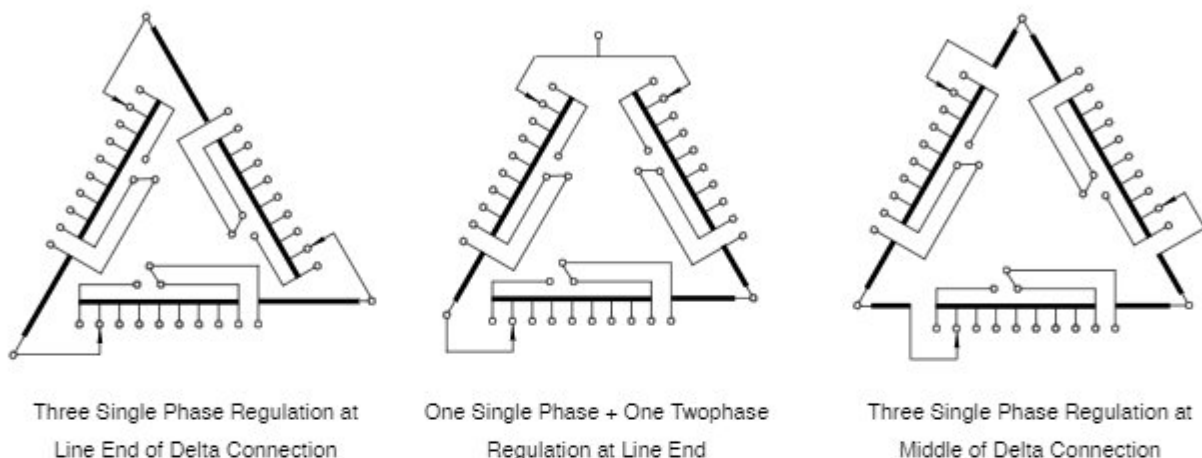


Fig. 10 Basic Connections for Delta-connected Transformer Winding

5.3 SHZV tap selector structures and application

5.3.1. Cylinder tap selector: It is suitable for the contact pitch of 10, 12 and 14 in linear, reversing and coarse and fine regulating.

5.3.2. Cage tap selector: SHZV cage tap selector is the same as the tap selector of type CM. Thus it can meet the technical data of CM tap selector. The maximum current of SHZV with cage tap selector is 600A for 3-phase and 1500A for single phase. Please refer to CM technical data for connection diagram or contact us.

5.3.3. Please contact us when choosing SHZV with cage tap selector for the drawings (in linear, reversing regulation or in other regulation) which are excluded in this SHZV technical data.

5.4 SHZV step protection and application

SHZV overvoltage between steps protection can be realized via spark gaps or varistor (ZnO). The spark gap protection mechanism has a simple structure and small size. When the overvoltage attacks, the breakdown first happens to the spark gap so as to protect the insulation between steps. After that, the spark gap must be replaced to ensure the tap-changer's normal operation. The varistor (ZnO) has the same function and better performance. The size, however, is bigger. So usually the spark gap is adopted. If the varistor (ZnO) is needed, please specify in the order specification.

6. Motor drive unit

SHZV OLTC may operate by SHM-D or CMA7 motor drive unit according to the requirement, please refer to table 4 for technical data.

Table 4 Technical Data of Motor Drive Unit

Motor drive unit		SHM-D	CMA7	
Motor	Rated power (W)	750	750	1100
	Rated voltage (V)	200-240 AC	380 3AC	
	Rated current (A)	4	2.0	2.8
	Rate frequency(Hz)	50 or 60	50 or 60	
	Rotate speed (r.p.m.)	330 direct drive	1400	
Rated torque on drive shaft (Nm)		35 static torque	18	26
Revolution of the drive shaft per switching operation		33	33	
Revolution of the hand crank per switching operation		33	33	
Running time per switching operation (s)		about 6	about 5	
Max. operation positions		107	107	
Voltage for control circuit and heater circuit (V)		~ 220	220/AC	
Heater power (W)		100	50	
A.C. voltage test to ground (kV/50Hz, 1min)		2	2	
Approx. weight (kg)		80	90	
Protective degree		IP66	IP56	

7. Operation controllers

7.1. SHM-K controller

SHM-K remote digital controller uses high-resolute color touch screen with a ARM9 chip as its CPU, which has a 400M clock speed, 256M FRASH, 64M RAM, a fast operation speed and huge memory space. The controller is connected with the SHM-D motor drive unit with cables for bilateral digital communication .

SHM-K main technical data is as below:

Working power: 85-264V AC, 50 or 50 Hz / 100-250V DC

Environment temperature: 0℃ to 50℃ Indoor

Comparative humidity: ≤RH95% (no condensation)

7.2. HMC-3C position indicator

HMC-3C OLTC position indicator is a support fitting for CMA7 motor drive unit, it can be used to indicate the OLTC position, and has the function of "1→N", "STOP", "N→1" control as well as remote control indicator lamp, its input is decimal code and output is BCD code. Please refer to HMC-3C manual for details.

HMC-3C technical data is as below. Working voltage: 220V AC

Power frequency: 50Hz

Maximum operation positions: 107

Environment temperature: -10℃ to 40℃ Indoor

Note: Please specify if special voltage required for motor, and control & heater circuit.

8. Accessories

8.1. Bevel gearbox

Bevel gearbox is used for the inter-connection of tap changer horizontal shaft and vertical shaft, in order to transfer the driving torque from motor drive unit to the tap changer. Please refer to Appendix 63 for the overall dimension.

8.2. Protective relay

Protective relay is the one of protective devices for oil-immersed on-load tap changer, when OLTC interior failure produces gas and oil surge, the protective relay contact acts, and switches on to the tripping circuit of the transformer circuit breaker, the transformer will be cut off at once.

Protective relay is mounted onto the connection pipe between OLTC head and oil conservator; make sure that protective relay marked with arrowhead side shall be connected to oil conservator. Please refer to Appendix 68 for the overall dimension.

8.3. Pressure relief device

Pressure relief valve and rupture disc are safety protection devices of the on-load tap changer. In case tap changer has an internal failure, which decomposes the oil in the compartment and produces large amount gas, the internal pressure inside the oil compartment will increase dramatically. If this pressure couldn't be released, tap changer will deform or even explode. Therefore, pressure relief devices can avoid the upgrade of the failure.

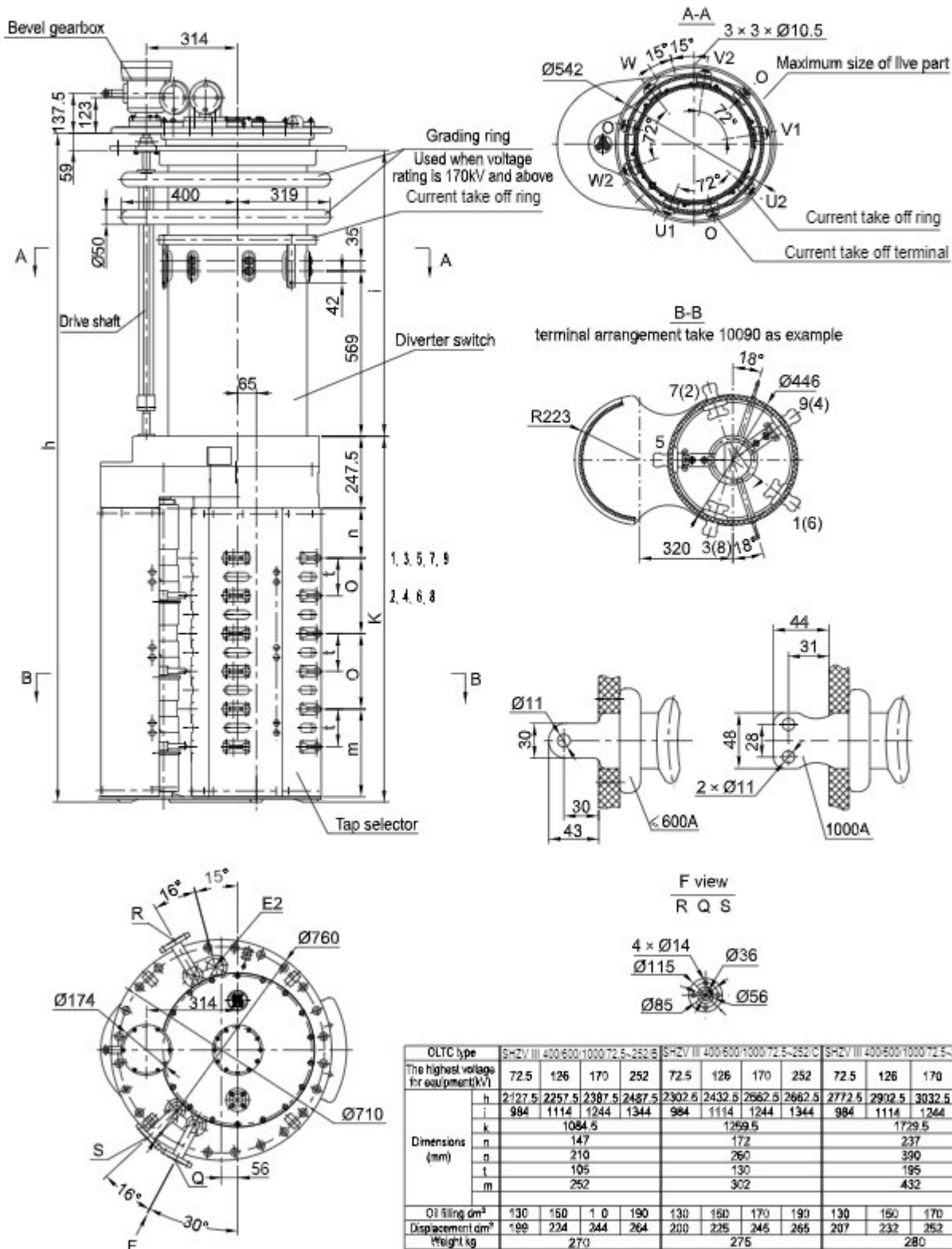
Pressure relief valve is a self-sealing relief valve. It opens the cover in case of over pressure and re-closes after the pressure is released, which can be used repeatedly and minimize the liquid loss during the operation.

The rupture disc is a weak point on the top cover of tap changer. Once the pressure in the oil compartment exceeds the set value, the disc will explode to release the over pressure of the compartment, as a result the oil compartment will be prevented from damage.

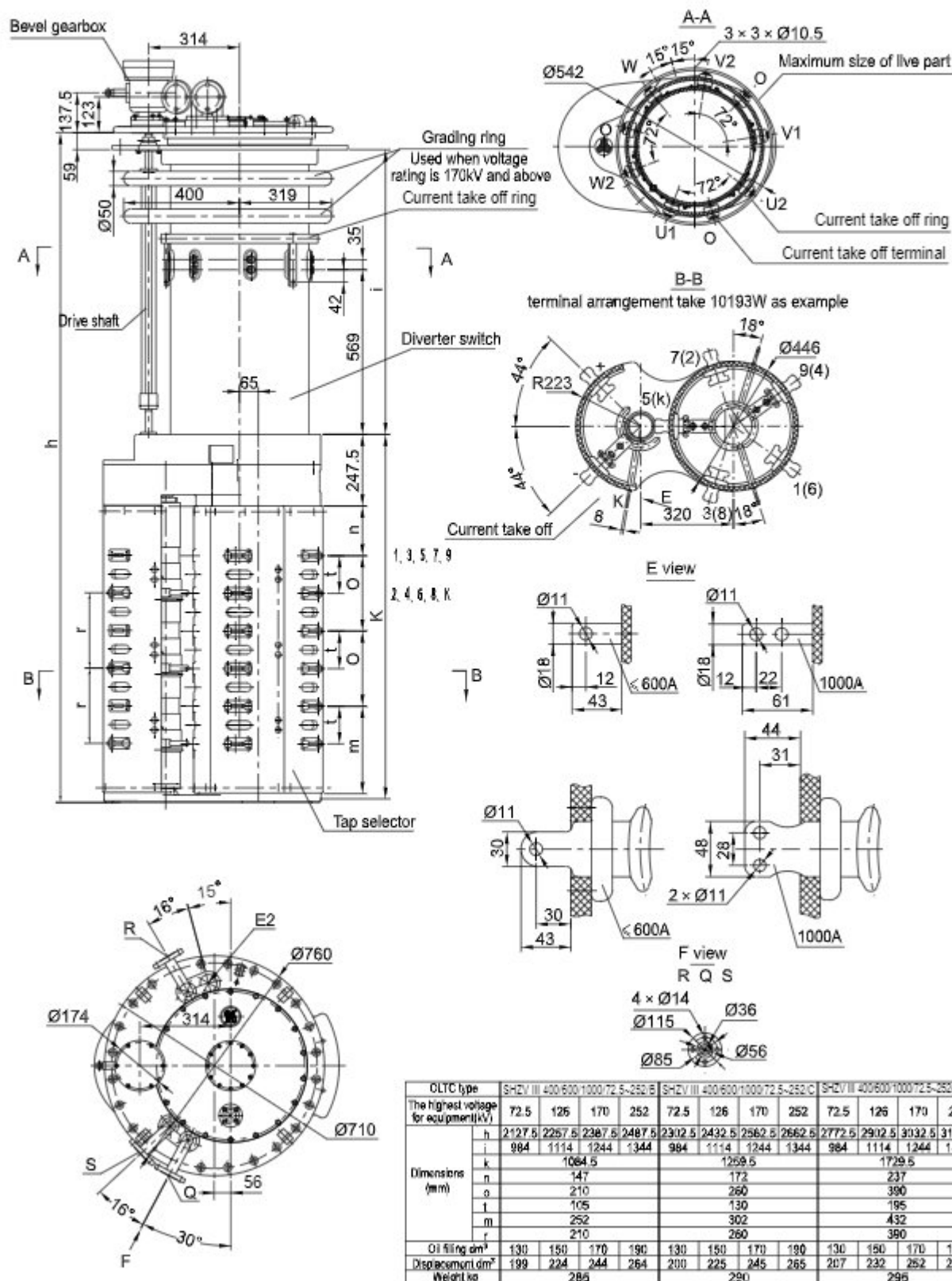
Pressure relief valve is a low-energy failure protection device. The rupture disc is a high-energy protection device. Tap changer failure usually tends to be high-energy failure. Hence, pressure relief valve is not recommended for tap changer, or use it as an assistant protection besides the rupture disc. Therefore, pressure relief valve is an optional accessory of tap changer for customer to select when ordering the tap changer.

9. Appendixes

**Appendix 1. SHZV III 400/600/1000A overall dimensions
without change-over selector, cylinder tap selector**

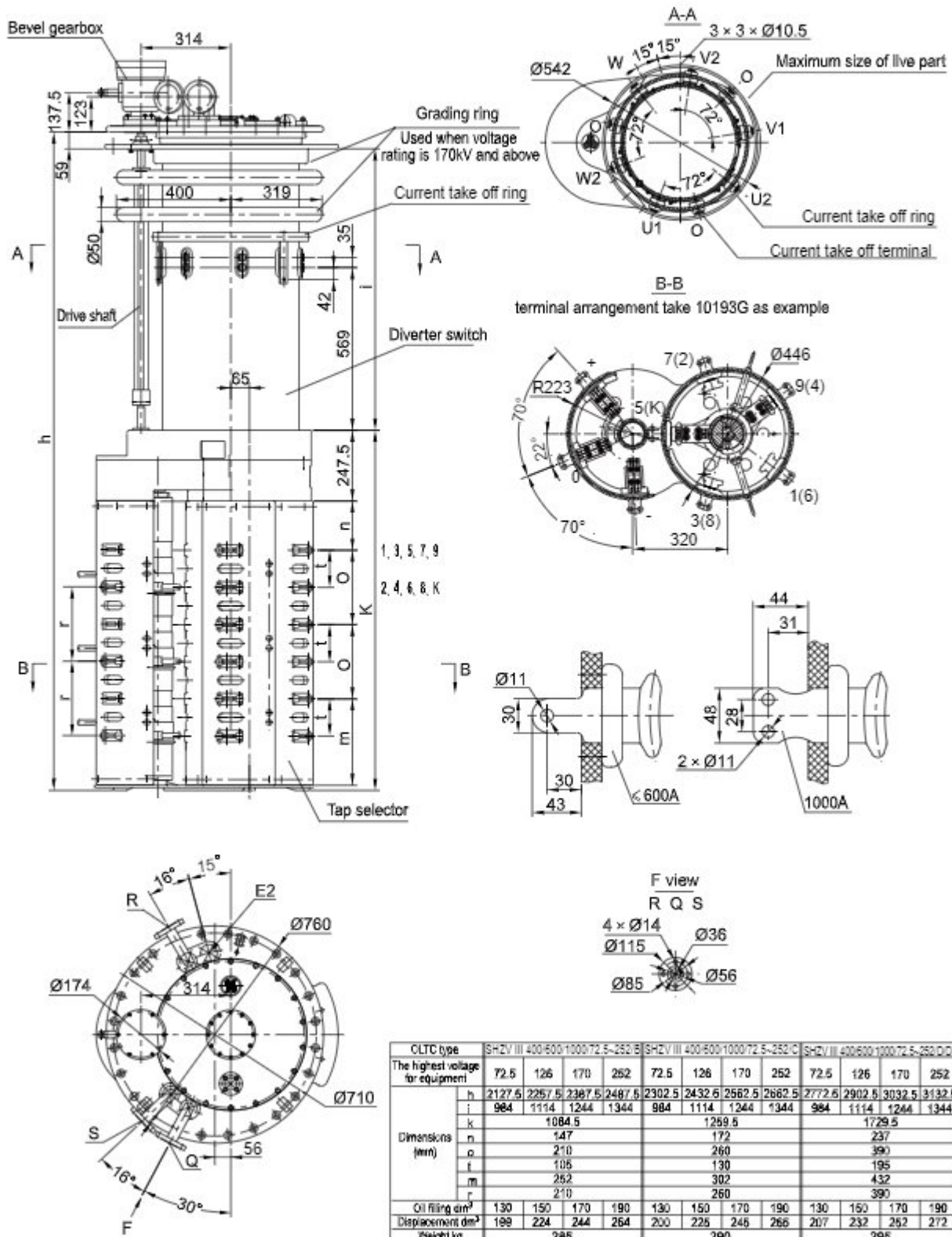


Appendix 2. SHZV III 400/600/1000A overall dimensions with reversing switch, cylinder tap selector



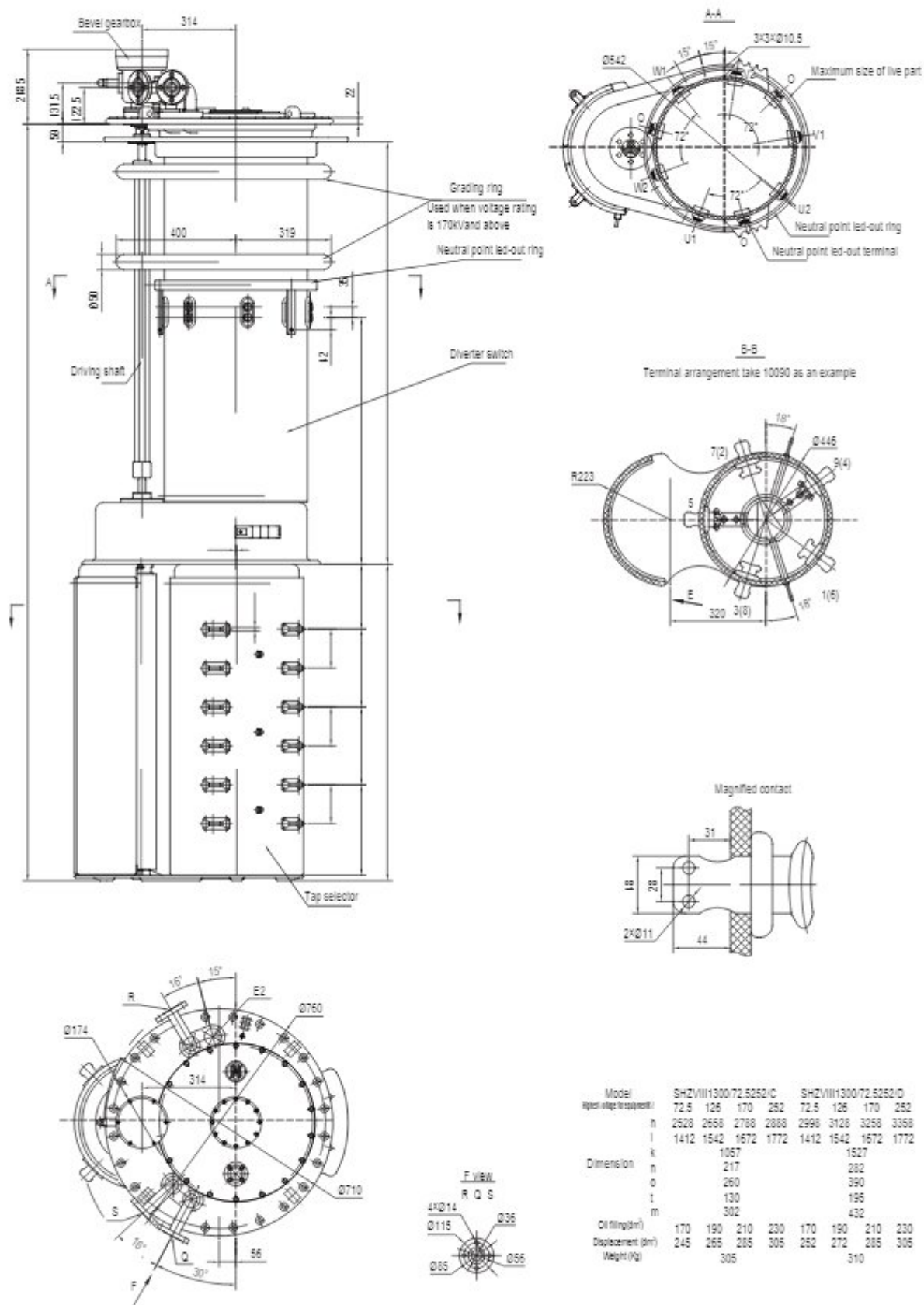
Unit: mm

Appendix 3. SHZV III 400/600/1000A overall dimensions with coarse/fine change-over selector, cylinder tap selector



Unit: mm

Appendix 4. SHZV III 1300A overall dimensions without change-over selector, cylinder type



Front View: Shows the main body of the terminal box with dimensions: 216.5, 131.5, 122.5, 314, 400, 319, 800, 12, 325, 13679, 2468K. Labels include: Bevel gearbox, Grading ring, Used when voltage rating is 170kV and above, Neutral point led-out ring, Diverter switch, Grooving shaft, Tap selector.

Top View: Shows the circular top of the terminal box with dimensions: 15°, 15°, 3x3xØ10.5, Ø542, W1, 72°, V1, U2, U1, Neutral point led-out ring, Neutral point led-out terminal. Label: Maximum size of live part.

Terminal Arrangement: Terminal arrangement take 10193W as an example. Dimensions: 44°, R223, 7(2), Ø445, 9(4), 16°, 1(6), 320, 3(8), 16°. Label: Led-out terminal K.

E view: Shows the end view of the terminal box with dimensions: Ø, B, 16°.

Magnified contact: Shows a detailed view of the contact mechanism with dimensions: 31, 18, 23, 2xØ11, 44.

Bottom View: Shows the bottom of the terminal box with dimensions: 16°, 15°, E2, Ø760, Ø174, 314, Ø710, 56, 30°, F, Q, S, 4xØ14, Ø115, Ø36, Ø65, Ø56.

F view: Shows the front view of the terminal box with dimensions: 4xØ14, Ø115, Ø36, Ø65, Ø56.

Unit: mm

Model

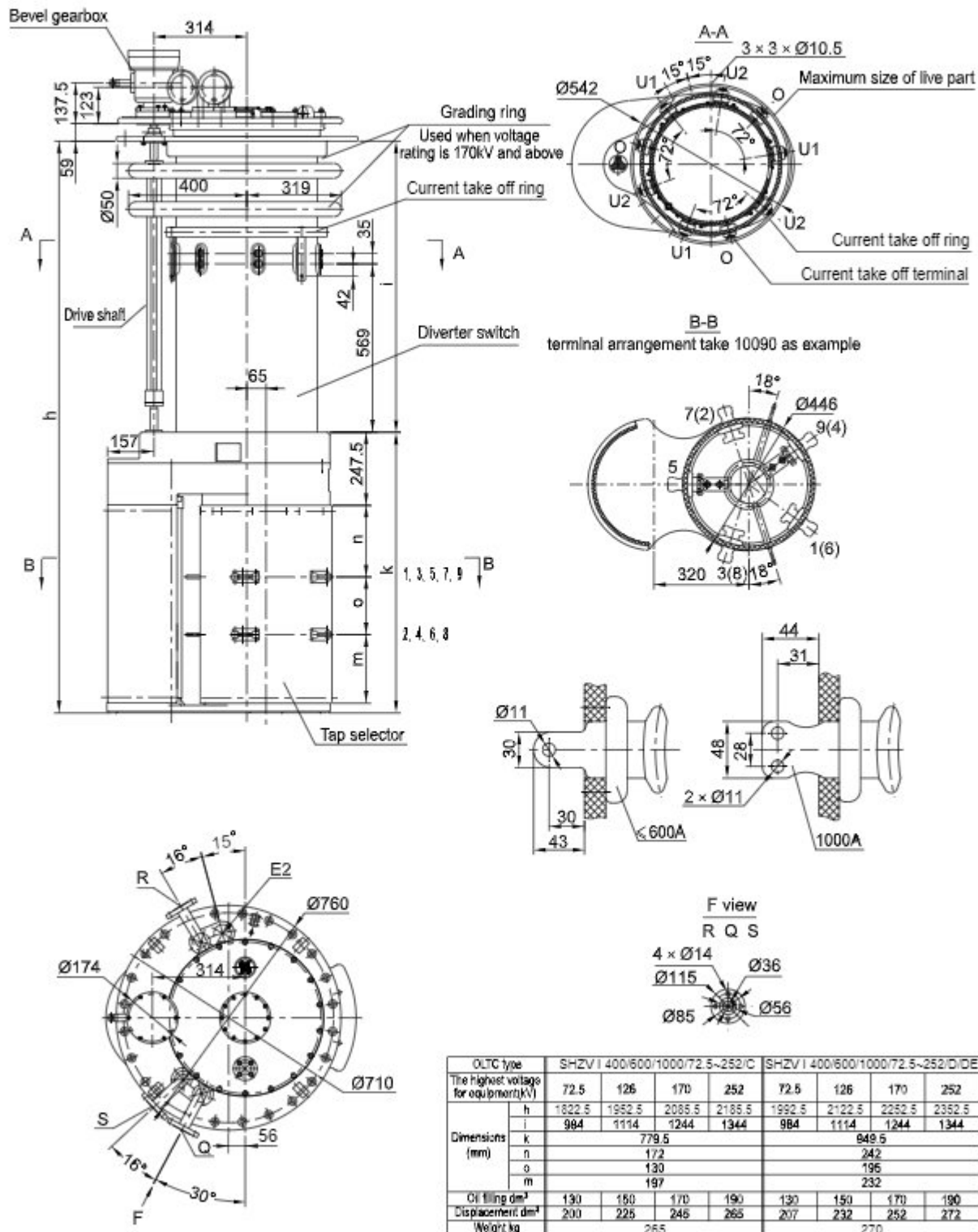
Highest voltage to apply (kV)	SHZVIII1300/72.5252/C	SHZVIII1300/72.5252/D
n	72.5	126
l	2528	2658
k	2768	2888
m	2990	3128
q	3258	3368
r	1412	1542
s	1672	1772
t	1527	1627
u	217	232
v	260	390
w	130	195
x	302	432
y	338	468
z	260	390
Oil filling (cm³)	170	190
Displacement (cm³)	245	265
Weight (kg)	305	310

Unit: mm

OLTC type	SHZV III 600/72.5-262B				SHZV III 600/72.5-252C				SHZV III 600/72.5-262D/E			
The highest voltage for equipment (kV)	72.5	126	170	252	72.5	126	170	252	72.5	126	170	252
h	2059	2169	2319	2419	2234	2364	2494	2594	2669	2819	2949	3049
i	1162	1292	1422	1522	1162	1292	1422	1522	1162	1292	1422	1522
k		897				1072					1527	
n		233				258					323	
o		190				240					370	
t		95				120					185	
m		197				247					377	
r						240					370	
q		255				305					435	
p		783				958					1413	
Oil filling dm ³	130	150	170	190	130	150	170	190	130	150	170	190
Displacement cm ³	199	224	244	264	200	225	245	265	207	232	252	272
Weight kg		280				285					290	

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Appendix 8. SHZV I 400/600/1000A overall dimensions without change-over selector, cylinder tap selector



Unit: mm

Bevel gearbox

Grading ring
Used when voltage rating is 170kV and above

Current take off ring

Drive shaft

Diverter switch

Tap selector

A-A
3 × 3 × Ø10.5
Maximum size of live part
U1 15° 15°
U2 72° 72°
U1
U2
Current take off ring
Current take off terminal

B-B
Terminal arrangement take 10193W as example
18°
Ø446
9(4)
7(2)
R223
44°
44°
5(k)
K
E
320 3(8) 18°
1(6)
8
Current take off

E view
Ø11
Ø18
12
43
≤ 600A
Ø11
Ø18
12
22
61
1000A

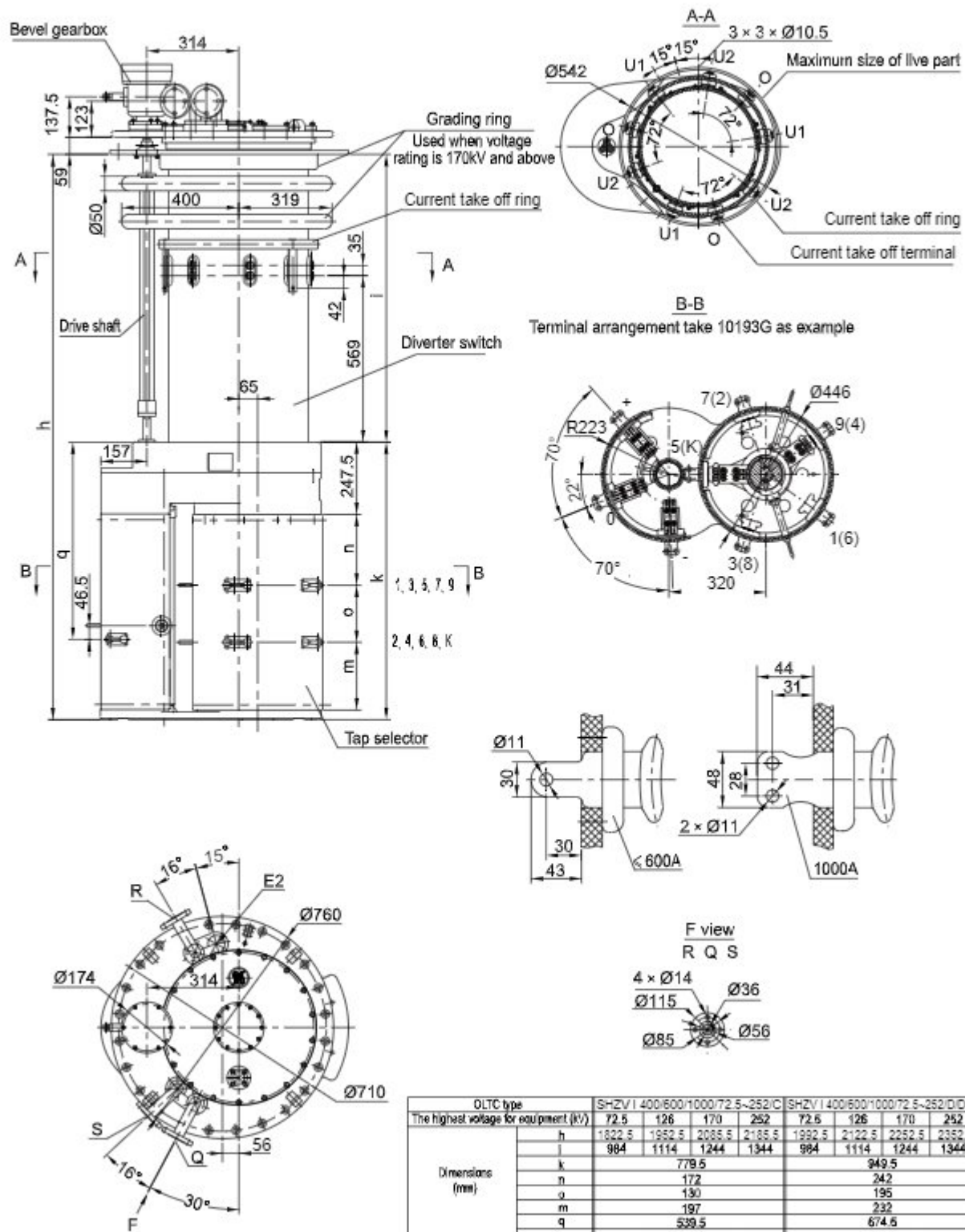
F view
R Q S
4 × Ø14
Ø115
Ø36
Ø85
Ø56

Dimensions (mm)

OLTC type	SHZV / 400/600/1000/72.5~252/C				SHZV / 400/600/1000/72.5~252/D/DE			
The highest voltage for equipment (kV)	72.5	126	170	252	72.5	126	170	252
h	1822.5	1952.5	2085.5	2185.5	1992.5	2122.5	2252.5	2352.5
i	964	1114	1244	1344	984	1114	1244	1344
k	779.5				949.5			
n	172				242			
c	130				195			
m	187				232			
q	539.5				674.5			
Oil filling dm³	130	150	170	190	130	150	170	190
Displacement dm³	200	225	245	265	207	232	252	272
Weight kg	270				275			

Unit: mm

Appendix 10. SHZV I 400/600/1000A overall dimensions with coarse/fine change-over selector, cylinder tap selector



Unit: mm

Bevel gearbox

314

137.5

123

59

Ø50

400

319

35

42

569

65

157

247.5

n

k

o

m

1, 3, 5, 7, 9

2, 4, 6, 8, K

Grading ring
Used when voltage rating is 170kV and above

Current take off ring

Drive shaft

Diverter switch

Tap selector

A-A

3 × 3 × Ø10.5

Ø542

U1 15° 15°

U2

Maximum size of live part

U1

U2

Current take off ring

Current take off terminal

B-B

Terminal arrangement take 10090 as example

18°

Ø446

7(2)

9(4)

3(8)

1(6)

320

327

1209.5

126

170

252

2082.5

2212.5

2342.5

2442.5

2252.5

2382.5

2512.5

2612.5

984

1114

1244

1344

44

31

48

28

2 × Ø11

K view

F view

R Q S

4 × Ø14

Ø115

Ø36

Ø85

Ø56

Ø11

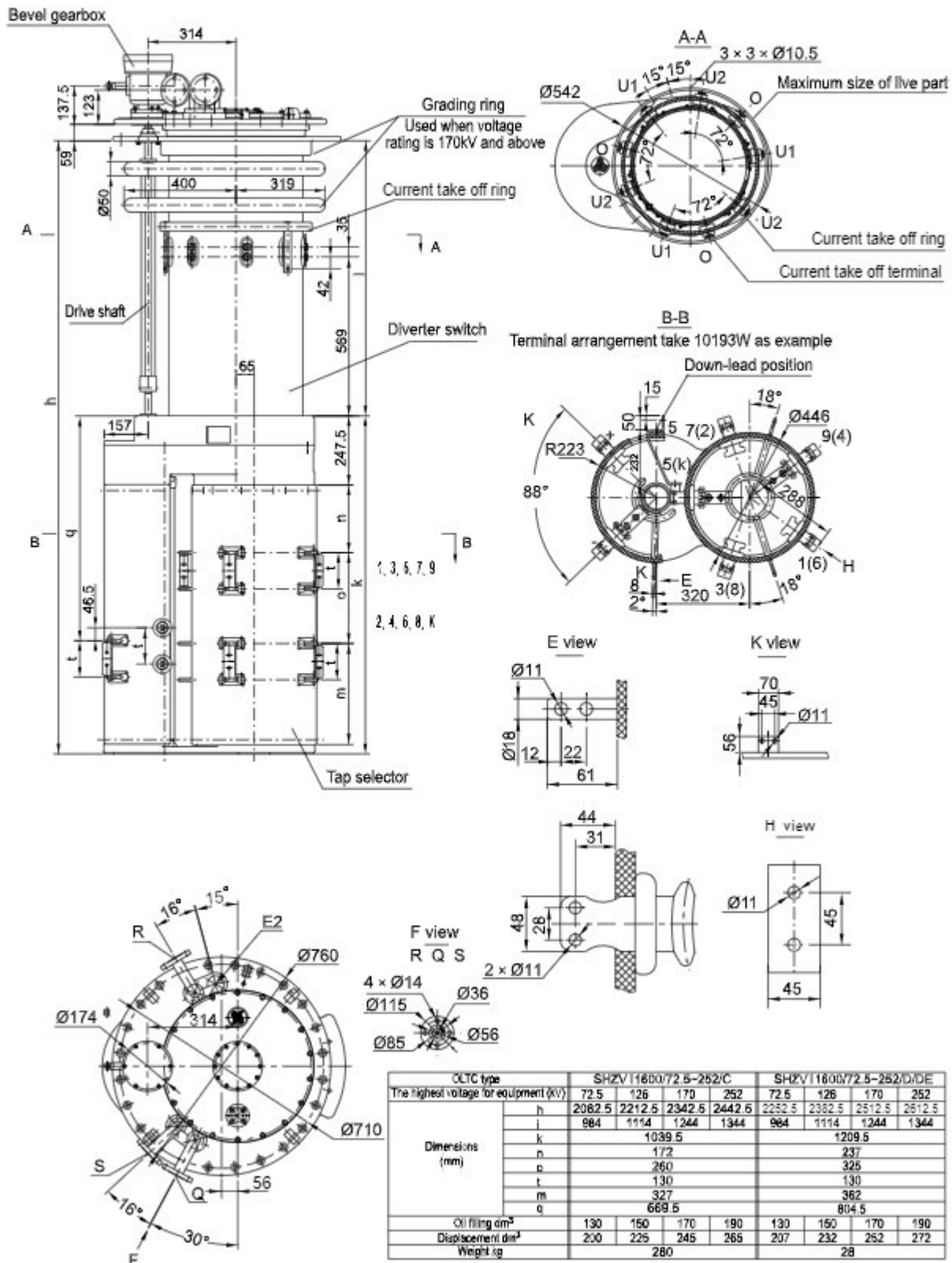
45

45

CLTC type		SHZV11800/72.5-252/C				SHZV11800/72.5-252/D/DE			
The highest voltage for equipment (kV)		72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	2082.5	2212.5	2342.5	2442.5	2252.5	2382.5	2512.5	2612.5
	l	984	1114	1244	1344	984	1114	1244	1344
	k	1039.5				1209.5			
	n	172				237			
	o	280				325			
	t	130				130			
Oil filling dm³		130	150	170	190	130	150	170	190
	Displacement dm³	200	225	245	265	207	232	252	272
	Weight kg	275				280			

Unit: mm

Appendix 12. SHZV I 1600A overall dimensions with reversing switch, cylinder tap selector



Unit: mm

Bevel gearbox

Grading ring

Used when voltage rating is 170kV and above

Current take off ring

Drive shaft

Diverter switch

Tap selector

A-A

314

137.5

123

59

400

319

42

569

65

157

247.5

46.5

11

13

5

7

9

2

4

6

8

K

B-B

Terminal arrangement take 10193G as example

Down lead position

Ø542

U1 15° 15° U2

Maximum size of live part

U1

U2

Current take off ring

Current take off terminal

F view

R Q S

4 × Ø14

Ø115

Ø36

Ø85

Ø56

Ø174

314

Ø760

Ø710

56

16°

15°

16°

30°

F

Q

S

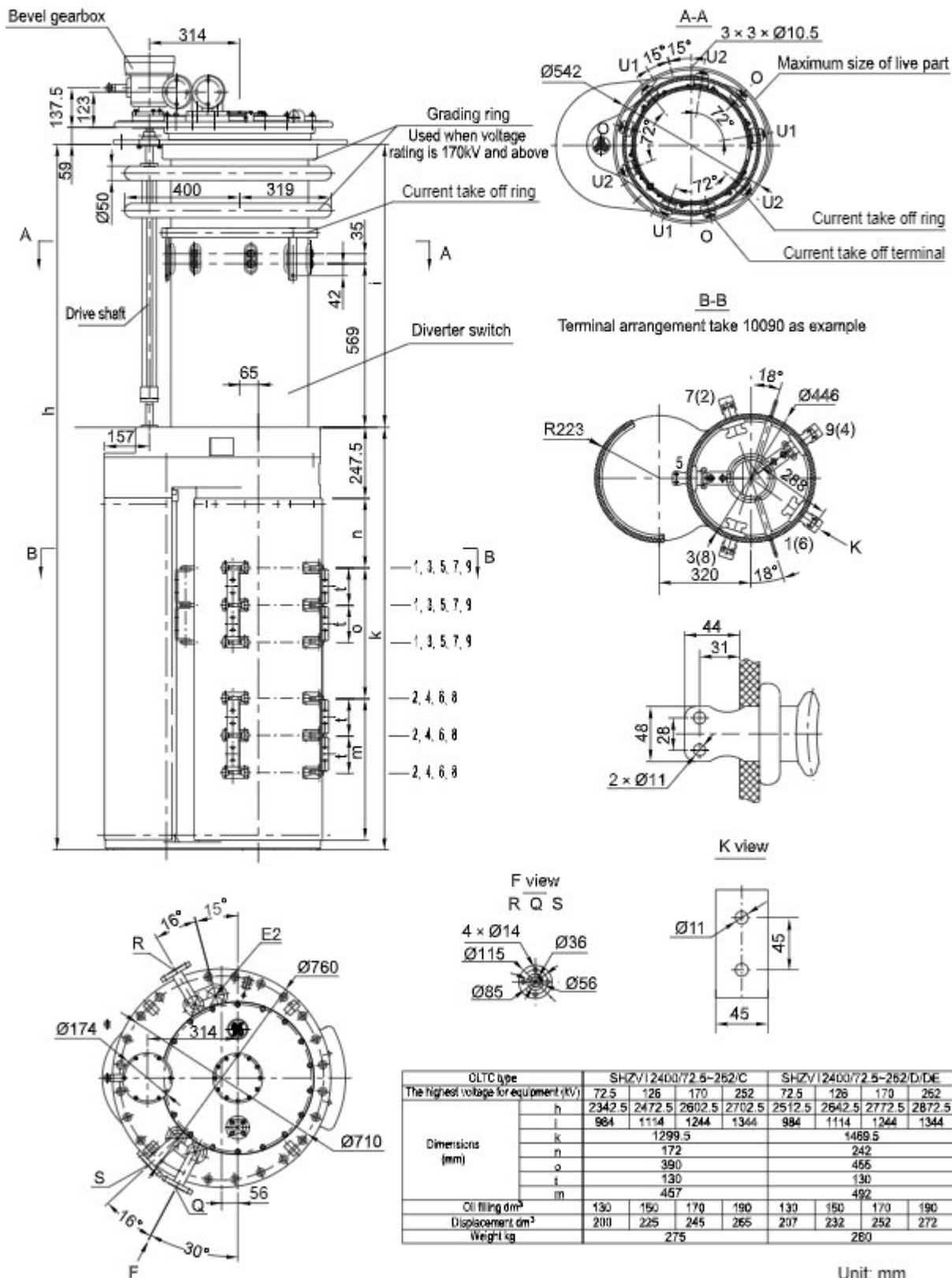
E2

R

CLTC type		SHZV 11600/72.5-252/C				SHZV 11600/72.5-252/D/DE			
The highest voltage for equipment (kV)		72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	2082.5	2212.5	2342.5	2442.5	2252.5	2382.5	2512.5	2612.5
	i	984	1114	1244	1344	984	1114	1244	1344
	k	1039.5				1209.5			
	n	172				237			
	o	260				325			
	p	130				130			
	t	327				362			
	m	689.5				804.5			
	q								
	Oil filling dm ³	130	150	170	190	130	150	170	190
Displacement dm ³	200	225	245	265	207	232	252	272	
Weight kg	280				285				

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**Appendix 14. SHZV I 2400A overall dimensions
without change-over selector, cylinder tap selector**



Bevel gearbox
314
137.5
123
59
Ø50
400
319
35
42
569
65
157
247.5
h
q
B
n
k
m
r
t
1.3, 5, 7, 9
1.3, 5, 7, 9
1.3, 5, 7, 9
2, 4, 6, 8, K
2, 4, 6, 8, K
2, 4, 6, 8, K

Grading ring
Used when voltage rating is 170kV and above

Current take off ring

Drive shaft

Diverter switch

A-A
3 × 3 × Ø10.5
Ø542
U1 15° 15°
U2
O
Maximum size of live part
U1
U2
Current take off ring
Current take off terminal

B-B
Terminal arrangement take 10193W as example
Down-lead position
K
R223
88°
15
50
5
7(2)
Ø446
9(4)
288
18°
H
1(6)
E
320
3(8)
18°
K view
70
45
Ø11
56
t
E view
Ø11
Ø18
12
22
61
H view
Ø11
45
45
44
31
48
28
2 × Ø11

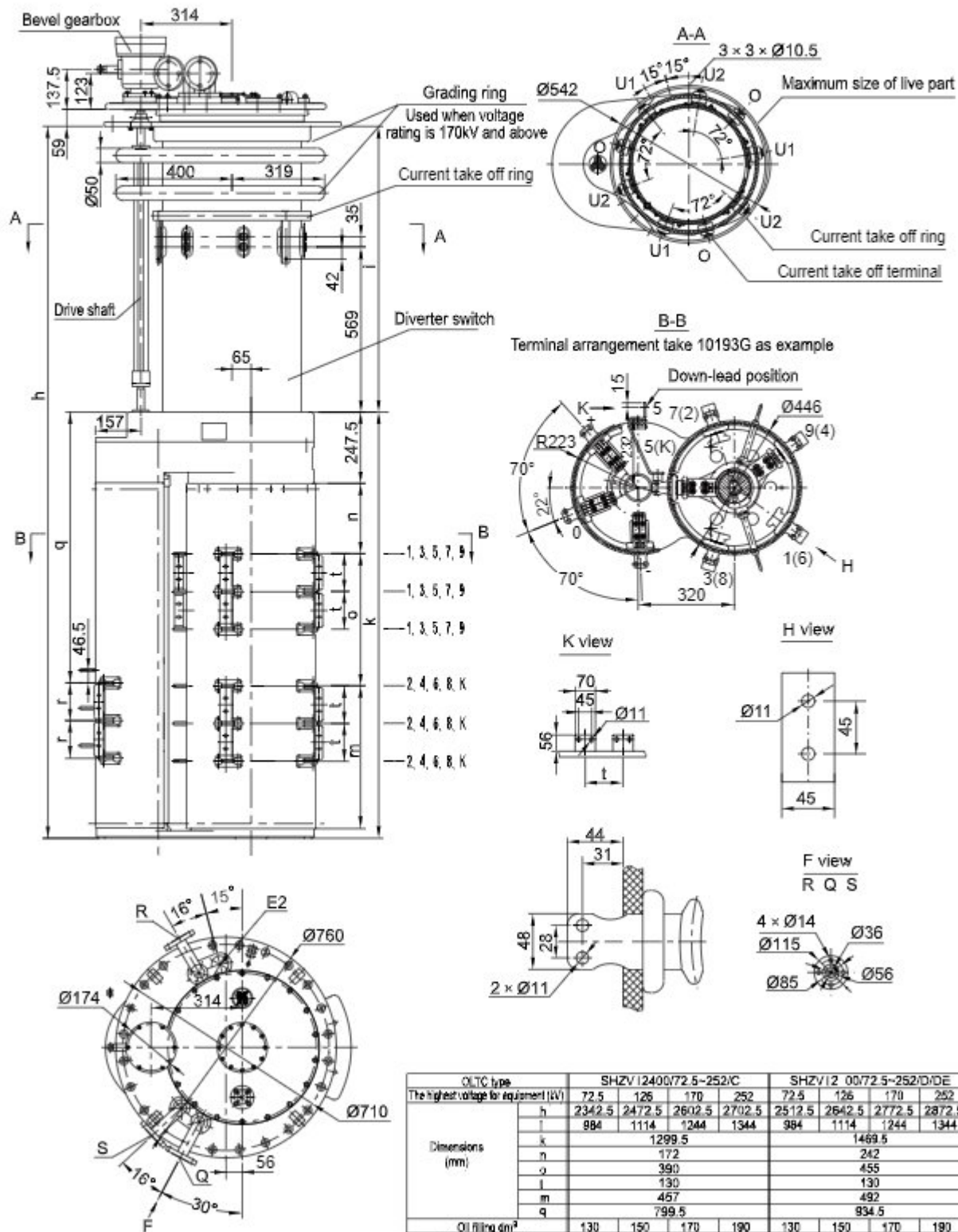
F view
R Q S
4 × Ø14
Ø36
Ø85
Ø56

Dimensions (mm)
h
k
n
q
t
m
q

OLTC type	SHZV 12400/72.5-252/C				SHZV 12400/72.5-252/D/DE			
The highest voltage for equipment (kV)	72.5	126	170	252	72.5	126	170	252
h	2342.5	2472.5	2602.5	2702.5	2512.5	2642.5	2772.5	2872.5
k	984	1114	1244	1344	984	1114	1244	1344
n		1299.5				1469.5		
q		172				242		
t		390				455		
m		130				130		
q		457				492		
q		799.5				934.5		
Oil filling dm ³	130	150	170	190	130	150	170	190
Displacement dm ³	200	225	245	265	207	232	252	272
Weight kg		290				295		

Unit: mm

**Appendix 16. SHZV I 2400A overall dimensions
with coarse/fine change-over selector, cylinder tap selector**



Unit: mm

Front View (A-A)

- Bevel gearbox
- Grading ring
- Neutral point led-out ring
- Driving shaft
- Diverter switch
- Tap selector

Top View (B-B)

- Terminal arrangement take 10090 as an example

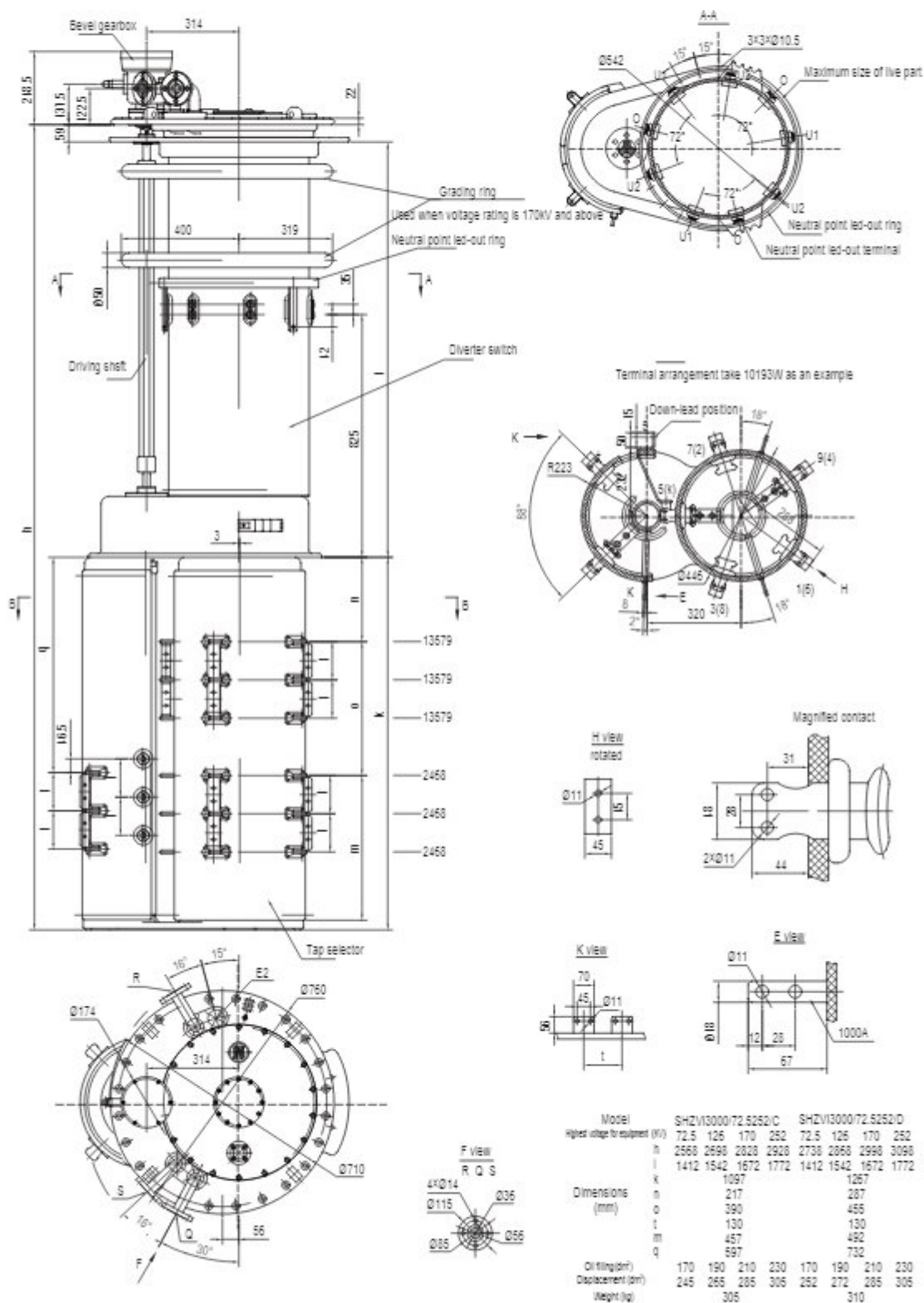
Magnified contact

K view

Model	SHZVI 3000/72.5kV/C				SHZVI3000/72.5kV/D			
Rated voltage for equipment (kV)	72.5	126	170	252	72.5	126	170	252
n	2668	2698	2828	2928	2738	2868	2998	3098
l	1412	1542	1672	1772	1412	1542	1672	1772
k			1097				1267	
n			217				287	
o			390				455	
t			130				130	
m			457				492	
Oil filling (dm ³)	170	190	210	230	170	190	210	230
Displacement (dm ³)	245	265	285	305	252	272	285	305
Weight (kg)			305				310	

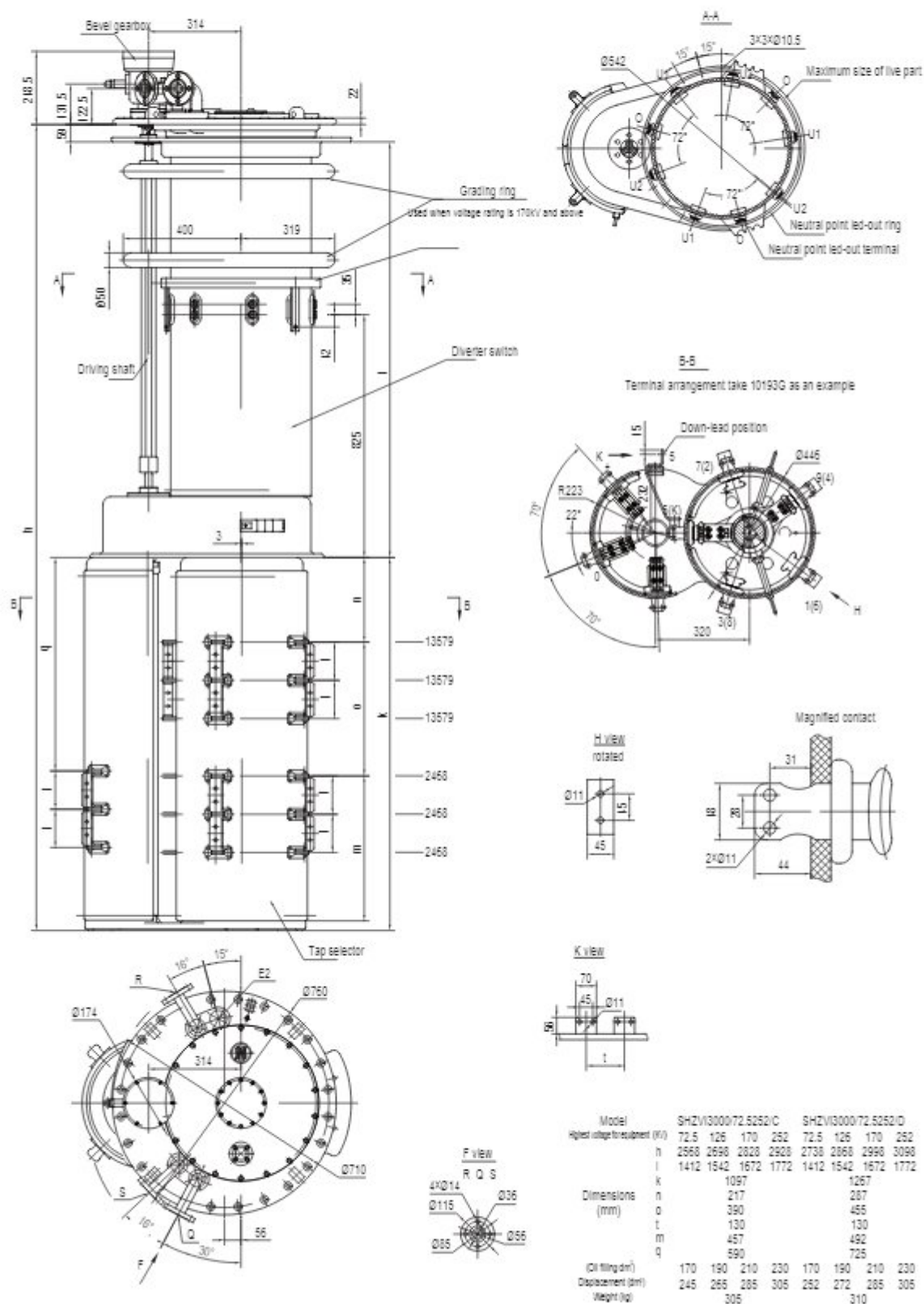
Unit: mm

Appendix 18. SHZV 1300A overall dimensions with reversing change-over selector, cylinder type



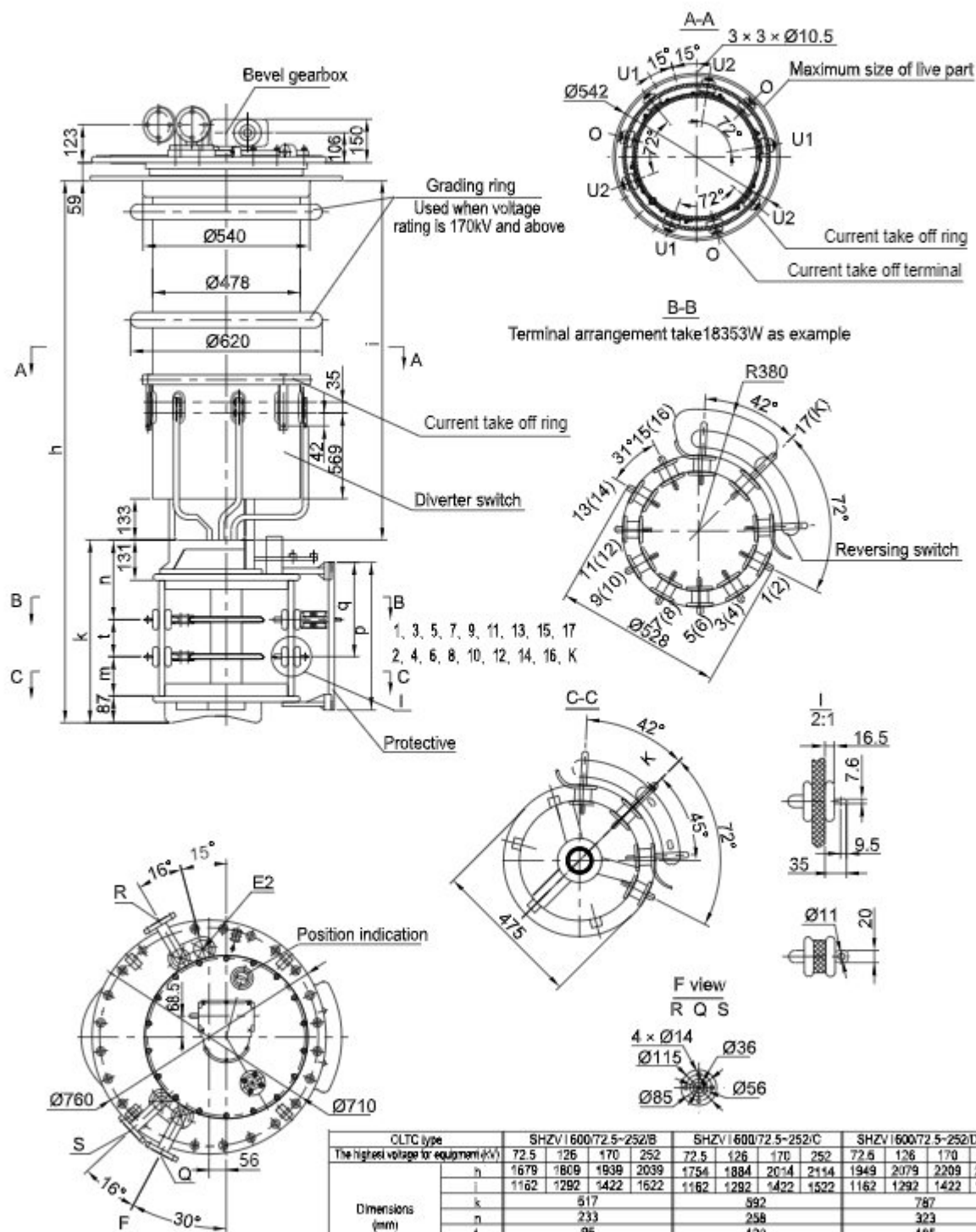
Unit: mm

Appendix 19. SHZV 3000A overall dimensions with coarse & fine change-over selector, cylinder type



Unit: mm

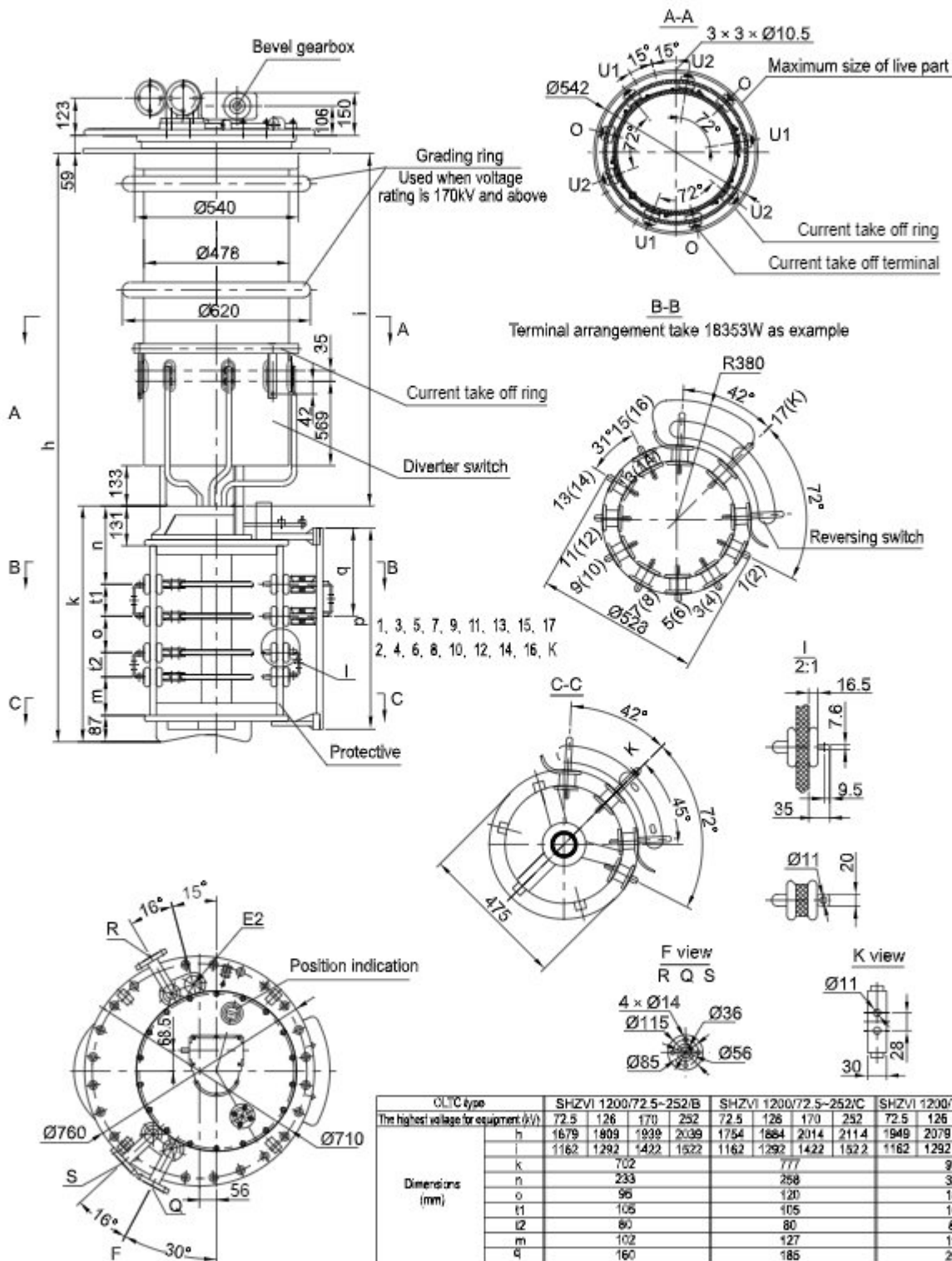
**Appendix 20. SHZV I 400/600A overall dimensions
with reversing switch, cage tap selector**



CLTC type		SHZV1600/72.5-252/B				SHZV1600/72.5-252/C				SHZV1600/72.5-252/D/E			
The highest voltage for equipment (kV)		72.5	126	170	252	72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	1679	1809	1939	2039	1754	1884	2014	2114	1949	2079	2209	2309
	i	1162	1292	1422	1522	1162	1292	1422	1522	1162	1292	1422	1522
	k			817				592				787	
	n			233				258				323	
	t			95				120				185	
	m			102				127				192	
	q			160				185				250	
	p			403				478				673	
Oil filling dm ³		130	160	170	190	130	160	170	190	130	160	170	190
Displacement dm ³		199	224	244	264	200	225	245	265	207	232	252	272
Weight kg		265				270				275			

Unit: mm

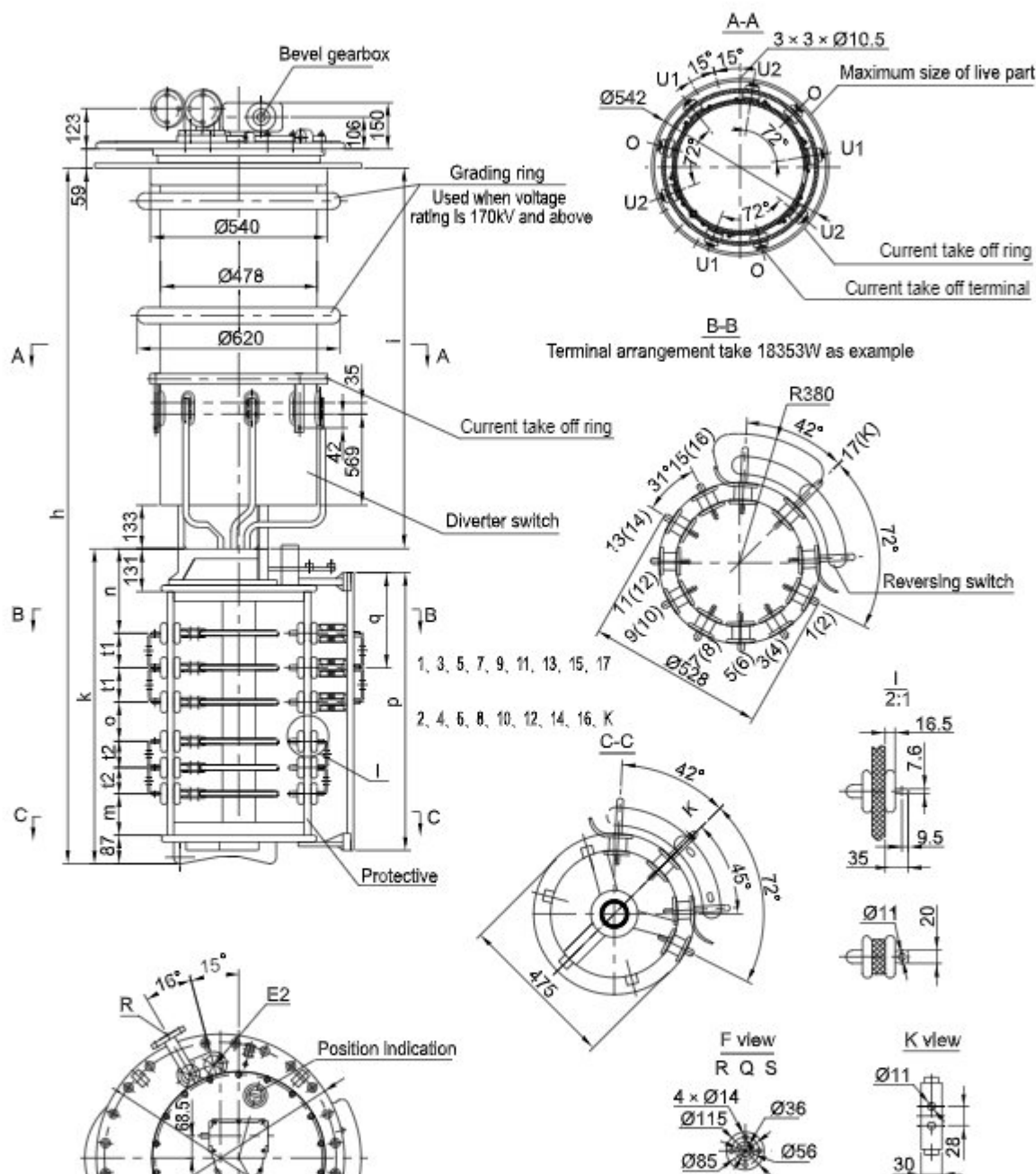
Appendix 21. SHZV I 1200A overall dimensions with reversing switch, cage tap selector



CLTC type		SHZVI 1200/72.5-252/B				SHZVI 1200/72.5-252/C				SHZVI 1200/72.5-252/D/E			
The highest voltage for equipment (kV)		72.5	126	170	252	72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	1679	1909	1939	2039	1754	1884	2014	2114	1949	2079	2209	2309
	i	1162	1292	1422	1522	1162	1292	1422	1522	1162	1292	1422	1522
	k	702				777				972			
	n	233				258				323			
	o	96				120				186			
	t1	106				106				106			
	t2	80				80				80			
	m	102				127				192			
	q	160				185				250			
	p	568				663				858			
Oil filling cm ³		130	150	170	190	130	150	170	190	130	150	170	190
Displacement cm ³		199	224	244	264	200	225	245	265	207	232	252	272
Weight kg		275				280				285			

Unit: mm

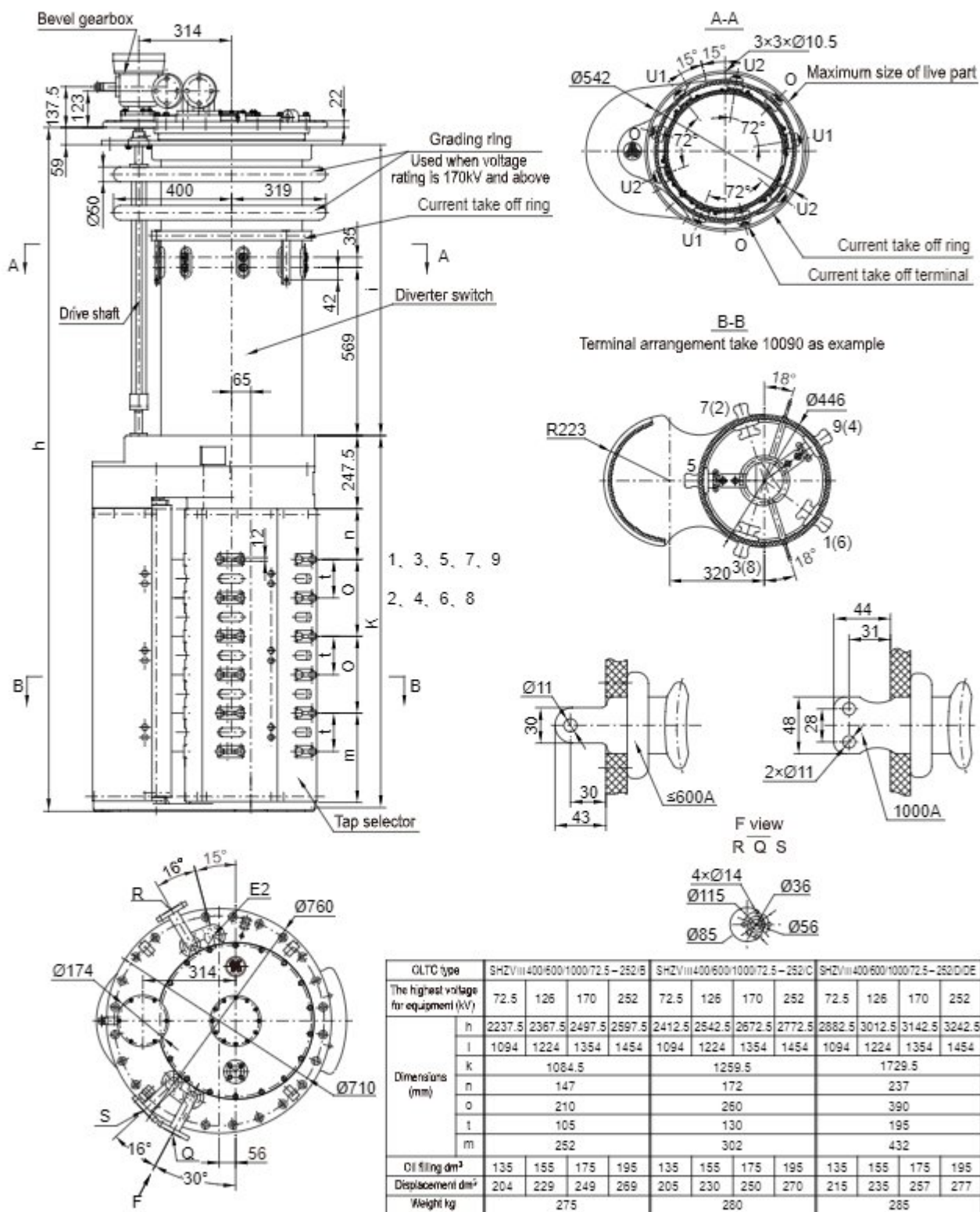
Appendix 22. SHZV I 1500A overall dimensions with reversing switch, cage tap selector



OLTC type	SHZV I 1500/72.5-252/B	SHZV I 1500/72.5-252/C	SHZV I 1500/72.5-252/D/DE
The highest voltage for equipment (kV)	72.5 126 170 252	72.5 126 170 252	72.5 126 170 252
h	2049 2179 2309 2409	2124 2264 2384 2484	2319 2449 2579 2679
i	1162 1292 1422 1522	1162 1292 1422 1522	1162 1292 1422 1522
k	887	962	1157
n	233	258	323
o	95	120	185
t1	105	105	105
t2	80	80	80
m	102	127	192
q	160	185	260
p	773	848	1043
Oil filling dm³	130 150 170 190	130 150 170 190	130 150 170 190
Displacement dm³	199 224 244 264	200 225 245 265	207 232 252 272
Weight kg	280	285	290

Unit: mm

Appendix 23. SHZV III 400/600/1000A (ZnO protection) overall dimensions without change-over selector, cylinder tap selector



E2: Bleeding of transformer tank

R: Connecting flange for protective relay

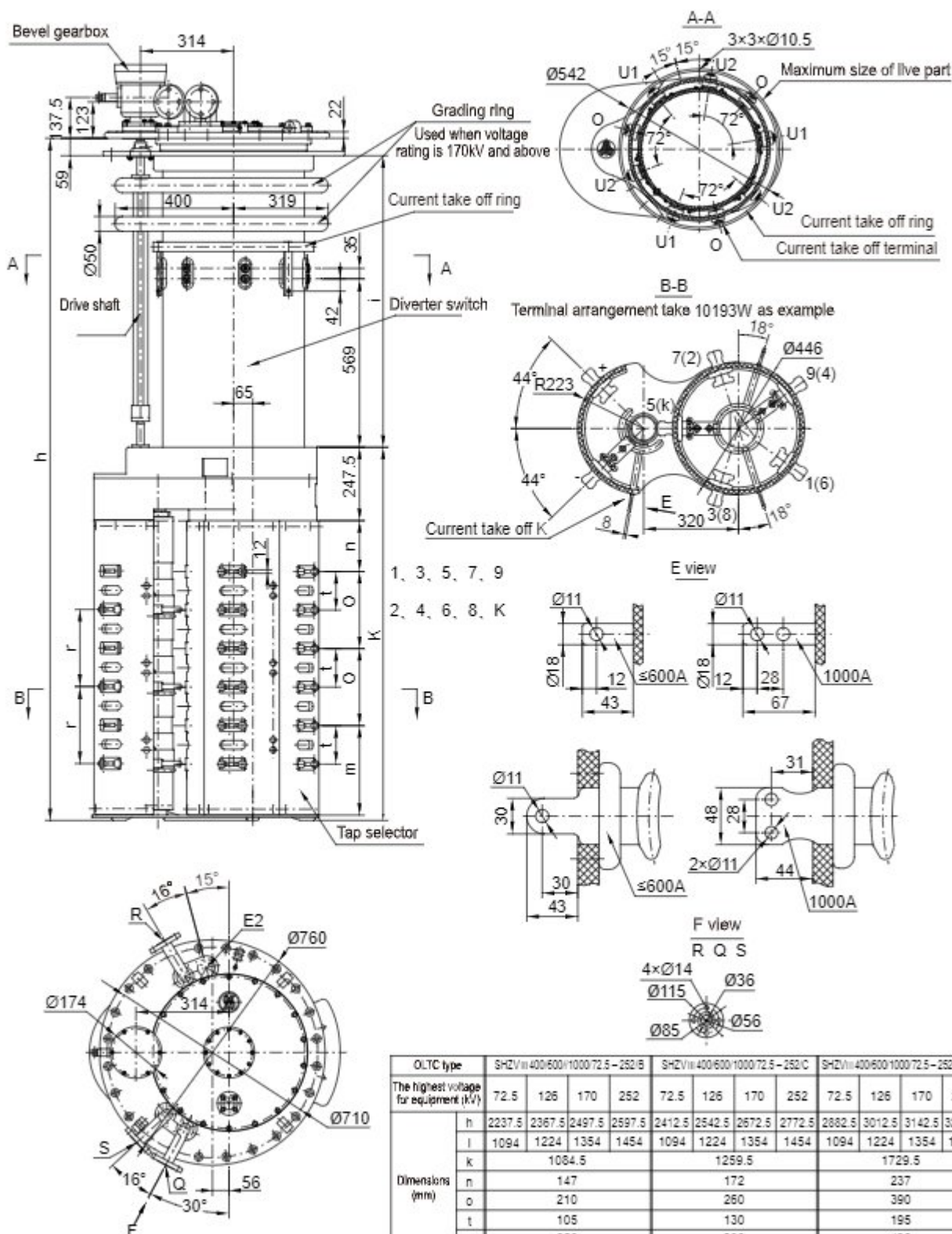
S: Connecting flange for suction pipe

Q: Connecting flange for oil return

Note: All connection pipe and upper gearbox can be turned in 360°

Unit: mm

Appendix 24. SHZV III 400/600/1000A (ZnO protection) overall dimensions with reversing switch, cylinder tap selector



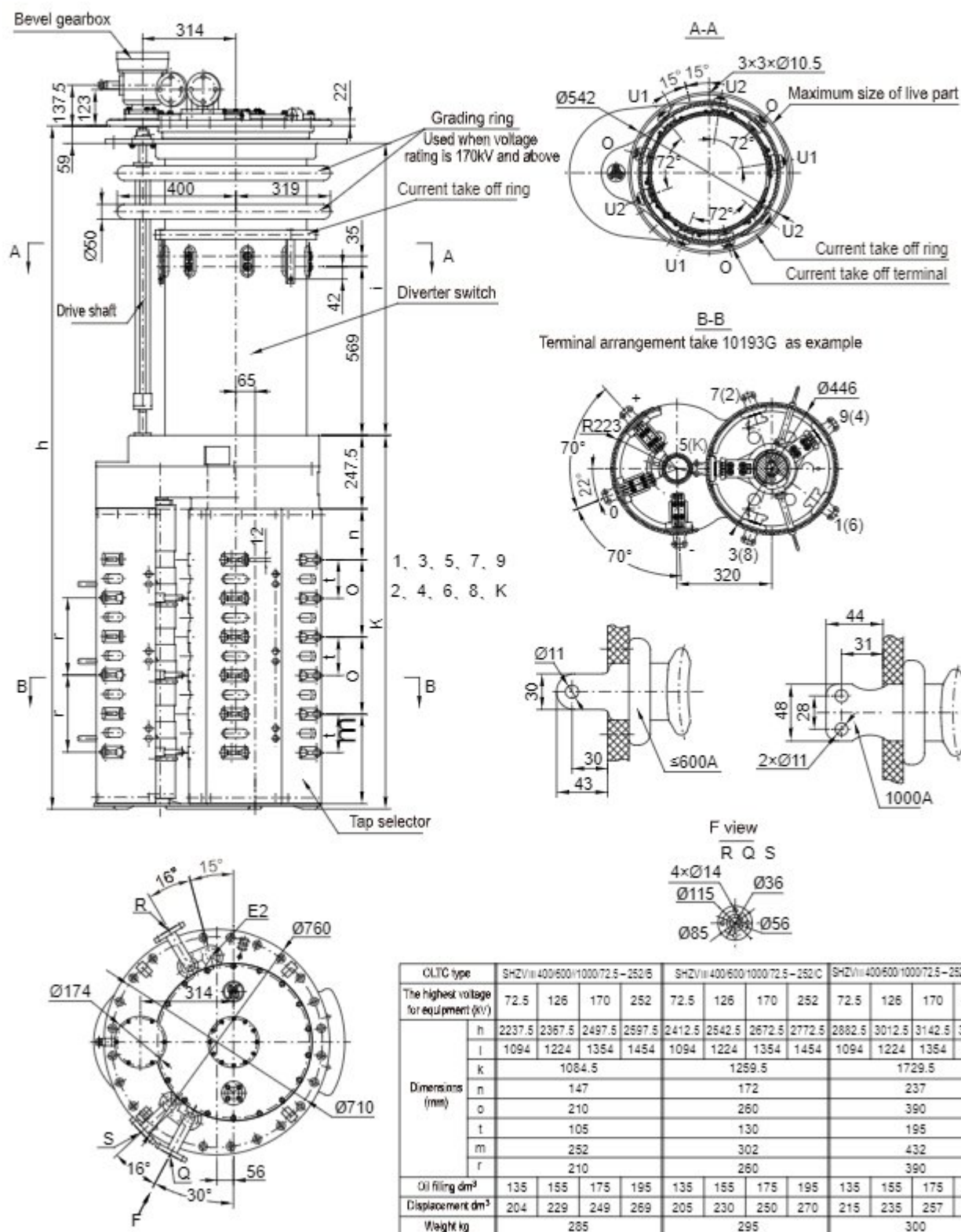
E2: Bleeding of transformer tank
R: Connecting flange for protective relay
S: Connecting flange for suction pipe
Q: Connecting flange for oil return

Note: All connection pipe and upper gearbox can be turned in 360°

OLTC type	SHZV/III 400/600/1000/72.5-252/B				SHZV/III 400/600/1000/72.5-252/C				SHZV/III 400/600/1000/72.5-252/D/DE				
The highest voltage for equipment (kV)	72.5	126	170	252	72.5	126	170	252	72.5	126	170	252	
Dimensions (mm)	h	2237.5	2367.5	2497.5	2597.5	2412.5	2542.5	2672.5	2772.5	2882.5	3012.5	3142.5	3242.5
	l	1094	1224	1354	1454	1094	1224	1354	1454	1094	1224	1354	1454
	k	1084.5				1259.5				1729.5			
	n	147				172				237			
	o	210				260				390			
	t	105				130				195			
	m	252				302				432			
r	210				260				390				
Oil filling dm ³	135	155	175	195	135	155	175	195	135	155	175	195	
Displacement dm ³	204	229	249	269	205	230	250	270	215	235	257	277	
Weight kg	285				295				300				

Unit: mm

Appendix 25. SHZV III 400/600/1000A (ZnO protection) overall dimensions with coarse/fine change-over selector, cylinder tap selector



E2: Bleeding of transformer tank R: Connecting flange for protective relay

S: Connecting flange for suction pipe Q: Connecting flange for oil return

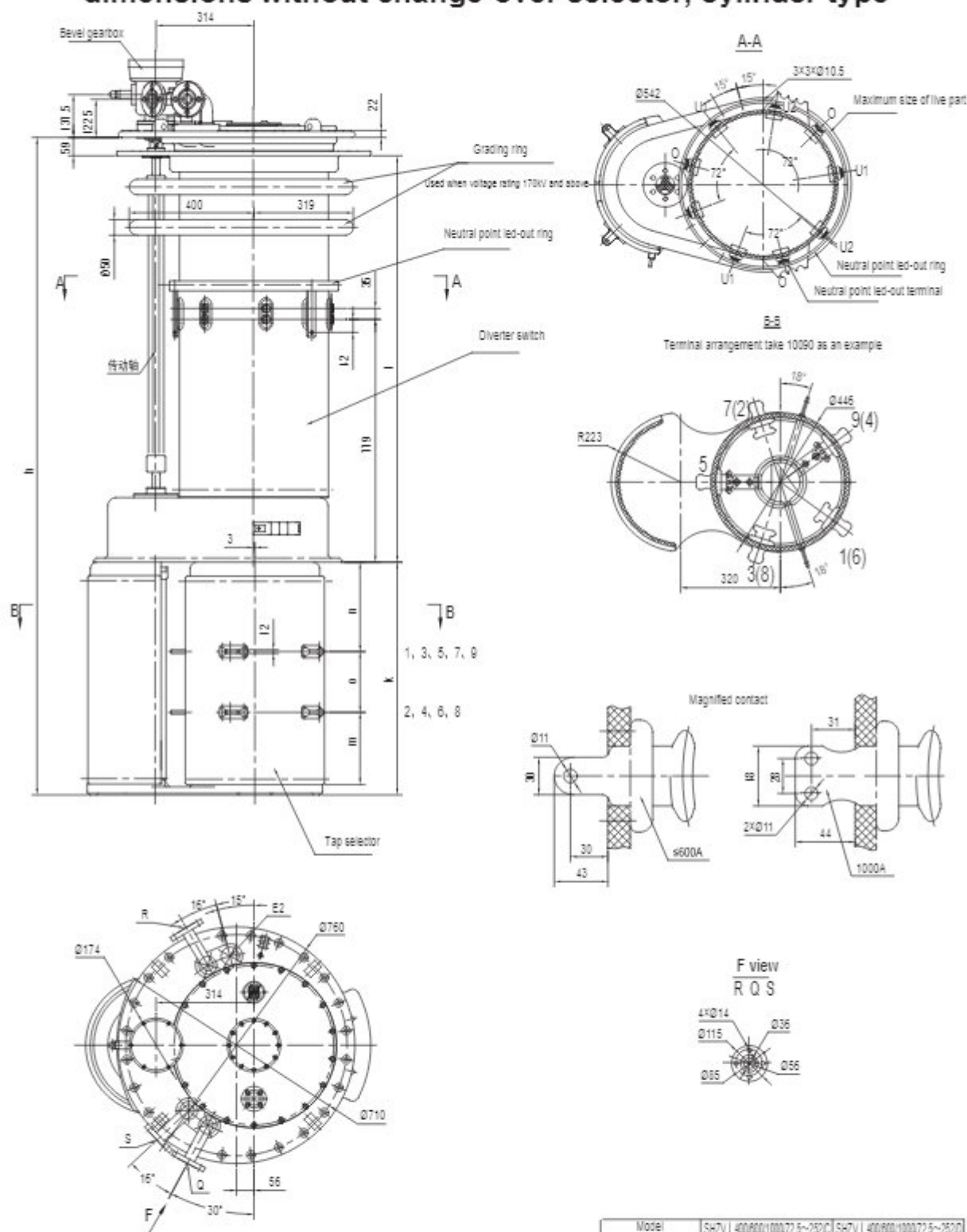
Note: All connection pipe and upper gearbox can be turned in 360°

Unit: mm

OLT type	SHZV III 350/600/600/72.5 – 252/B				SHZV III 350/600/600/72.5 – 252/C				SHZV III 350/600/600/72.5 – 252/D/DE				
The highest voltage for equipment (kV)	72.5	126	170	252	72.5	126	170	252	72.5	126	170	252	
Dimensions (mm)	h	2169	2299	2429	2529	2344	2474	2604	2704	2799	2929	3059	3159
	l	1272	1402	1532	1622	1272	1402	1532	1622	1272	1402	1532	1622
	k	897				1072				1527			
	n	233				258				323			
	o	190				240				370			
	t	95				120				185			
	m	197				247				377			
	r	190				240				370			
	q	255				305				435			
	p	783				958				1413			
Oil filling dm³	135	155	175	195	135	155	175	195	135	155	175	195	195
Displacement dm³	204	229	249	269	205	230	250	270	212	237	257	277	277
Weight kg	285				290				295				

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Appendix 27. SHZV I 1000A (ZnO protection) overall dimensions without change-over selector, cylinder type

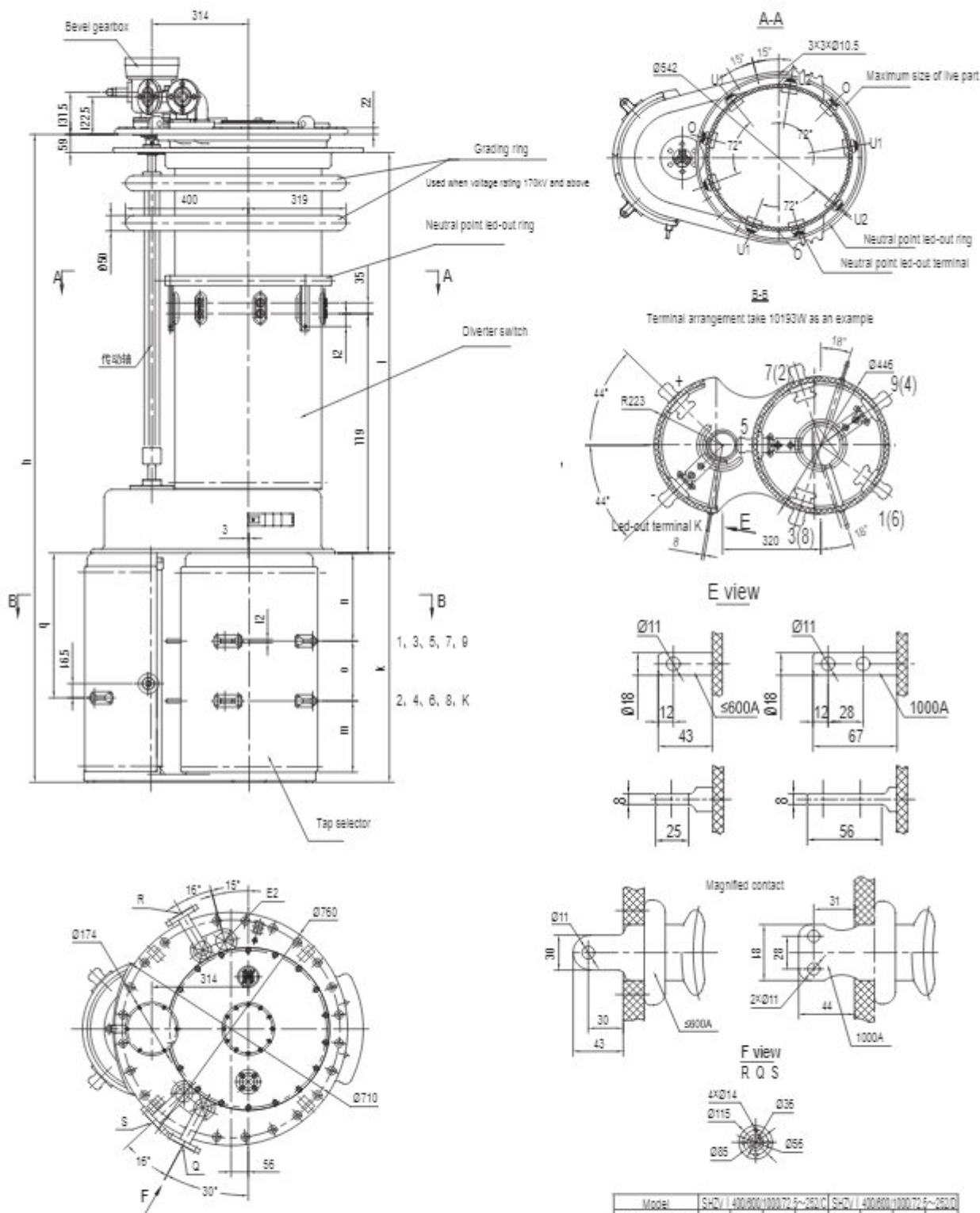


E2: Bleeding of transformer tank
R: Connecting flange for protective relay
S: Connecting flange for suction pipe
Q: Connecting flange for oil return
Note: All connection pipes and upper gearbox can be turned in 360 degrees.

Model		SHV / 400/600/1000/225~252C				SHV / 400/600/1000/225~252D			
Signal voltage to equipment (V)		72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	1940	2070	2200	2300	2110	2240	2370	2470
	i	1304	1434	1584	1684	1304	1434	1584	1684
	k		577				747		
	n		217				287		
	o		130				195		
	m		197				232		
Drilling (mm)		130	150	170	190	130	150	170	190
Displacement (mm)		200	225	245	265	207	232	252	272
Weight (kg)			230				275		

Unit: mm

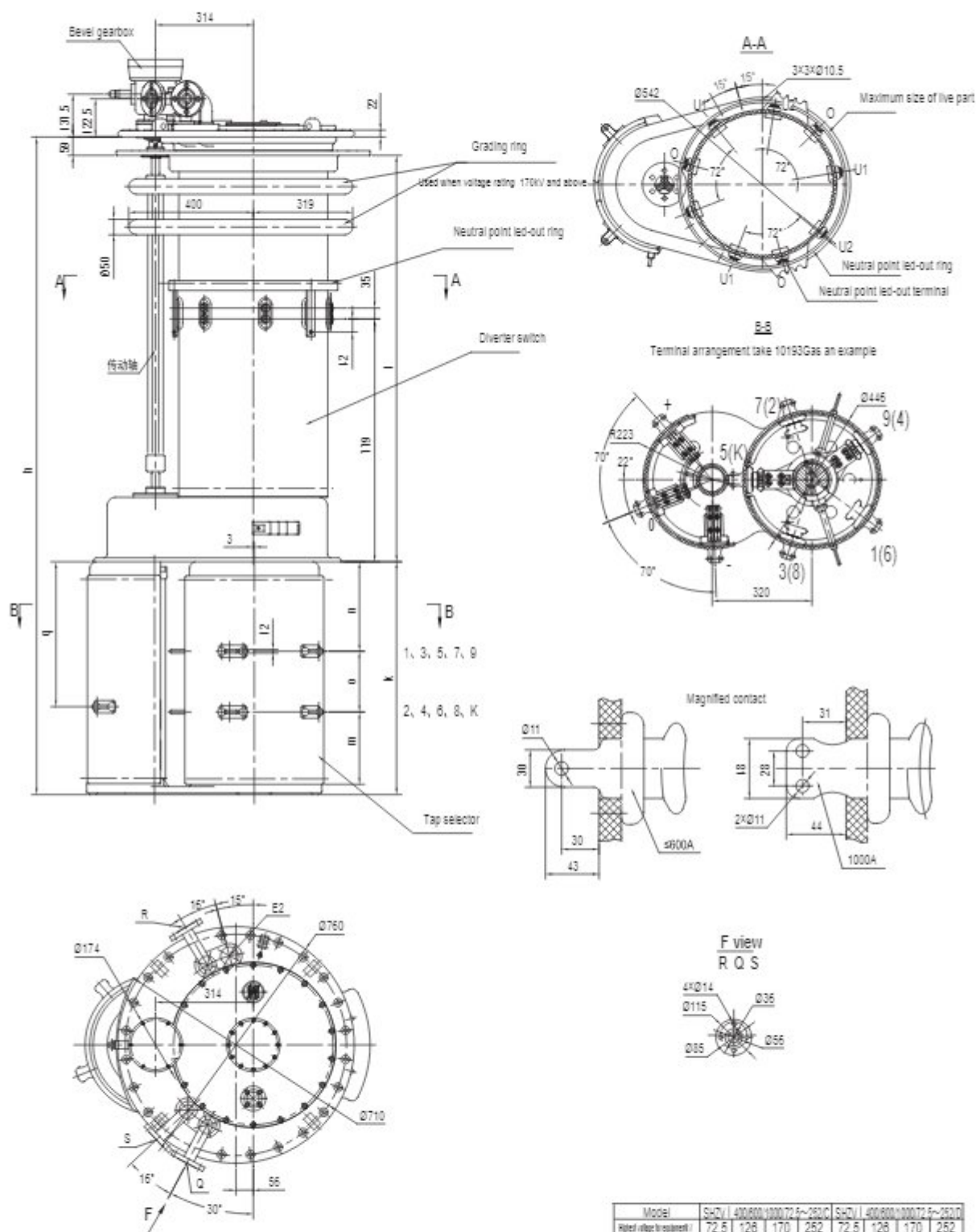
Appendix 28. SHZV I 1000A (ZnO protection) overall dimensions with reversing switch, cylinder type



Model	SHZV I 400/600/1000/1250-252/C				SHZV I 400/600/1000/1250-252/C			
Rated voltage (kV)	72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	1940	2070	2200	2300	2110	2240	2370
	i	1304	1434	1584	1684	1304	1434	1584
	k		577				747	
	n		217				287	
	o		130				165	
	m		197				232	
q		330				472		
Oil filling (dm)	130	160	170	180	130	160	170	180
Displacement (dm)	200	226	245	265	207	232	252	272
Weight (kg)			270				275	

Unit: mm

Appendix 29. SHZV I 1000A (ZnO protection) overall dimensions with coarse & fine change-over selector, cylinder type

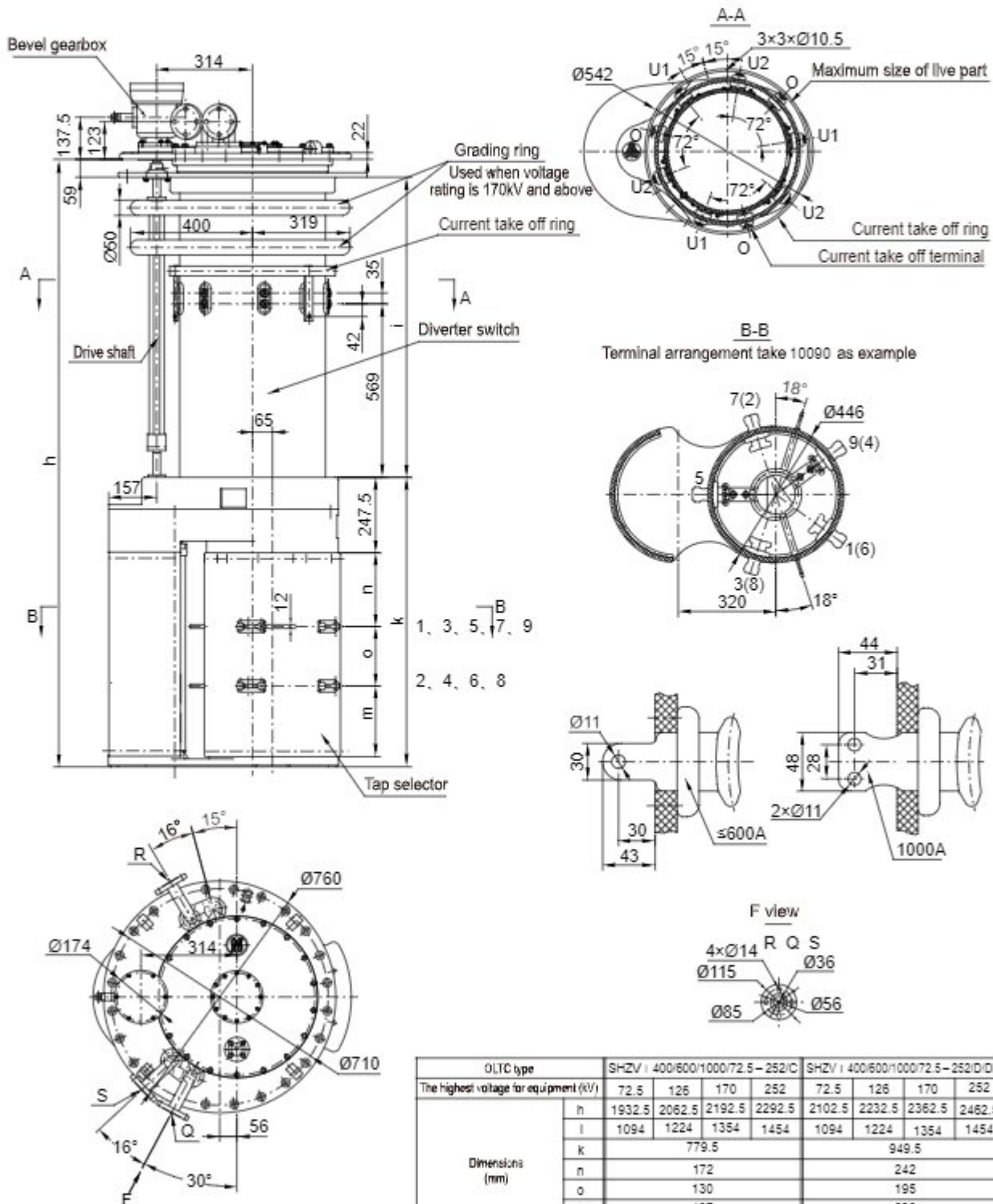


E2: Bleeding of transformer tank
R: Connecting flange for protective relay
S: Connecting flange for suction pipe
Q: Connecting flange for oil return
Note: All connection pipes and upper gearbox can be turned in 360 degrees.

Model		SiC ₁ /400/600/1000/25~28°C				SiC ₂ /400/600/1000/25~28°C			
Rated voltage (kV/cm)		72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	1840	2070	2200	2300	2110	2240	2370	2470
	i	1304	1434	1584	1684	1304	1434	1584	1684
	k		577				747		
	n		217				287		
	o		130				165		
	p		197				232		
	q		330				465		
Drilling (cm)		130	150	170	190	130	150	170	190
Displacement (g/r)		200	225	245	265	207	232	252	272
Weight (kg)		370				375			

Unit: mm

Appendix 30. SHZV I 400/600/1000A (ZnO protection) overall dimensions without change-over selector, cylinder tap selector



E2: Bleeding of transformer tank
 R: Connecting flange for protective relay
 S: Connecting flange for suction pipe
 Q: Connecting flange for oil return

Note: All connection pipe and upper gearbox can be turned in 360°

Unit: mm

OLTC type	SHZV I 400/600/1000/72.5-252/C				SHZV I 400/600/1000/72.5-252/D/DE			
The highest voltage for equipment (kV)	72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	1932.5	2062.5	2192.5	2292.5	2102.5	2232.5	2362.5
	l	1094	1224	1354	1454	1094	1224	1354
	k	779.5				949.5		
	n	172				242		
	o	130				195		
	m	197				232		
Oil filling dm³	135	155	175	195	135	155	175	195
Displacement dm³	205	230	250	270	212	237	257	277
Weight kg	270				275			

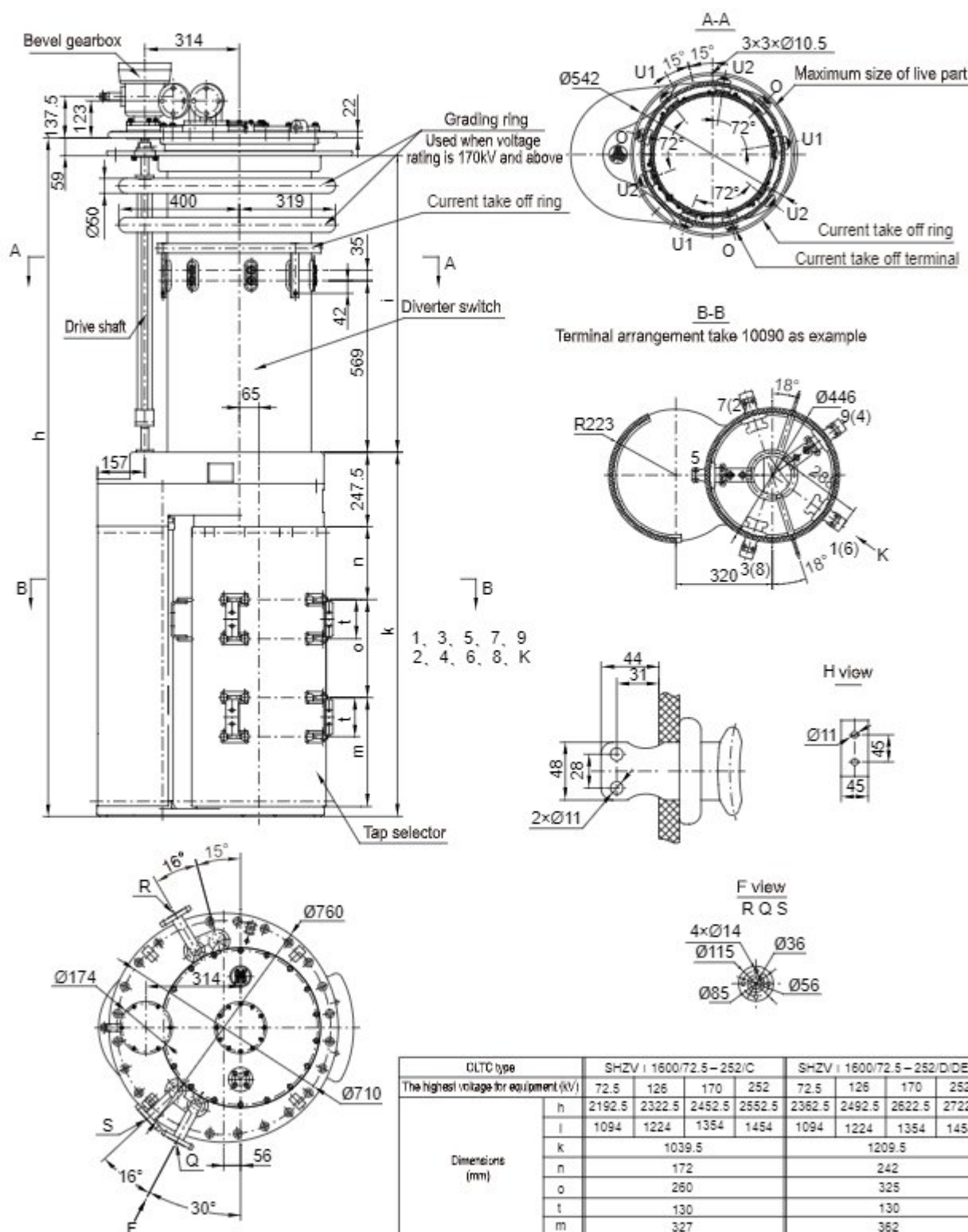
Note: All connection pipe and upper gearbox can be turned in 360°

OLTC type		SHZV 400/600/1000/72.5 – 252/C				SHZV 400/600/1000/72.5 – 252/D/DE			
The highest voltage for equipment (kV)		72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	1932.5	2062.5	2192.5	2292.5	2102.5	2232.5	2362.5	2462.5
	l	1094	1224	1354	1454	1094	1224	1354	1454
	k	779.5				949.5			
	n	172				242			
	o	130				195			
	m	197				232			
	q	539.5				674.5			
Oil filling dm³		135	155	175	195	135	155	175	195
Displacement dm³		205	230	250	270	212	237	257	277
Weight kg		270				275			

Unit: mm

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Appendix 33. SHZV I 1600A (ZnO protection) overall dimensions without change-over selector, cylinder tap selector

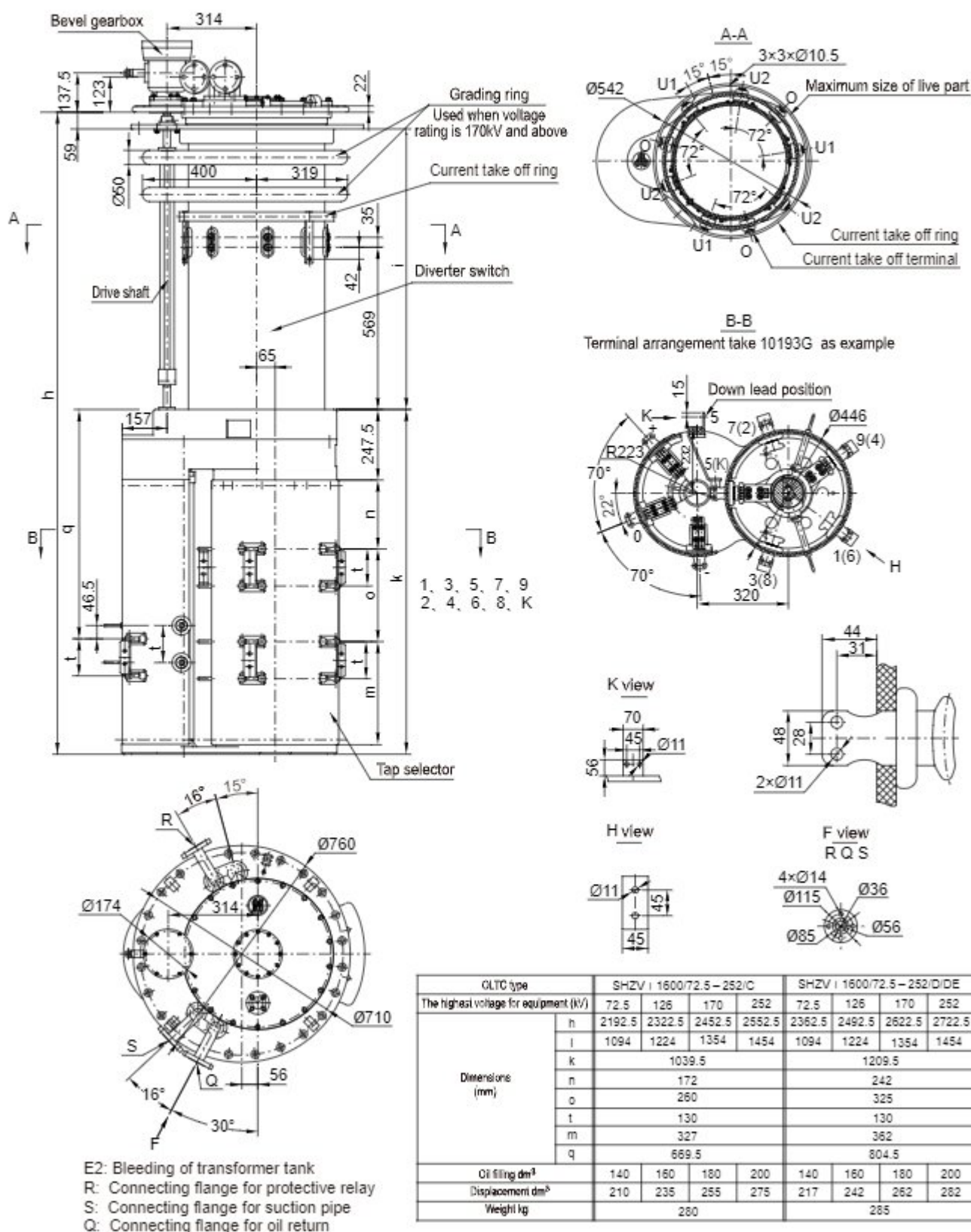


Unit: mm

Technical drawing of a circular flange with the following dimensions: 4xØ14, Ø115, Ø36, Ø85, and Ø56.

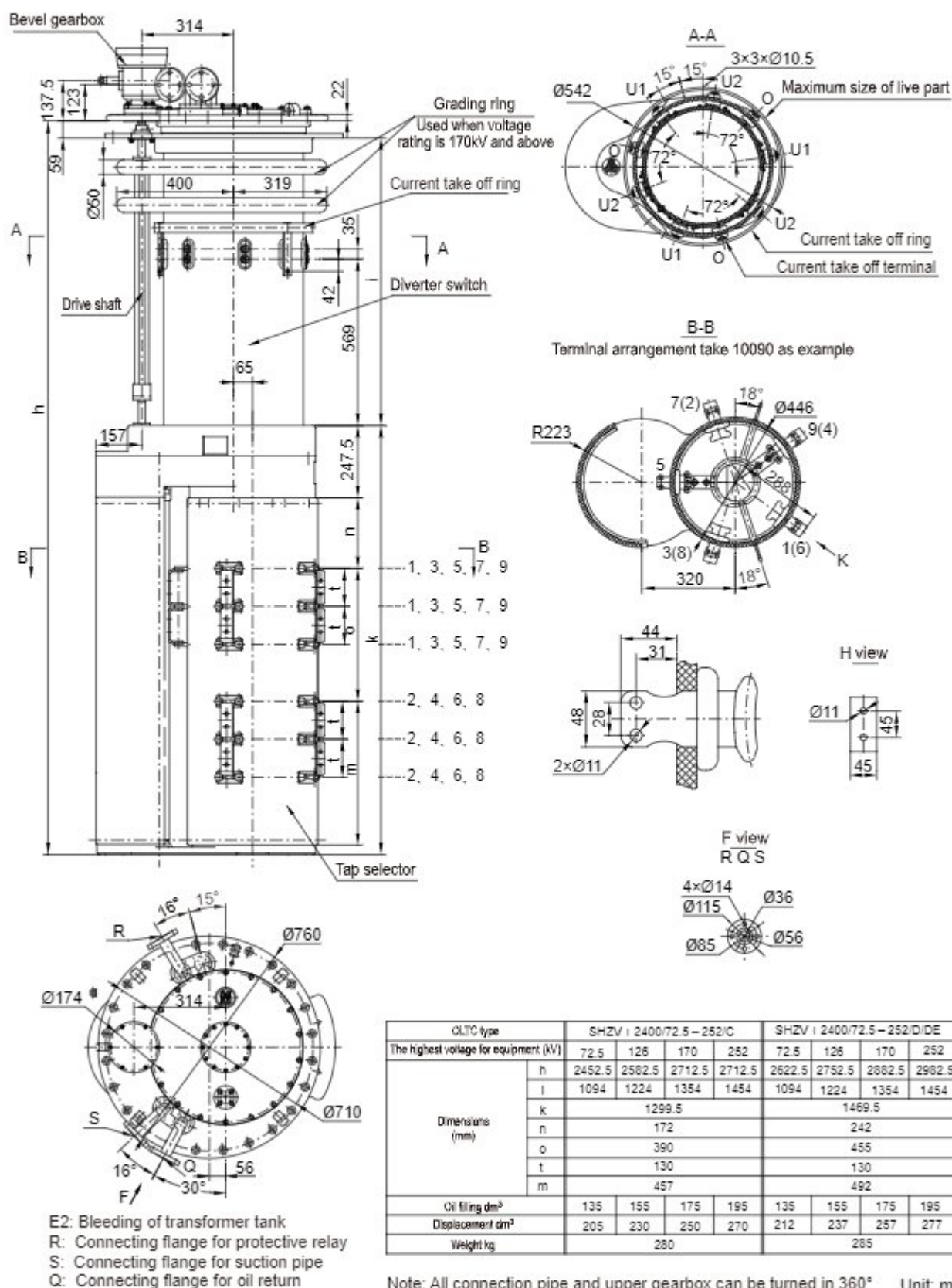
50 | SHZV Vacuum Tap-changer Technical Data

Appendix 35. SHZV I 1600A (ZnO protection) overall dimensions with coarse/fine change-over selector, cylinder tap selector

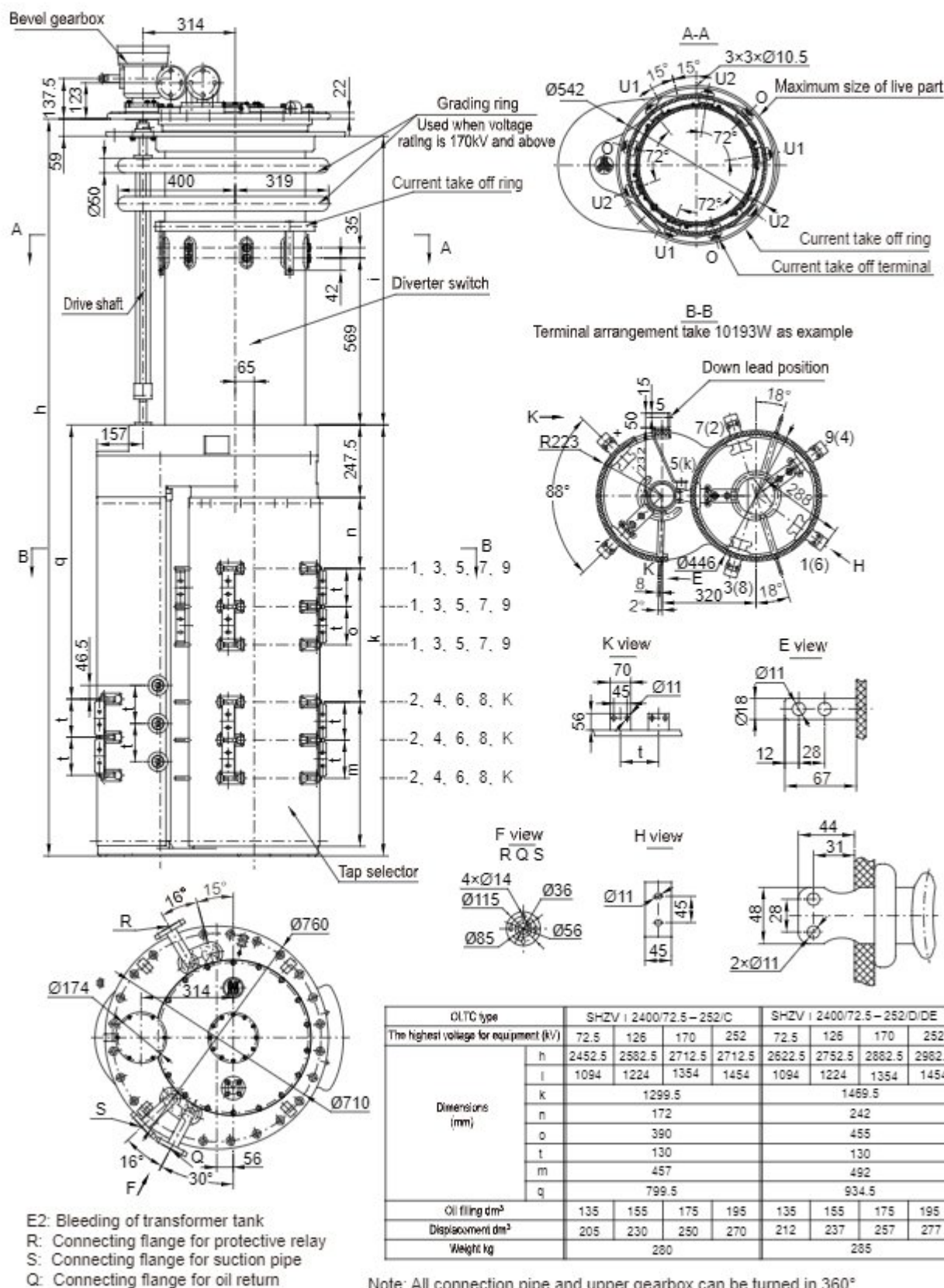


Unit: mm

Appendix 36. SHZV I 2400A (ZnO protection) overall dimensions without change-over selector, cylinder tap selector



Appendix 37. SHZV I 2400A (ZnO protection) overall dimensions with reversing switch, cylinder tap selector



Dimensions (mm):

Symbol	Value
h	247.5
q	46.5
t	157
m	247.5
n	46.5
k	157
q	46.5

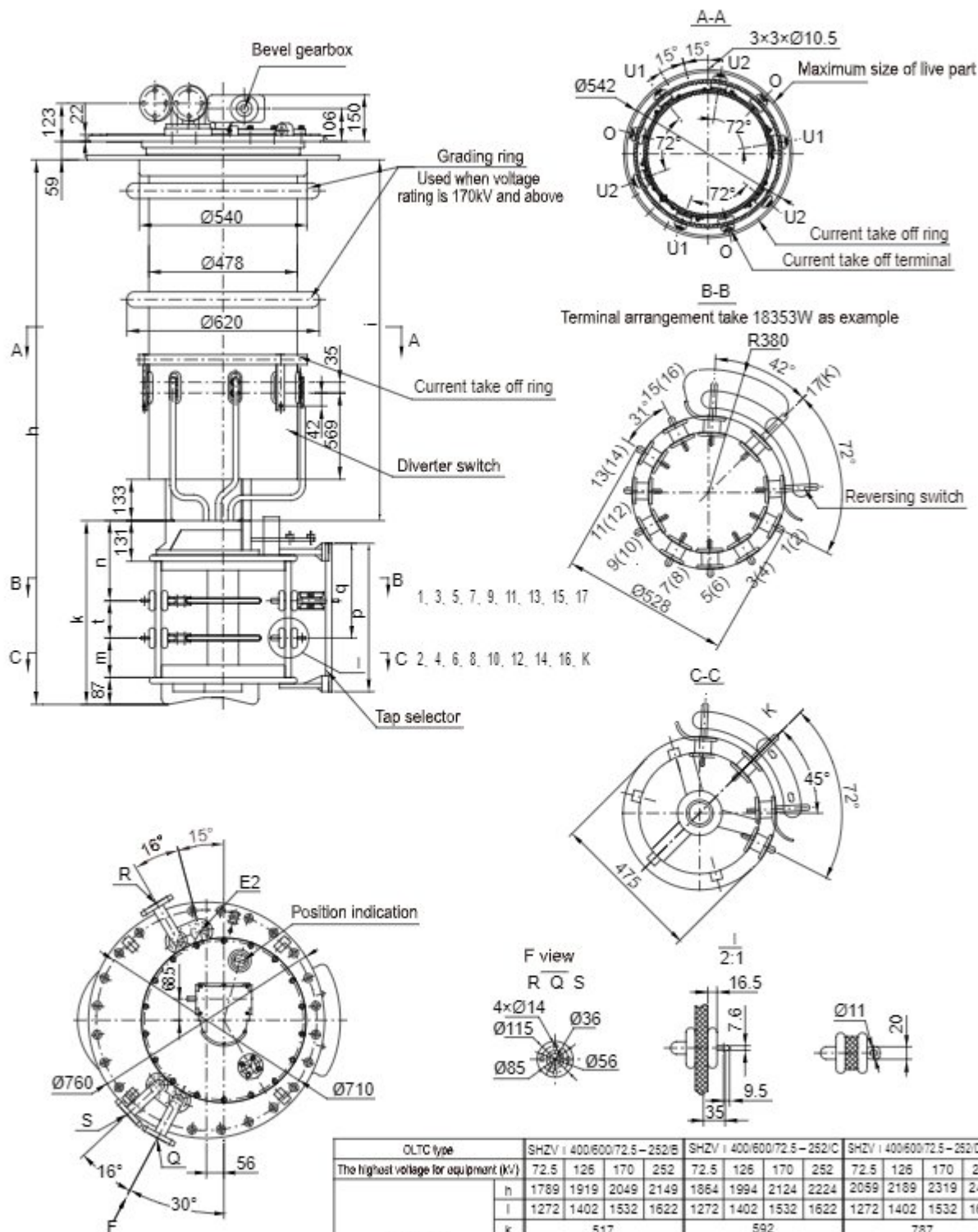
Technical Specifications:

CLTC type	SHZV / 2400/72.5 - 252/C				SHZV / 2400/72.5 - 252/D/DE			
The highest voltage for equipment (kV)	72.5	126	170	252	72.5	126	170	252
h	2452.5	2582.5	2712.5	2712.5	2622.5	2752.5	2882.5	2982.5
l	1094	1224	1354	1454	1094	1224	1354	1454
k	1299.5				1469.5			
n	172				242			
o	390				455			
t	130				130			
m	457				492			
q	799.5				934.5			
Oil/gil m ³	135	155	175	195	135	155	175	195
Displacement dm ³	205	230	250	270	212	237	257	277
Weight kg	280				285			

Note: All connection pipe and upper gearbox can be turned in 360°

Unit: mm

Appendix 39. SHZV I 400/600A (ZnO protection) overall dimensions, with reversing switch, cage tap selector

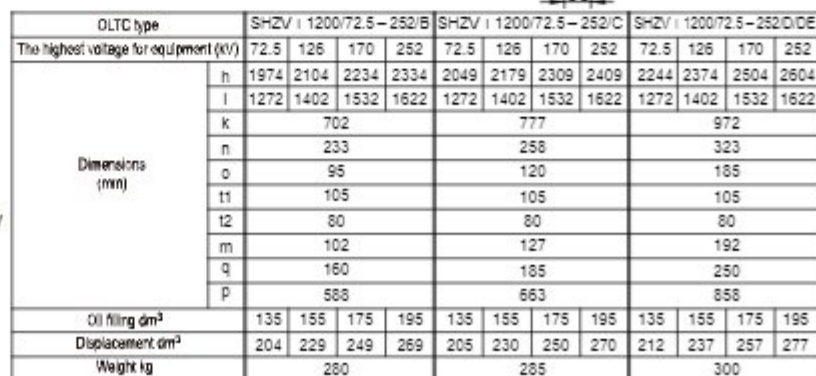


E2: Bleeding of transformer tank
R: Connecting flange for protective relay
S: Connecting flange for suction pipe
Q: Connecting flange for oil return
Note: All connection pipe and upper gearbox can be turned in 360°

OLTC type		SHZV I 400/600/72.5-252/B				SHZV I 400/600/72.5-252/C				SHZV I 400/600/72.5-252/D/DE			
The highest voltage for equipment (kV)		72.5	126	170	252	72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	1789	1919	2049	2149	1864	1994	2124	2224	2059	2189	2319	2419
	i	1272	1402	1532	1622	1272	1402	1532	1622	1272	1402	1532	1622
	k	517				592				787			
	n	233				258				323			
	t	95				120				185			
	m	102				127				192			
	q	160				185				250			
	p	403				478				673			
Oil filling dm ³		135	155	175	195	135	155	175	195	135	155	175	195
Displacement dm ³		204	229	249	269	205	230	250	270	212	237	257	277
Weight kg		270				275				280			

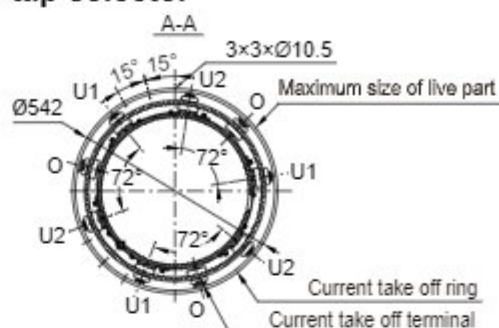
Unit: mm

CCL
A-A

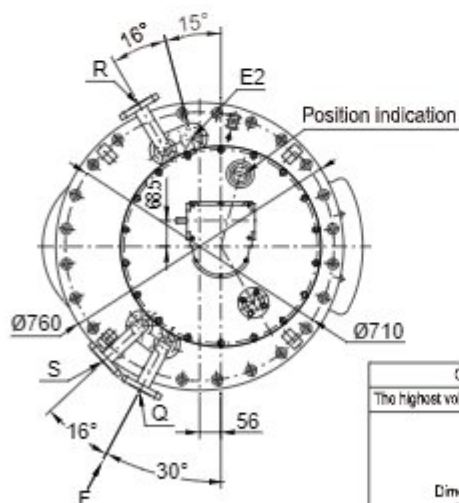
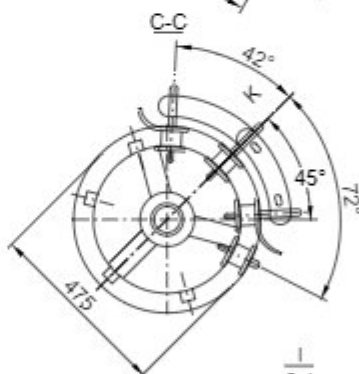
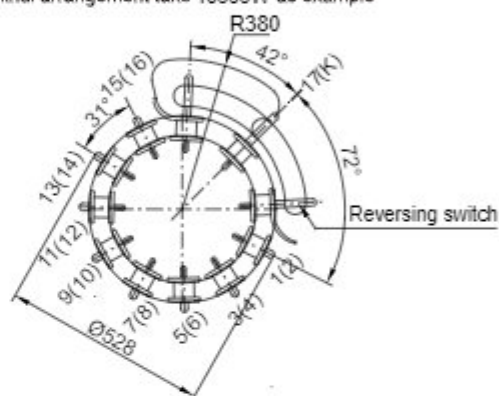


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with reversing switch, cage tap selector

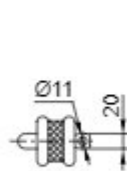
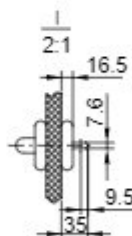
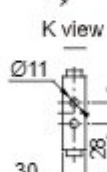
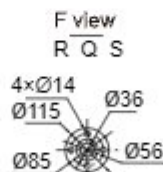


B-B
Terminal arrangement take 18353W as example



E2: Bleeding of transformer tank
R: Connecting flange for protective relay
S: Connecting flange for suction pipe
Q: Connecting flange for oil return

Note: All connection pipe and upper gearbox can be turned in 360°

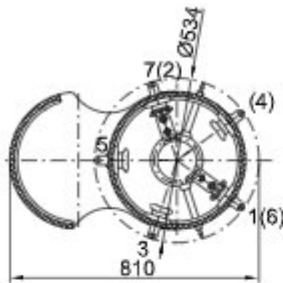


OLTC type		SHZV / 1500/72.5 - 252/B				SHZV / 1500/72.5 - 252/C				SHZV / 1500/72.5 - 252/D/E			
The highest voltage for equipment (kV)		72.5	126	170	252	72.5	126	170	252	72.5	126	170	252
Dimensions (mm)	h	2159	2289	2419	2519	2234	2364	2494	2594	2429	2559	2689	2789
	l	1272	1402	1532	1622	1272	1402	1532	1622	1272	1402	1532	1622
	k	887				962				1157			
	n	233				258				323			
	o	95				120				185			
	t1	105				105				105			
	t2	80				80				80			
	m	102				127				192			
	q	160				185				250			
p	773				848				1043				
Oil filling dm³		135	155	175	195	135	155	175	195	135	155	175	195
Displacement dm³		204	229	249	269	205	230	250	270	212	237	257	277
Weight kg		285				290				295			

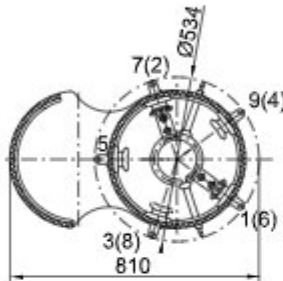
Unit: mm

Appendix 42. SHZV tap selector contacts arrangement, cylinder typ selector

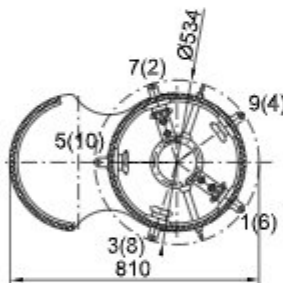
Without change-over selector



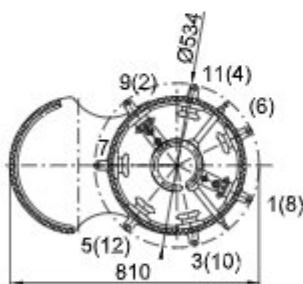
10070



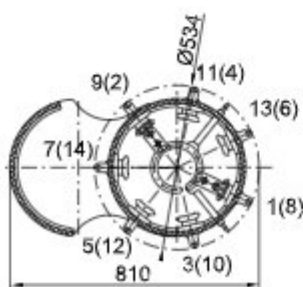
10090



10100

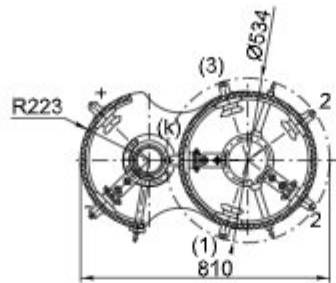


14120

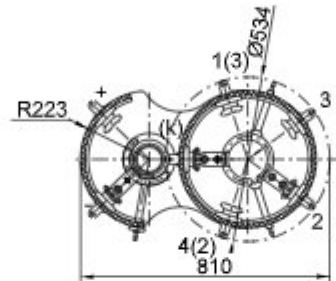


14140

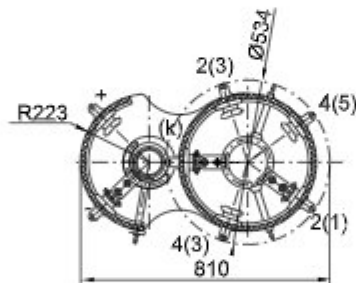
With reversing switch



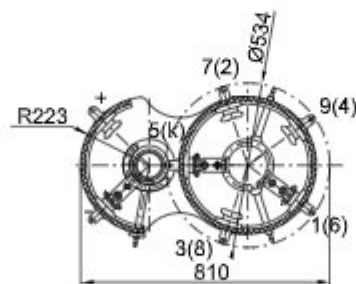
10051W



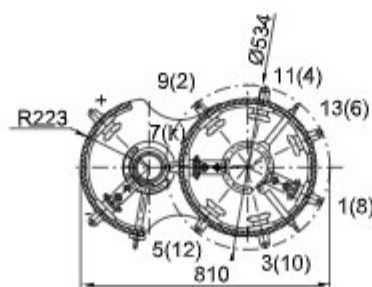
10071W



10091W

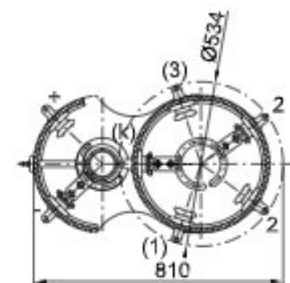


10191W

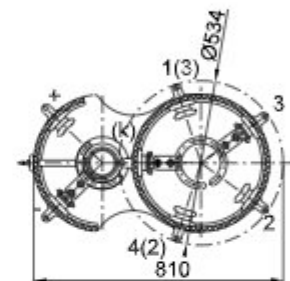


14271W

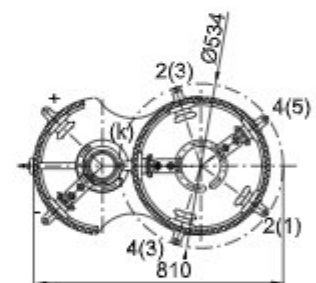
With coarse/fine change-over selector



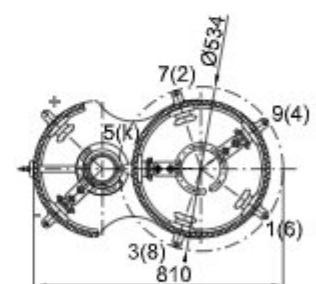
10051G



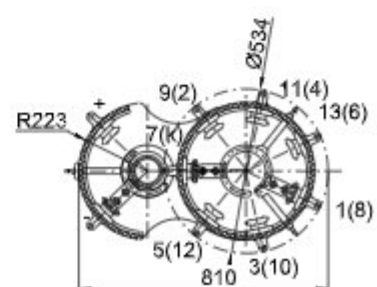
10071G



10091G



10191G

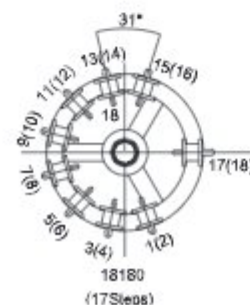
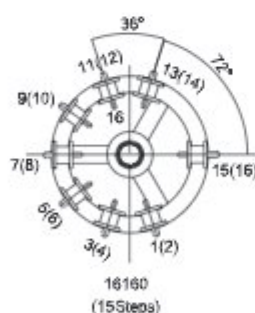
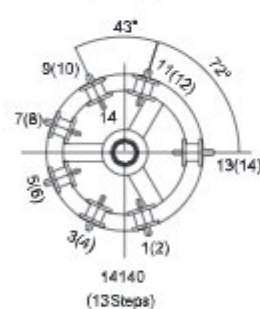
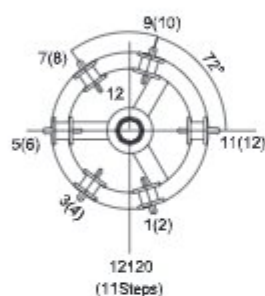
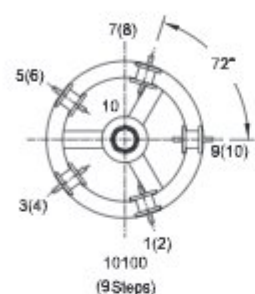


14271G

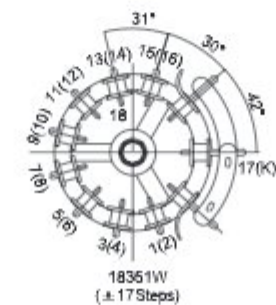
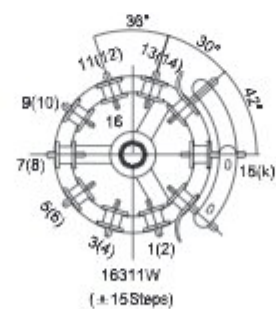
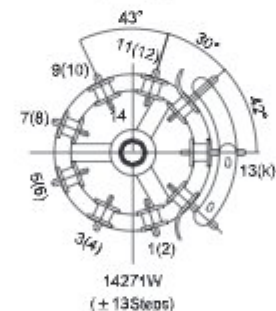
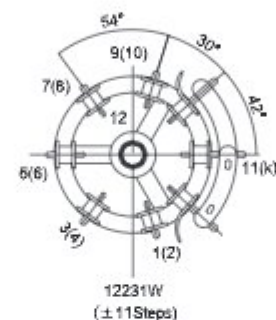
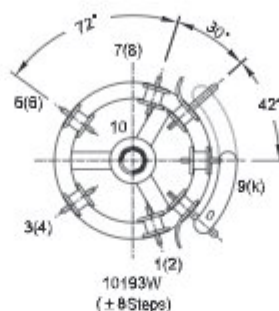
Unit: mm

Appendix 43. SHZV tap selector contacts arrangement, cage tap selector

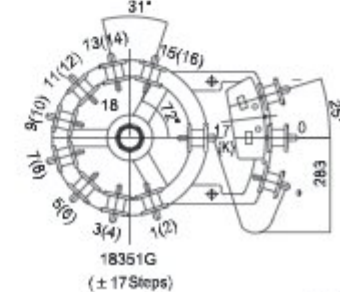
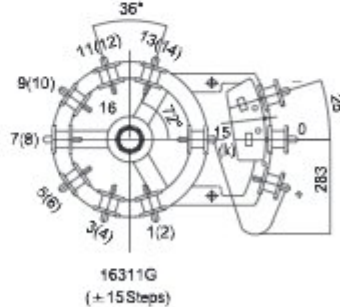
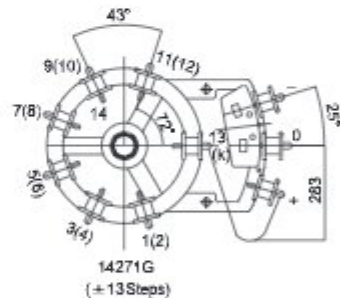
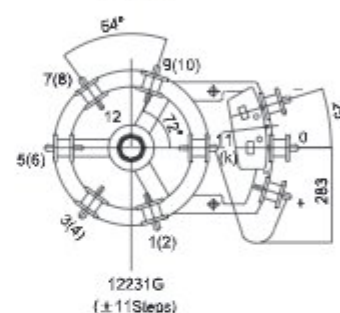
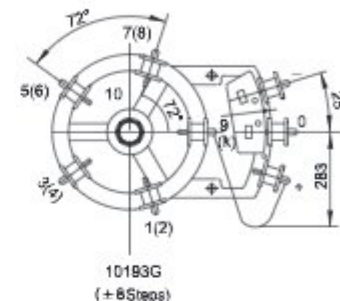
Without change-over selector



With reversing switch

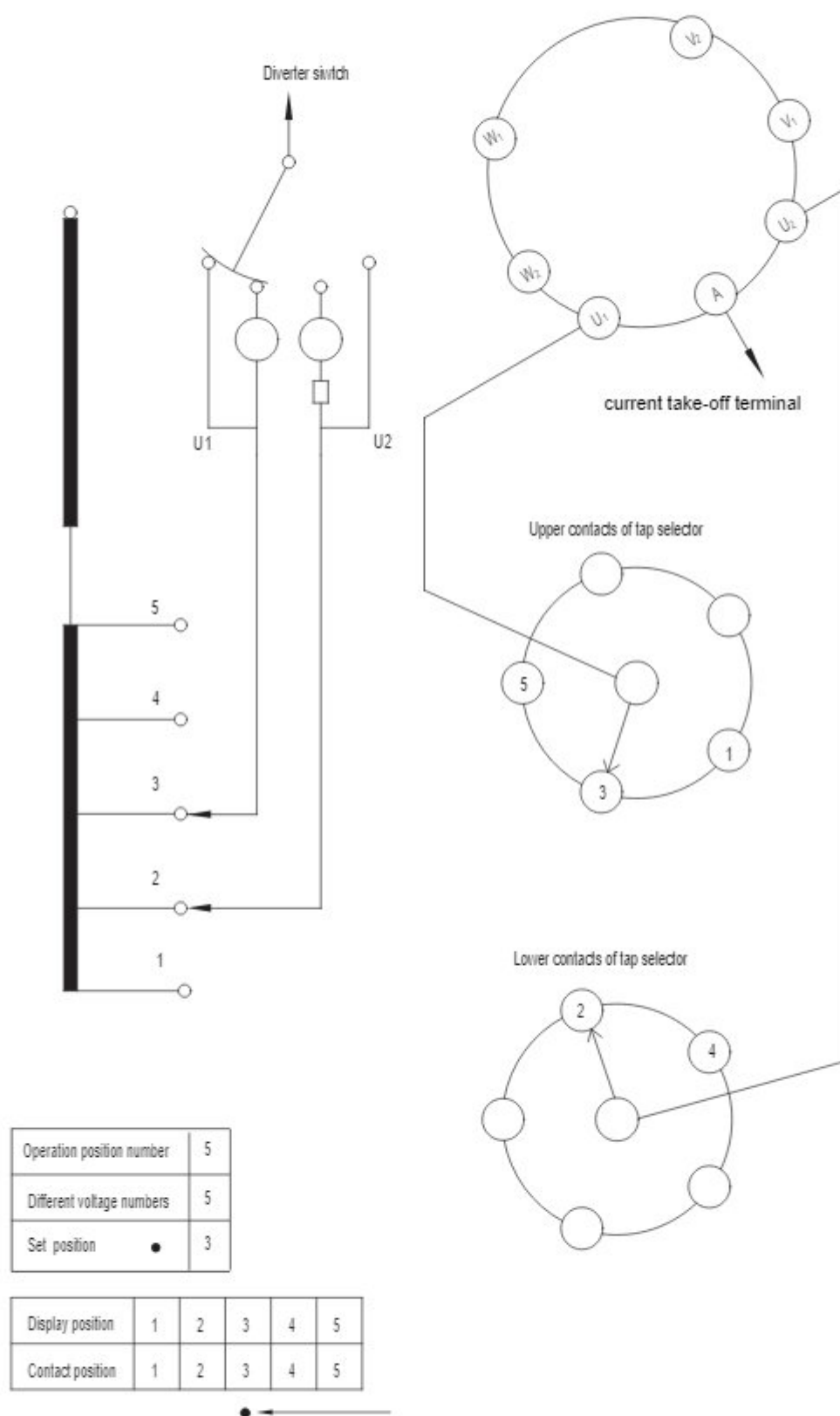


With coarse/fine change-over selector

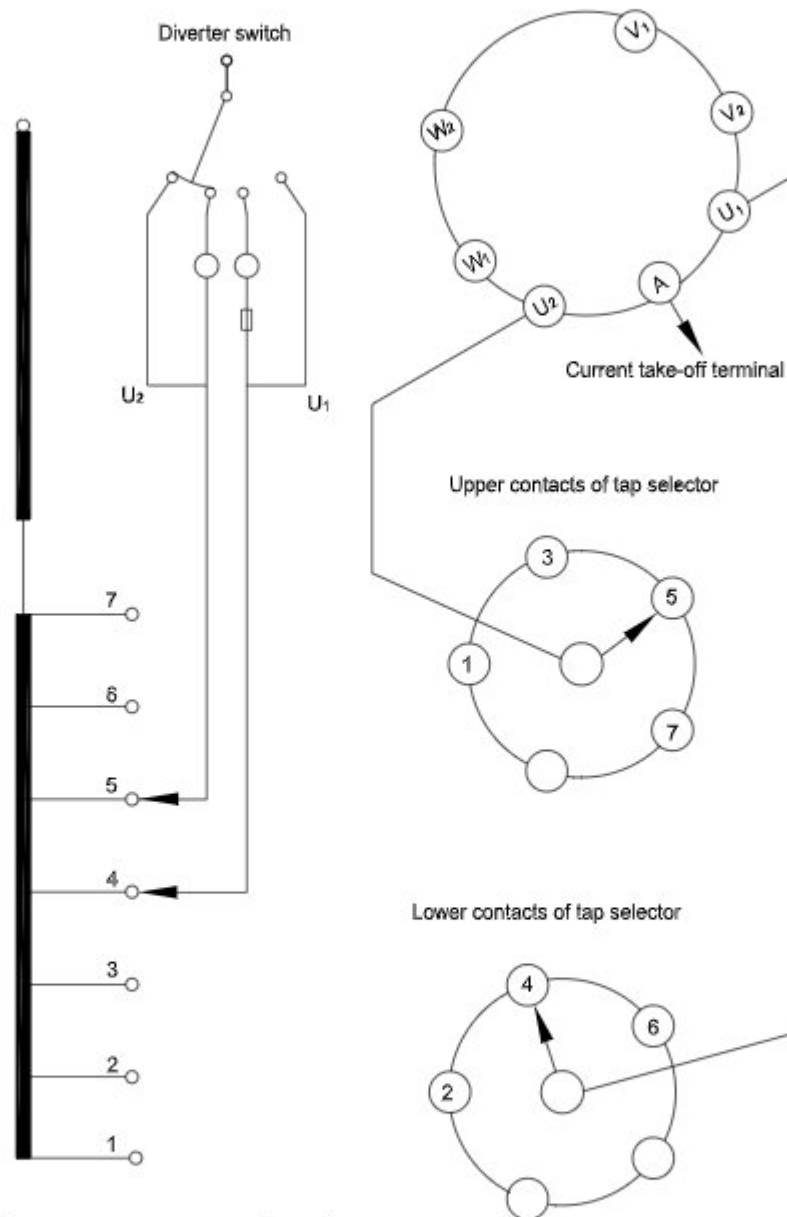


Unit: mm

Appendix 44. SHZV (10050) operating position table and connection diagram



Appendix 45. SHZV (10070) operating position table and connection diagram



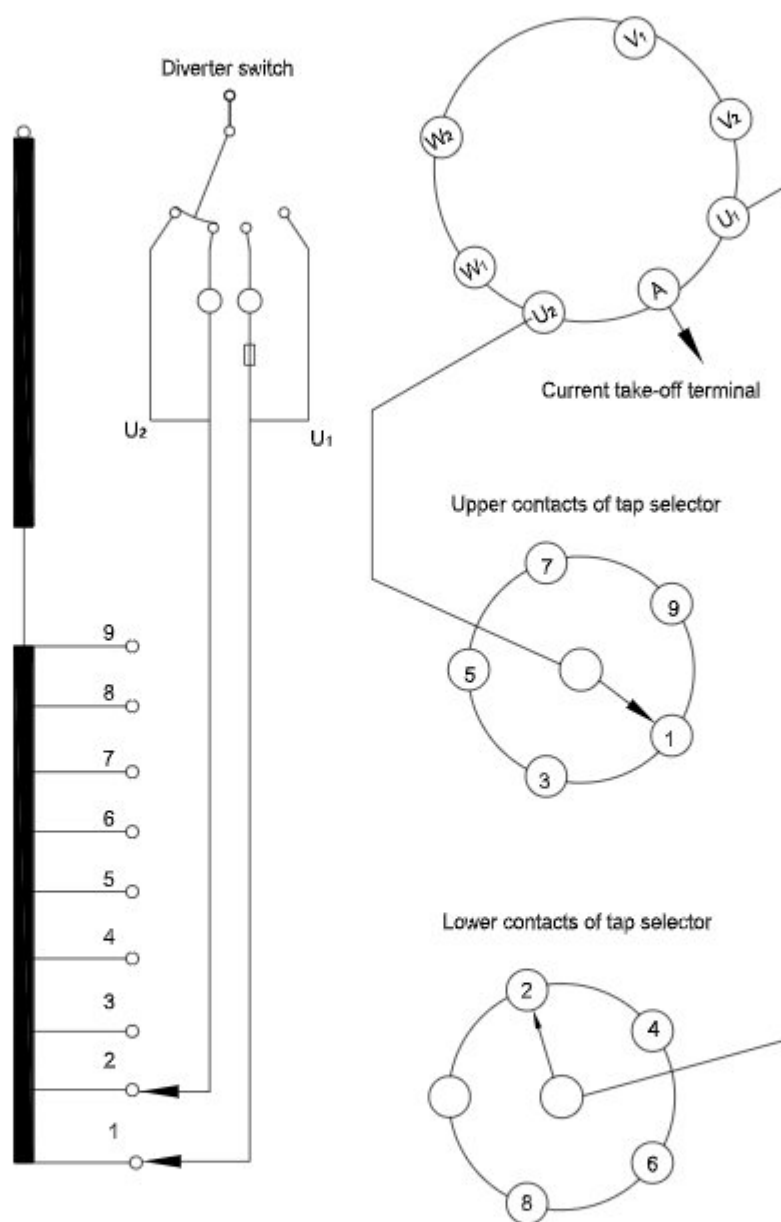
Operation position number	7
Different voltage number	7
Set position ●	4

Display position	1	2	3	4	5	6	7
Tap selector contact position	1	2	3	4	5	6	7



● Drawing is shown at the set position

Appendix 46. SHZV (10090) operating position table and connection diagram



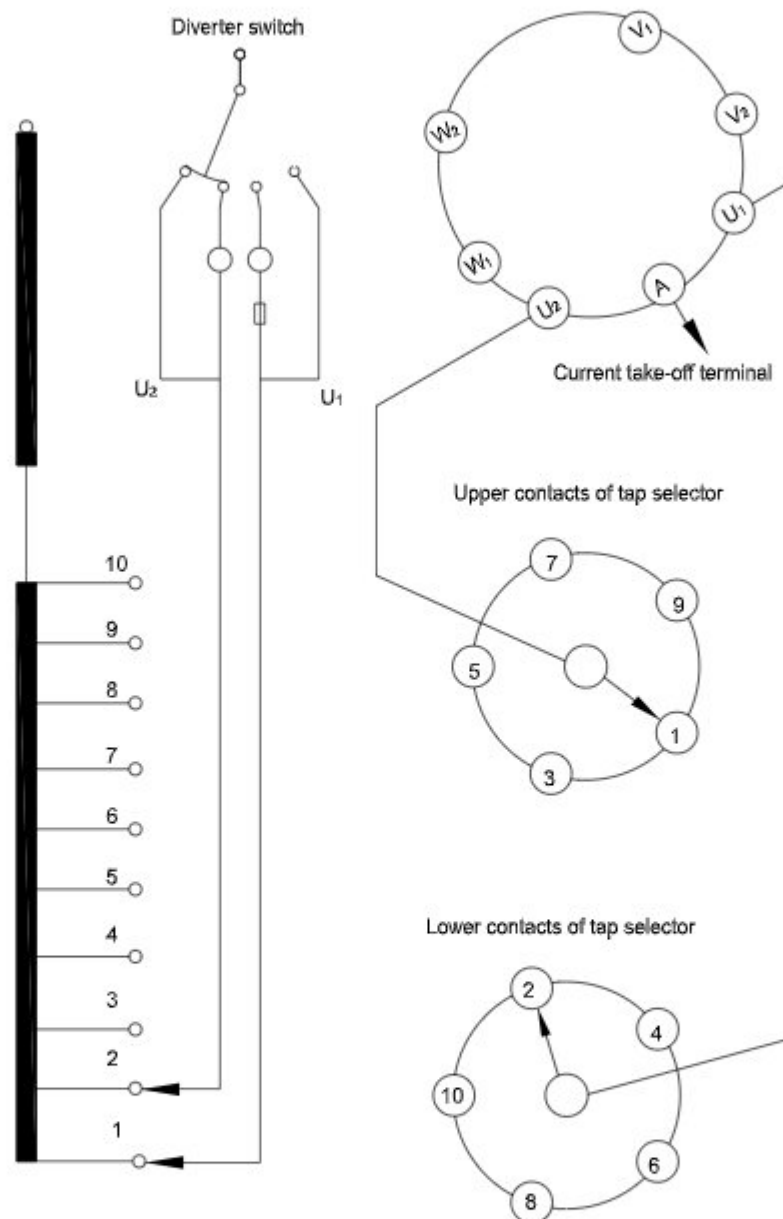
Operation position number	9
Different voltage number	9
Set position ●	5

Display position	1	2	3	4	5	6	7	8	9
Tap selector contact position	1	2	3	4	5	6	7	8	9

● ←

● Drawing is shown at the set position

Appendix 47. SHZV (10100) operating position table and connection diagram



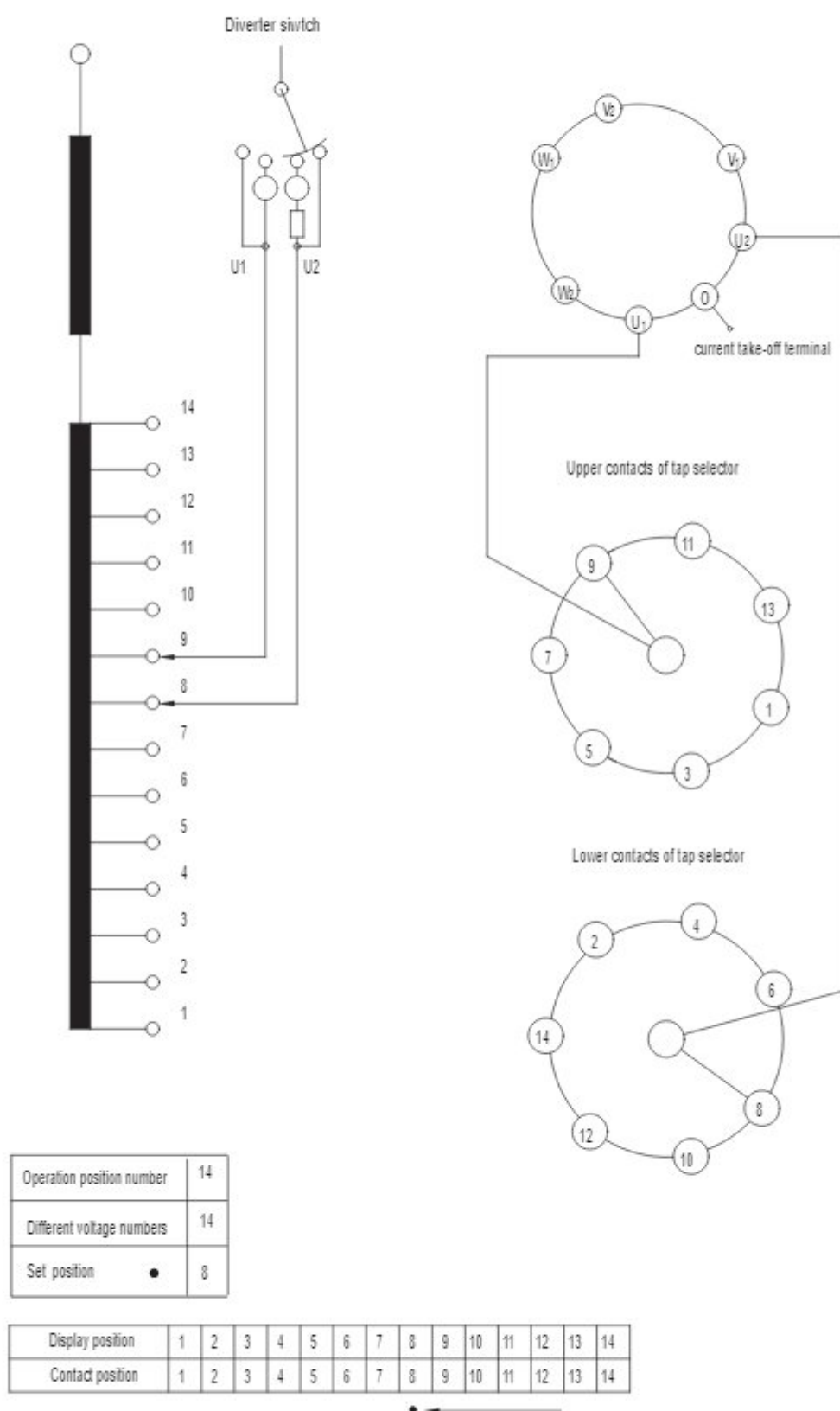
Operation position number	10
Different voltage number	10
Set position ●	6

Display position	1	2	3	4	5	6	7	8	9	10
Tap selector contact position	1	2	3	4	5	6	7	8	9	10

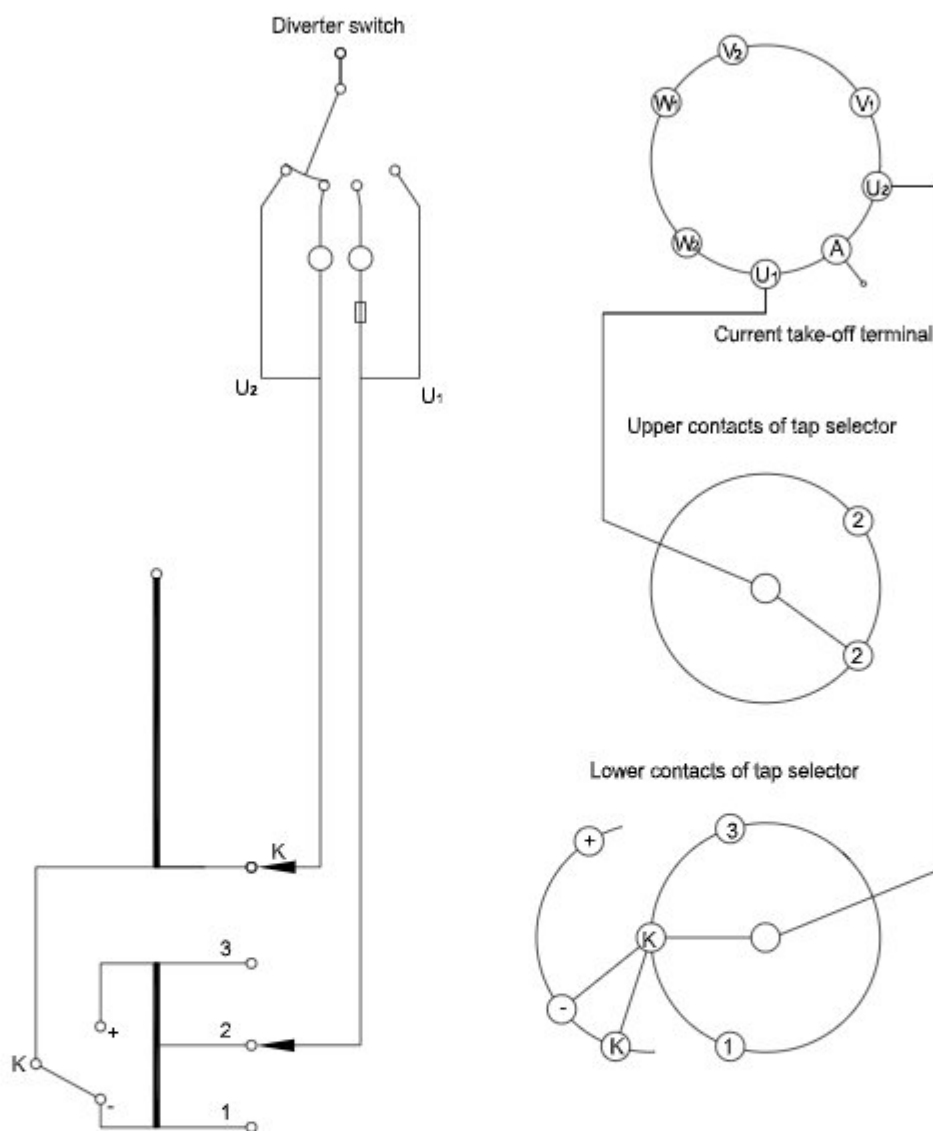


● Drawing is shown at the set position

Appendix 48. SHZV (14140) operating position table and connection diagram



Appendix 49. SHZV (10051W) operating position table and connection diagram



Please connect terminal 1 and "-", 3 and "+", 2 and 2 in the same phase.

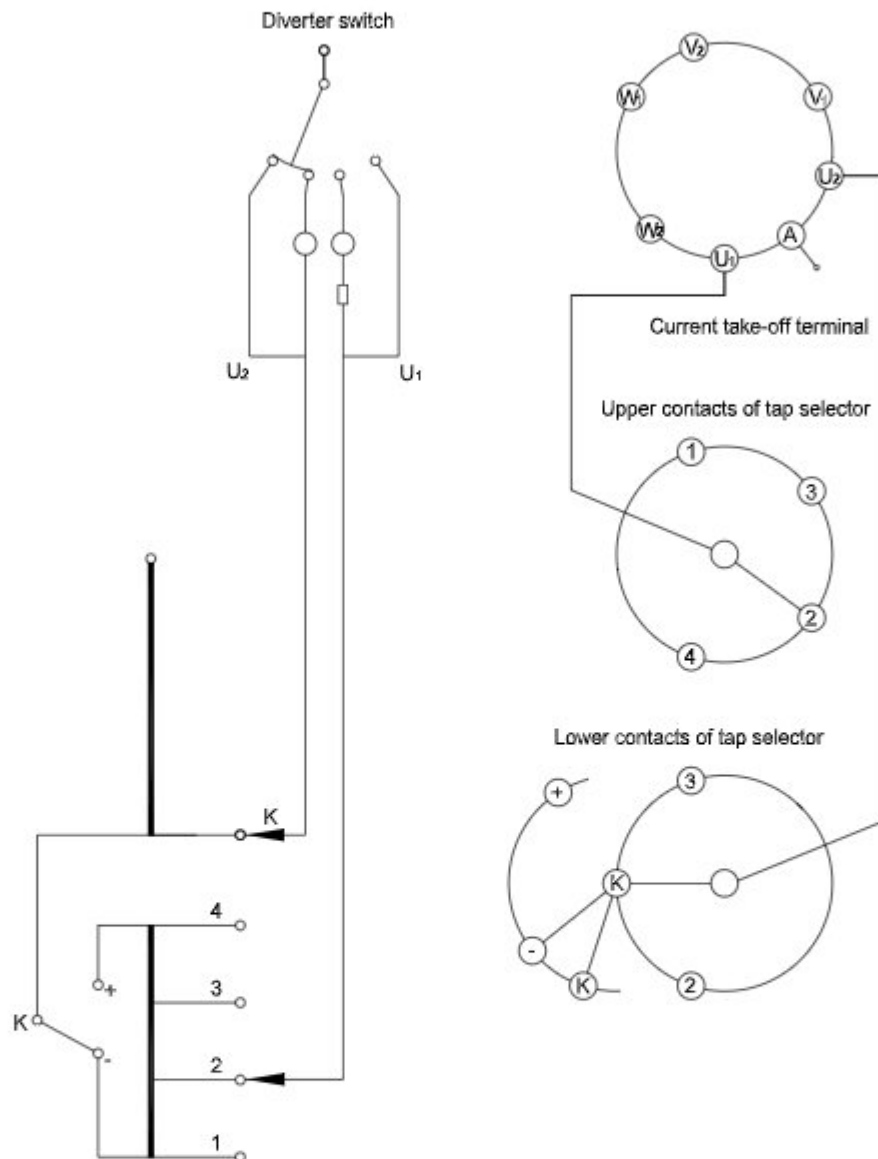
Operation position number	5
Different voltage number	5
Set position ●	3

Change-over selector	← K + → K -				
Change-over selector location	1	2	K	2	3
Display position	1	2	3	4	5



● Drawing is shown at the set position

Appendix 50. SHZV (10071W) operating position table and connection diagram



Please connect terminal 1 and "V", 4 and "W", 2 and 2, 3 and 3 in the same phase.

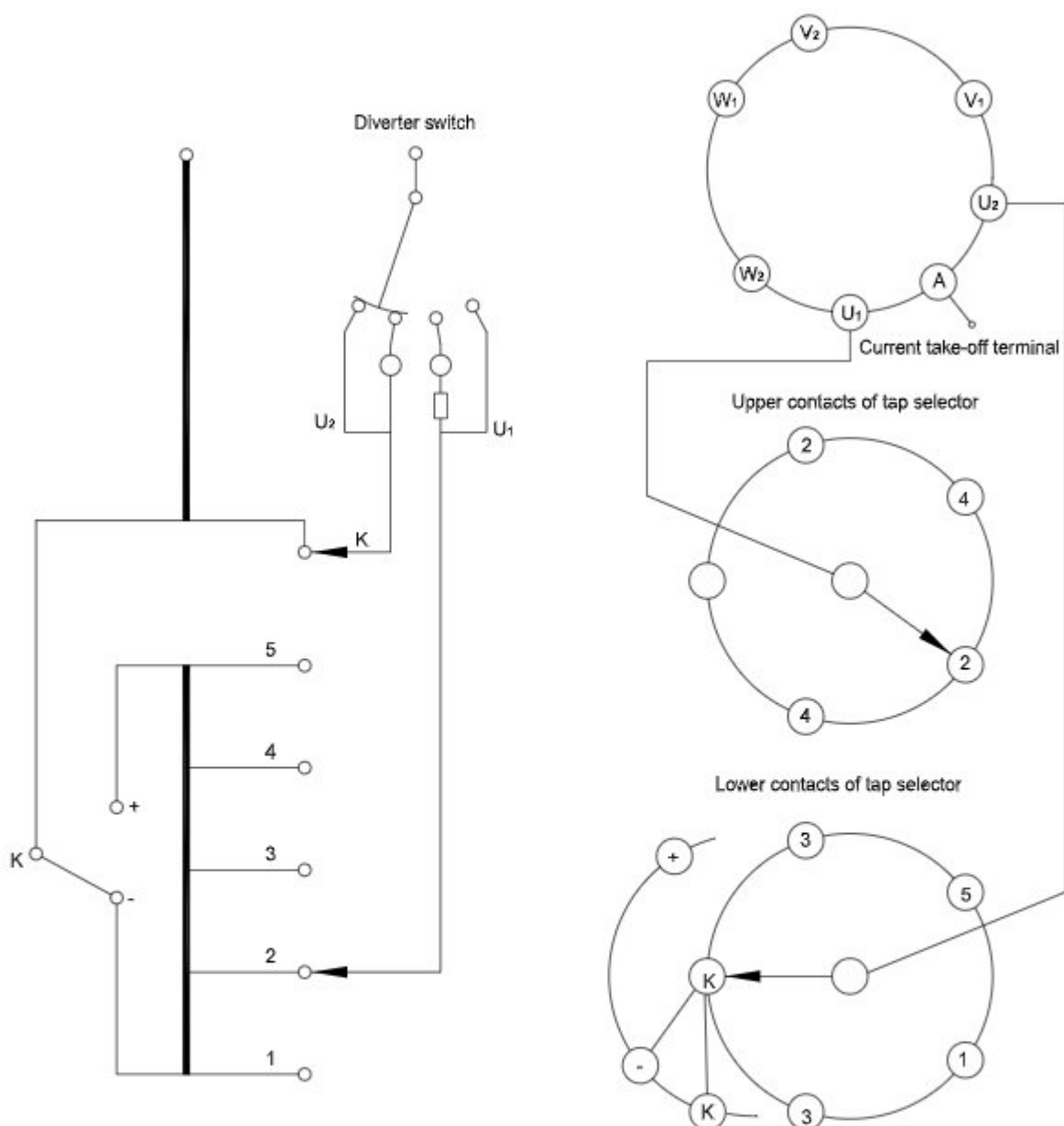
Operation position number	7
Different voltage number	7
Set position ●	4

Change-over selector	← K+ → K-						
Change-over selector location	1	2	3	K	2	3	4
Display position	1	2	3	4	5	6	7



● Drawing is shown at the set position

Appendix 51. SHZV (10091W) operating position table and connection diagram



Please connect terminal 1 and "-", 5 and "+", 2 and 2, 3 and 3, 4 and 4 in the same phase.

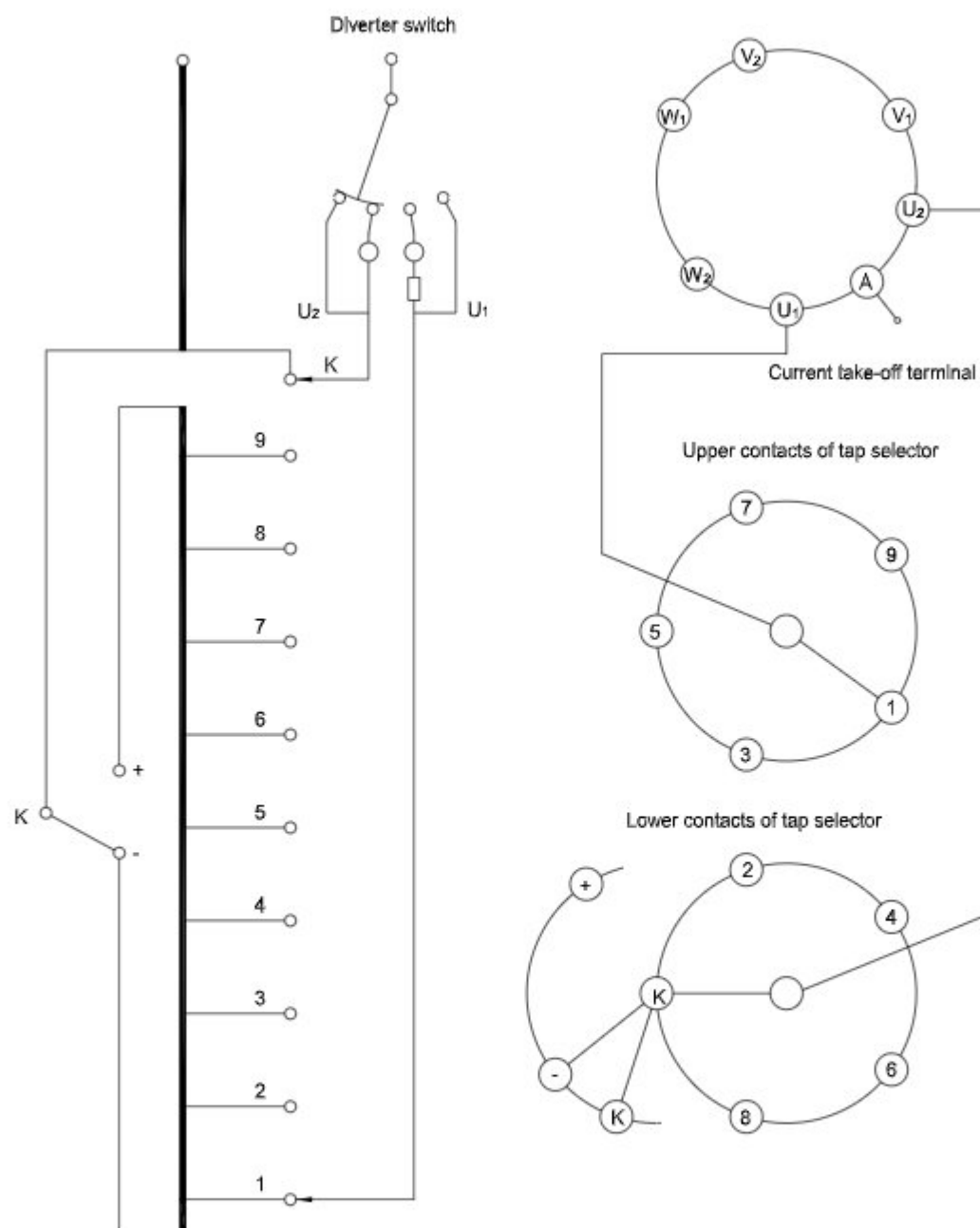
Operation position number	9
Different voltage number	9
Set position ●	5

Change-over selector									
Change-over selector location	1	2	3	4	K	2	3	4	5
Display position	1	2	3	4	5	6	7	8	9

● ←

● Drawing is shown at the set position

Appendix 52. SHZV (10191W) operating position table and connection diagram

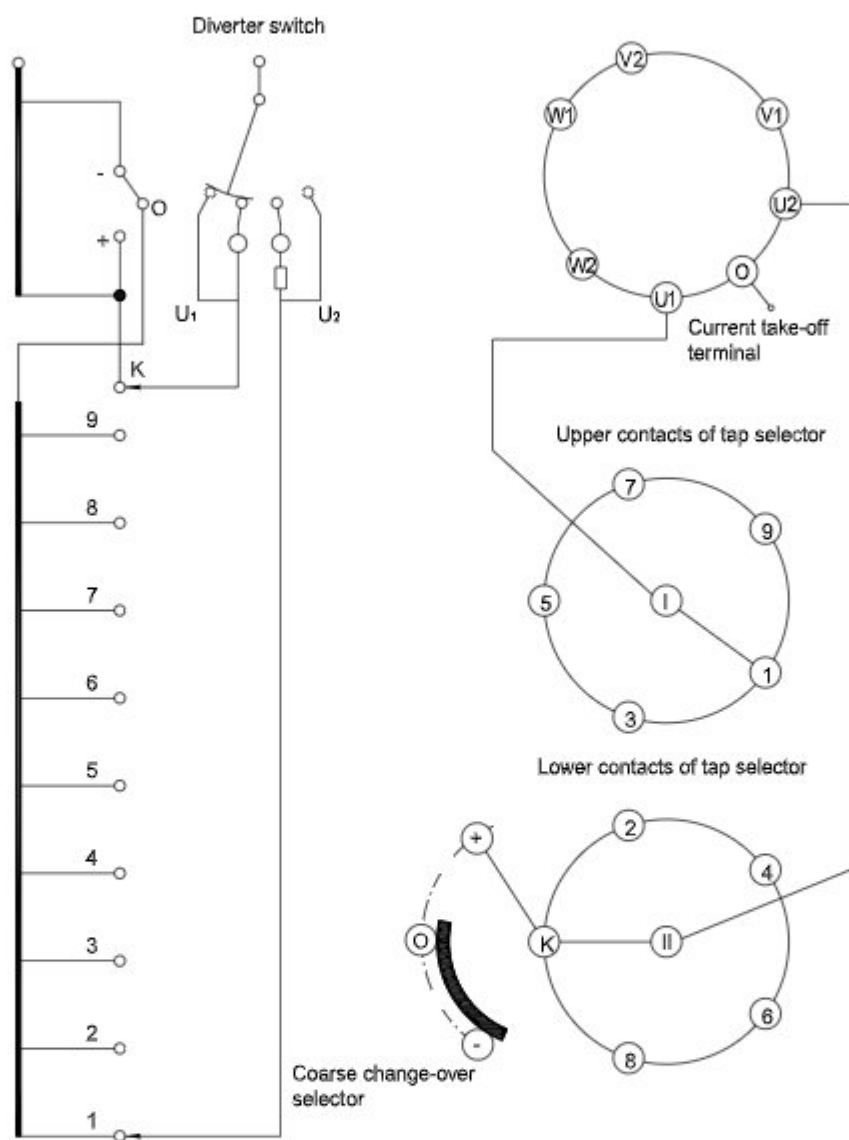


Operation position number	19
Different voltage number	19
Set position ●	10

Change-over selector	K+										K-								
Change-over selector location	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

● Drawing is shown at the set position

Appendix 53. SHZV (10191G) operating position table and connection diagram



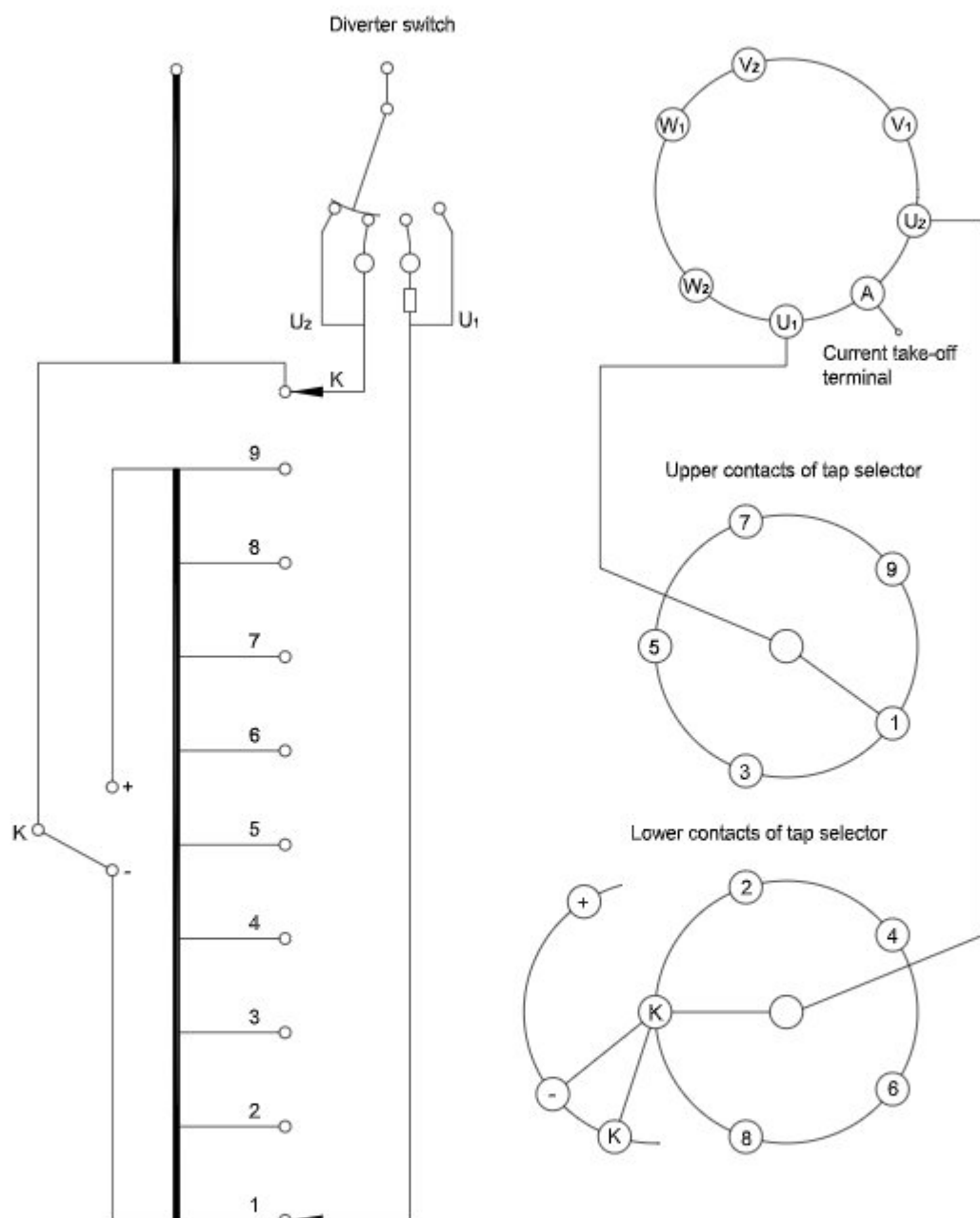
Operation position number	19
Different voltage number	19
Set position ●	10

Change-over selector	O+										O-								
Change-over selector location	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19



● Drawing is shown at the set position

Appendix 54. SHZV (10193W) operating position table and connection diagram



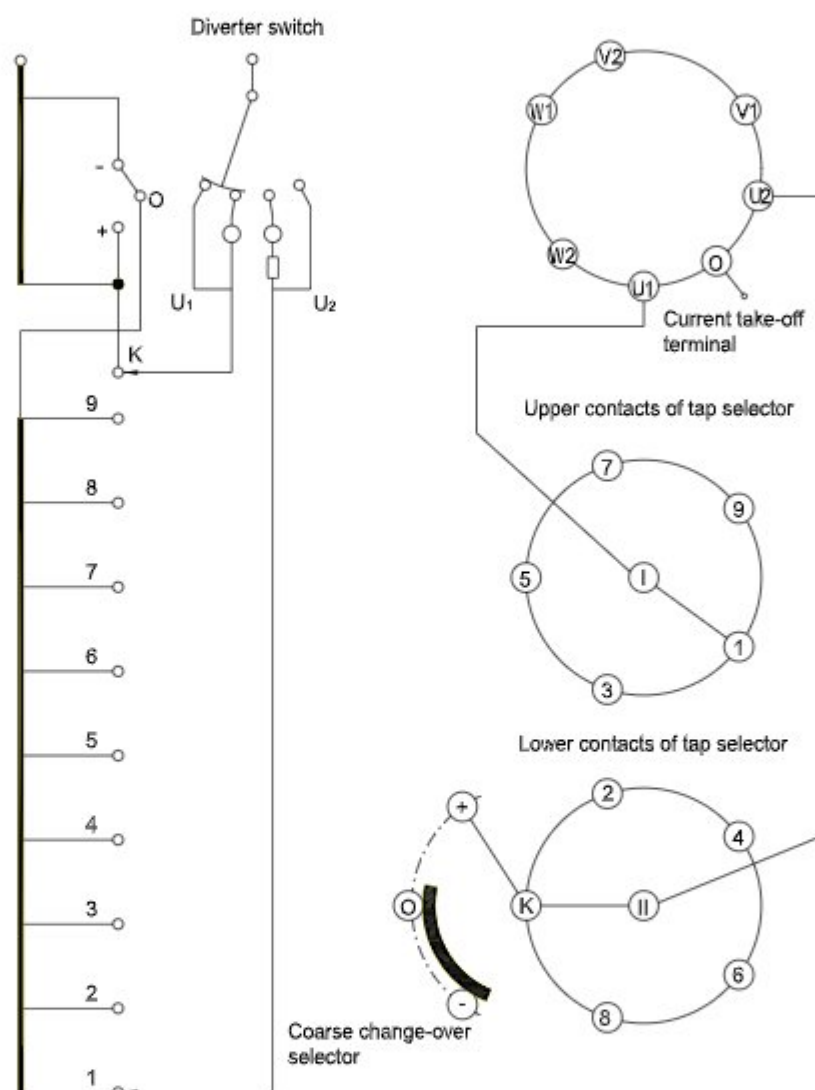
Operation position number	19
Different voltage number	17
Set position ●	9b

Please connect terminal 1 and "-", 9 and "+" in the same phase.

Change-over selector																			
Change-over selector location	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9
Display position	1	2	3	4	5	6	7	8	9a	9b	9c	10	11	12	13	14	15	16	17

● Drawing is shown at the set position

Appendix 55. SHZV (10193G) operating position table and connection diagram

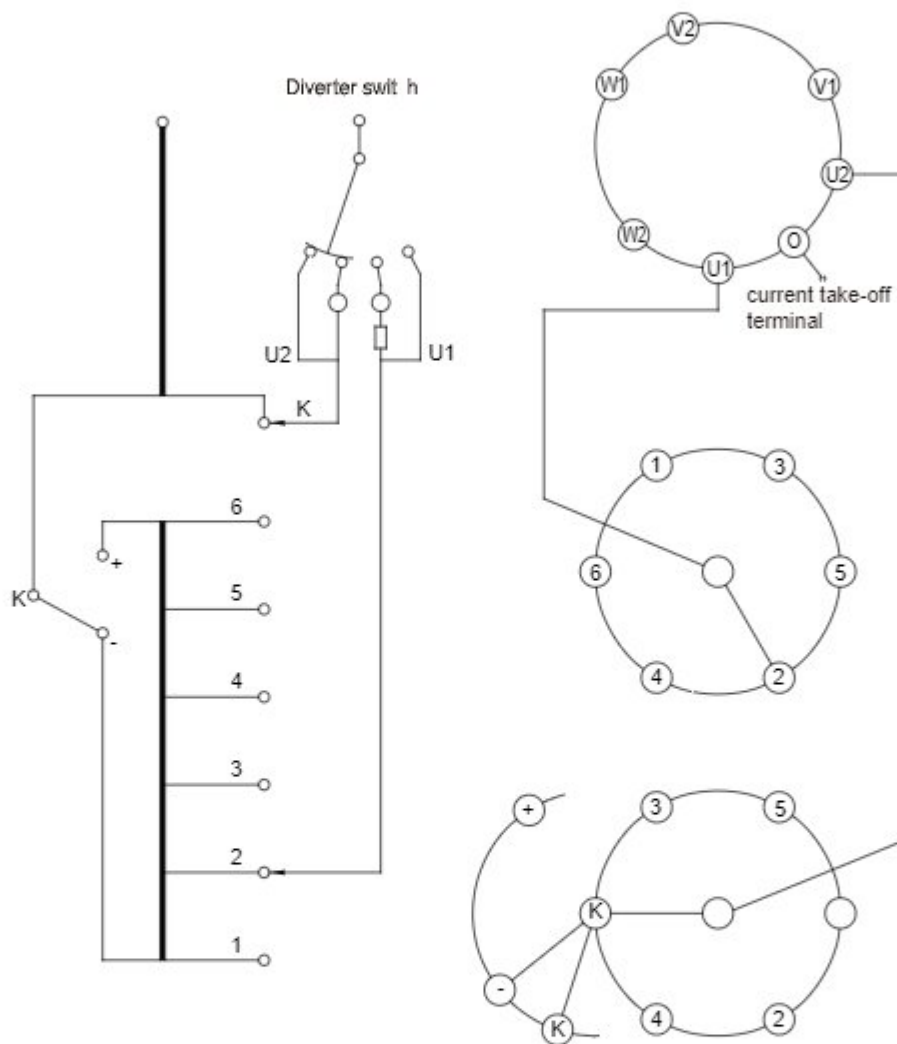


Operation position number	19
Different voltage number	17
Set position ●	9b

Change-over selector																			
Change-over selector location	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9
Display position	1	2	3	4	5	6	7	8	9a	9b	9c	10	11	12	13	14	15	16	17

● Drawing is shown at the set position

Appendix 56. SHZV (12111W) operating position table and connection diagram



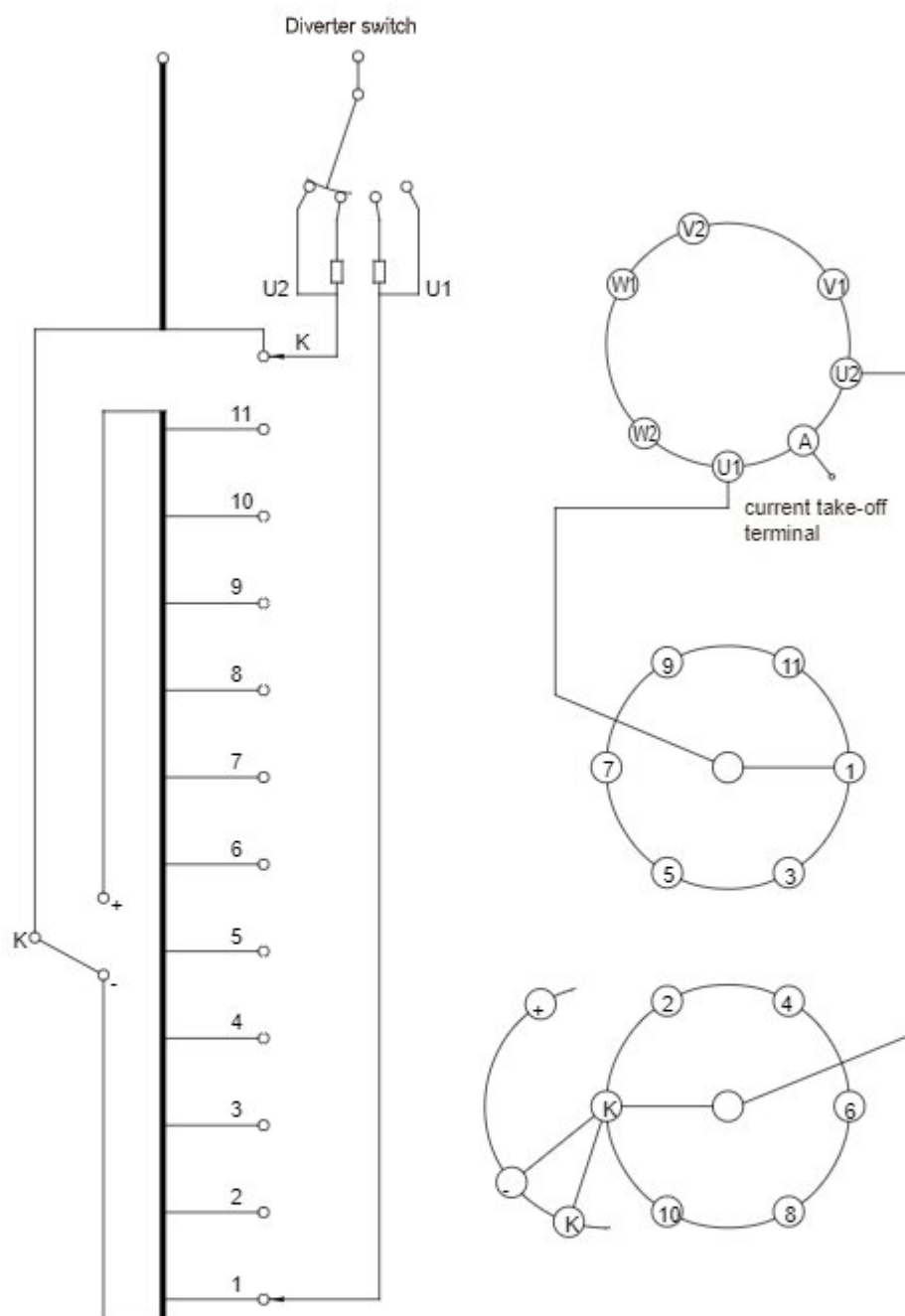
Please connect terminal 1 and "-", 6 and "+" 3 and 3, 4 and 4, 5 and 5 in the same phase

Operation position number	11
Different voltage number	11
Set position ●	6

Change-over selector											
Operation position	1	2	3	4	5	6	7	8	9	10	11
Change-over selector location	1	2	3	4	5	K	2	3	4	5	6
Display position	1	2	3	4	5	6	7	8	9	10	11

● ←

Appendix 57. SHZV (12231W) operating position table and connection diagram

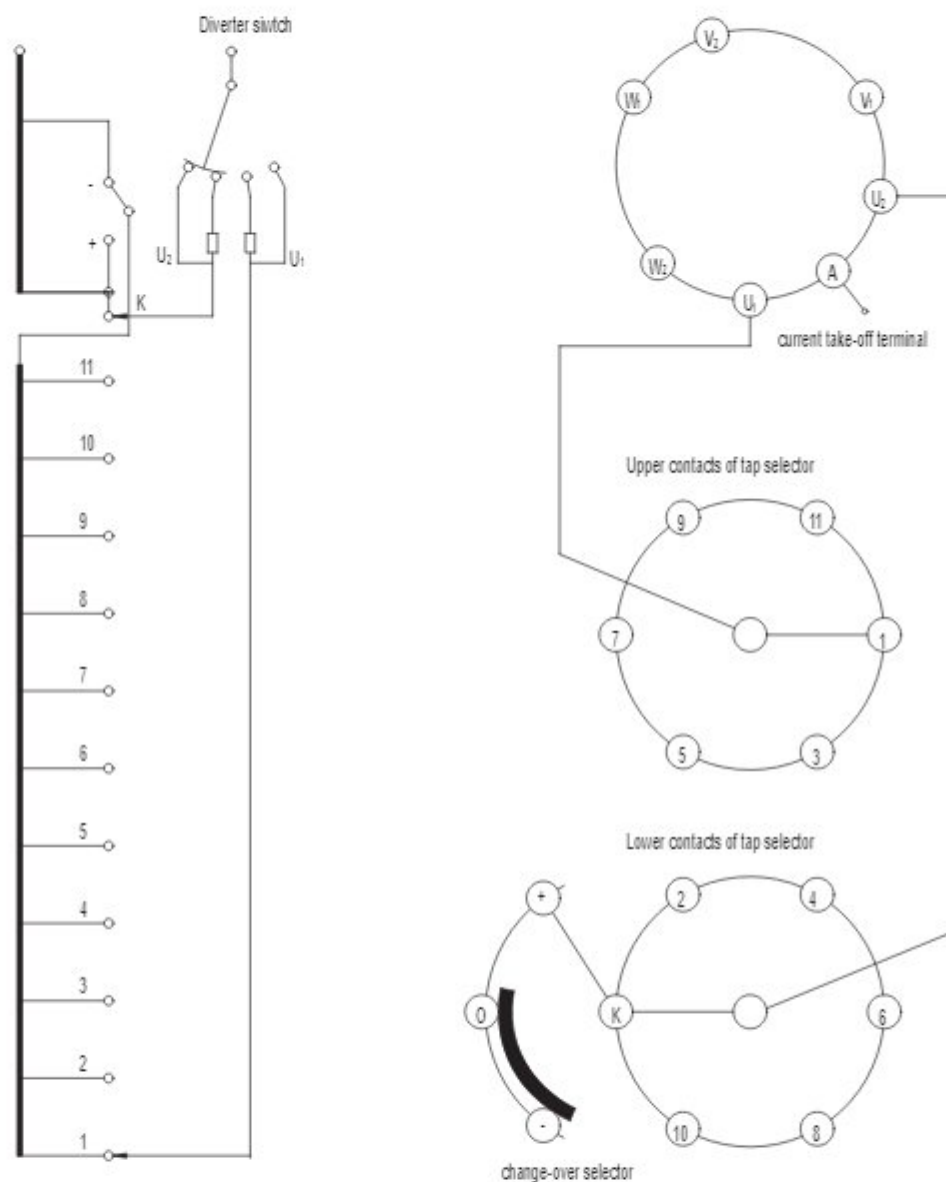


Operation position number	23
Different voltage number	23
Set position ●	12

Change-over selector	K+											K-											
Operation position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Change-over selector location	1	2	3	4	5	6	7	8	9	10	11	K	1	2	3	4	5	6	7	8	9	10	11
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23

● Drawing is shown at the set position

Appendix 58. SHZV (12231G) operating position table and connection diagram

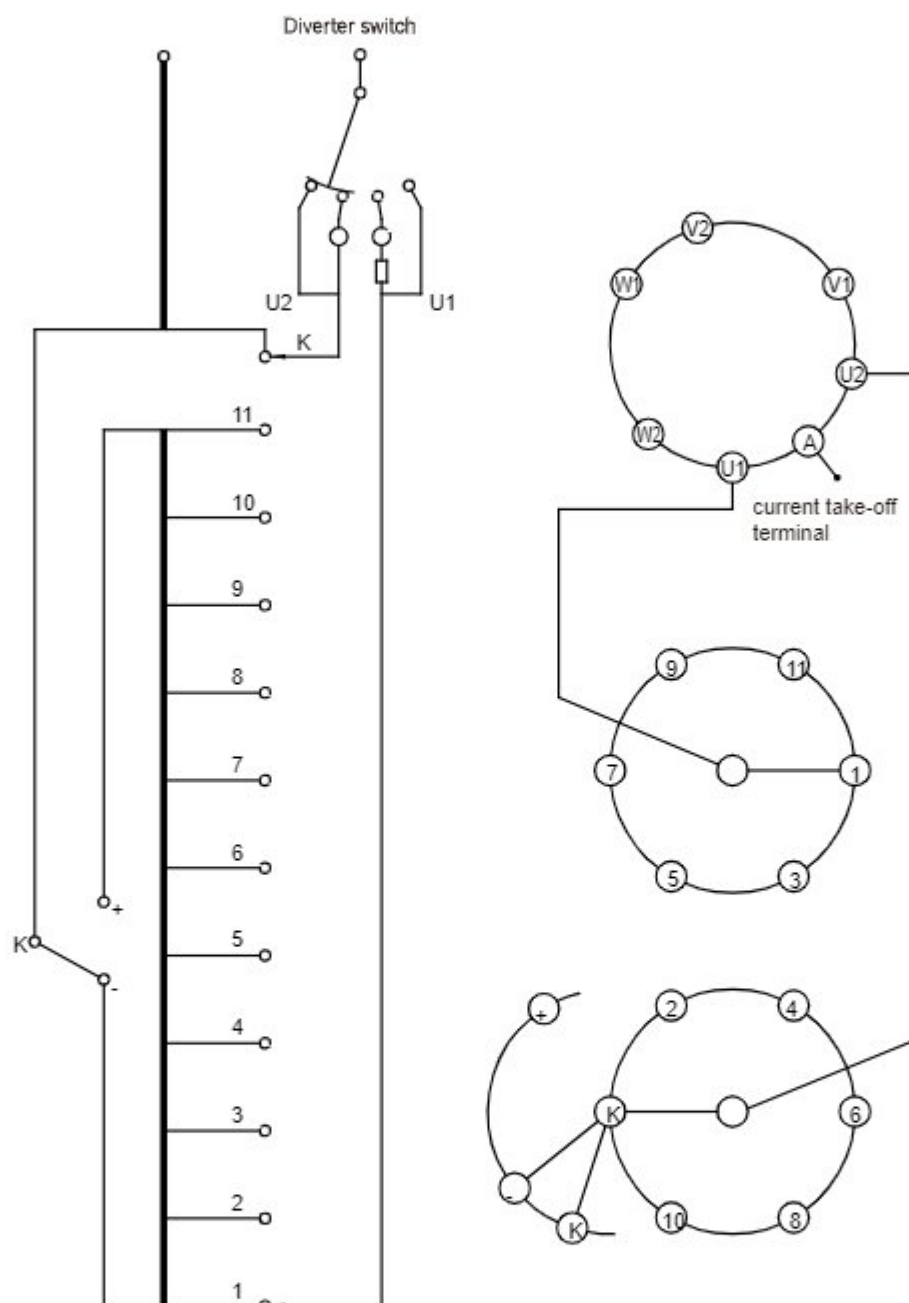


Operation position number	23
Different voltage numbers	23
Set position	● 12

change-over selector	K+											K-											
tap-changer position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Contact position	1	2	3	4	5	6	7	8	9	10	11	K	1	2	3	4	5	6	7	8	9	10	11
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23

• Drawing is shown at the set position

Appendix 59. SHZV (12233W) operating position table and connection diagram



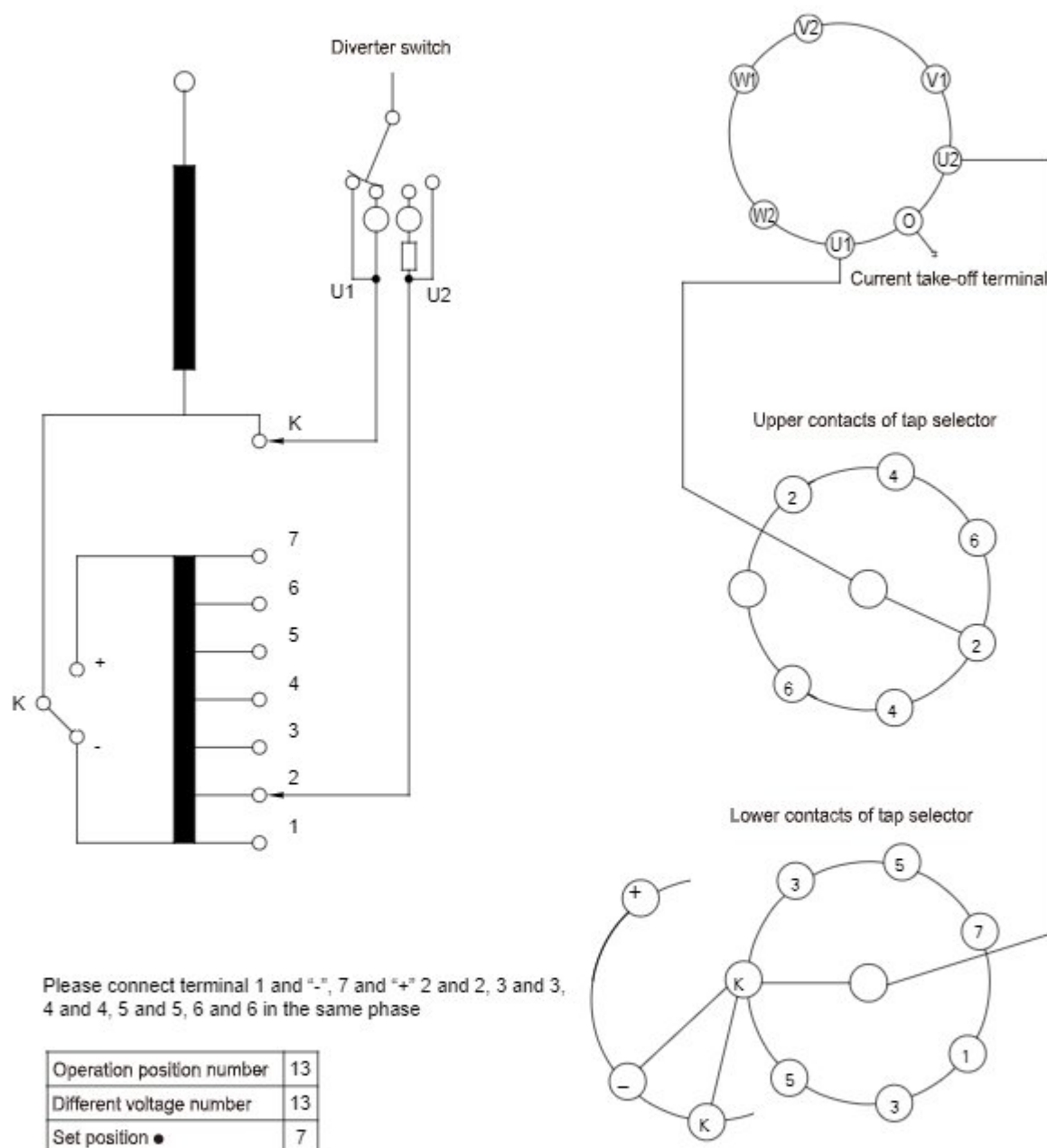
Please connect terminal 1 and "-", 11 and "+" in the same phase

Operation position number	23
Different voltage number	21
Set position ●	11b

Change-over selector	K+											K-											
Operation position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Change-over selector location	1	2	3	4	5	6	7	8	9	10	11	K	1	2	3	4	5	6	7	8	9	10	11
Display position	1	2	3	4	5	6	7	8	9	10	11a	11b	11c	12	13	14	15	16	17	18	19	20	21

● Drawing is shown at the set position

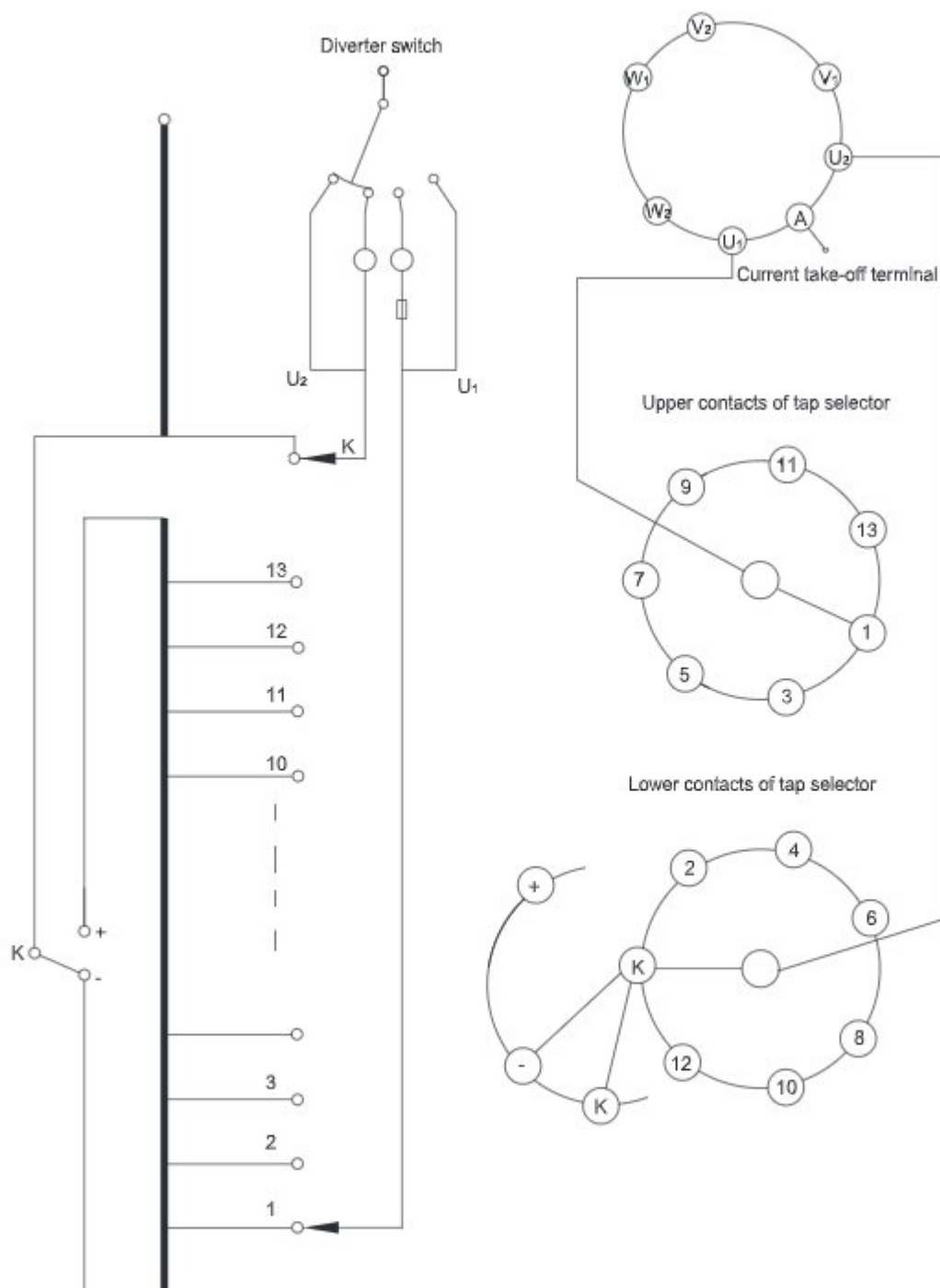
Appendix 60. SHZV (14131W) operating position table and connection diagram



Change-over selector													
Operation position	1	2	3	4	5	6	7	8	9	10	11	12	13
Change-over selector location	1	2	3	4	5	6	K	2	3	4	5	6	7
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13

● Drawing is shown at the set position

Appendix 61. SHZV (14271W) operating position table and connection diagram



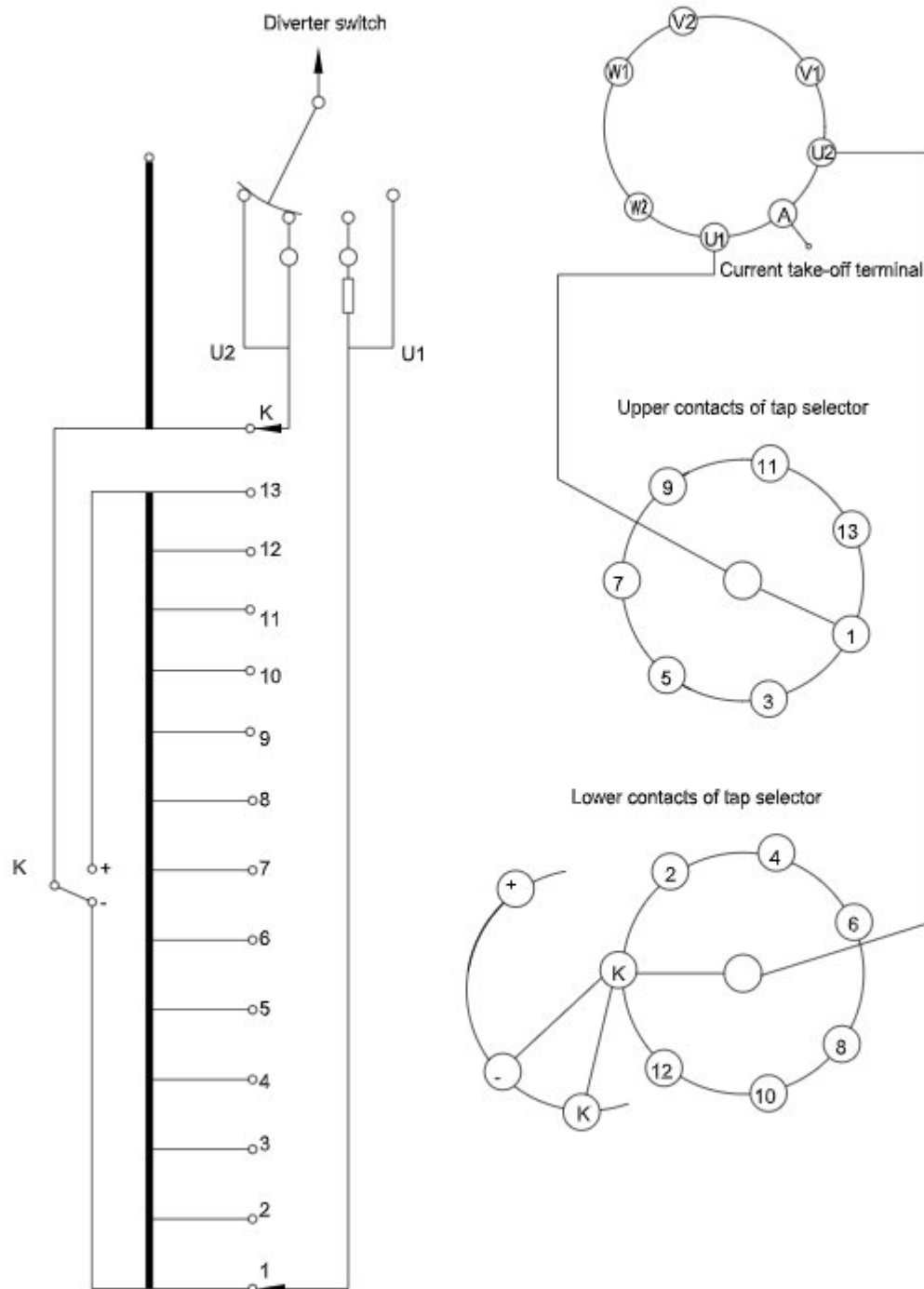
Operation position number	27
Different voltage number	27
Set position ●	14

Change-over selector																											
Change-over selector location	1	2	3	4	5	6	7	8	9	10	11	12	13	K	1	2	3	4	5	6	7	8	9	10	11	12	13
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27



● Drawing is shown at the set position

Appendix 62. SHZV (14273W) operating position table and connection diagram



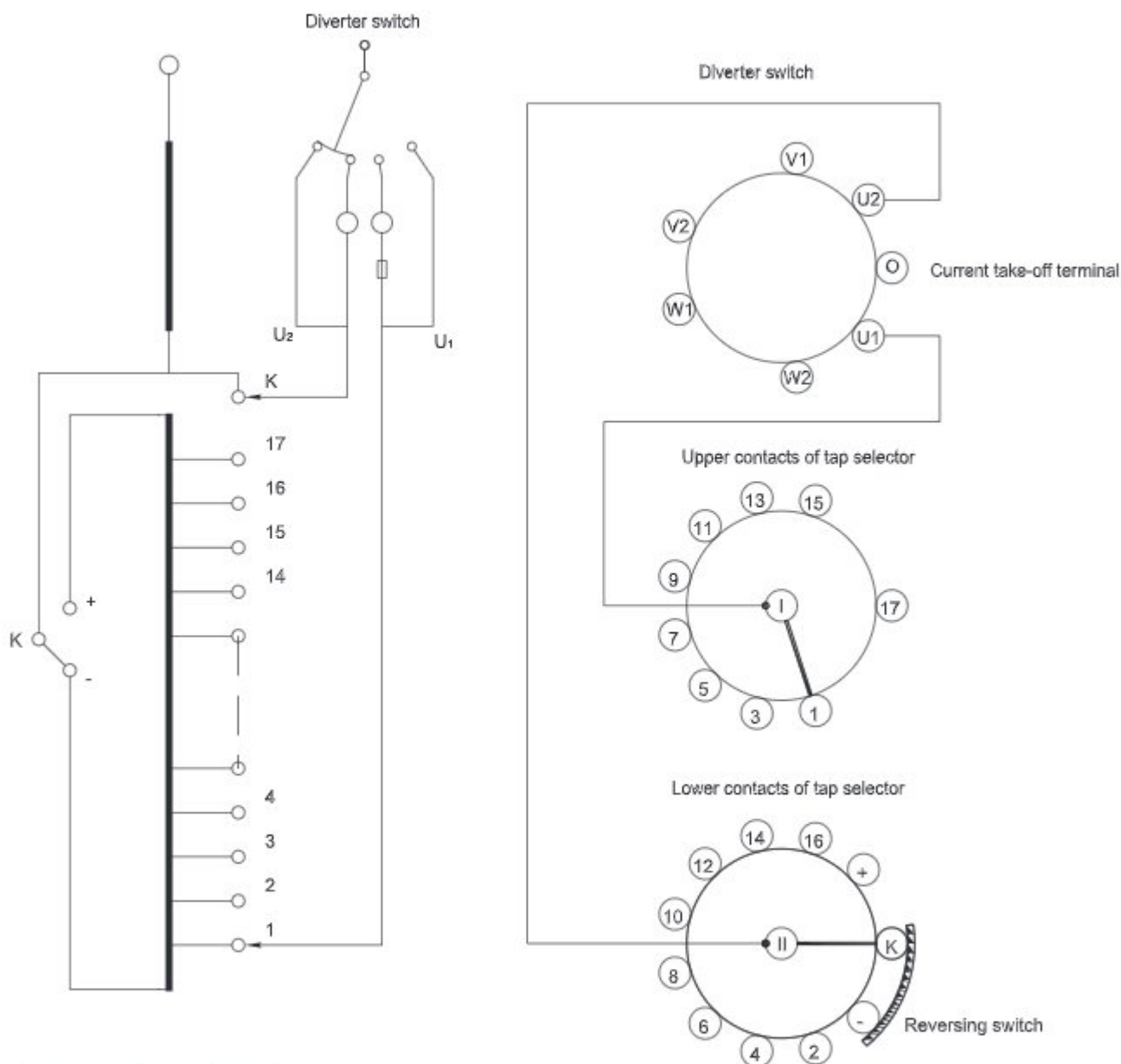
Operation position number	27
Different voltage number	25
Set position ●	13b

Please connect terminal 1 and "-", 13 and "+" in the same phase.

Change-over selector	K+													K-													
Change-over selector location	1	2	3	4	5	6	7	8	9	10	11	12	13	K	1	2	3	4	5	6	7	8	9	10	11	12	13
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13a	13b	13c	14	15	16	17	18	19	20	21	22	23	24	25

● Drawing is shown at the set position

Appendix 63. SHZV (18351W) operating position table and connection diagram



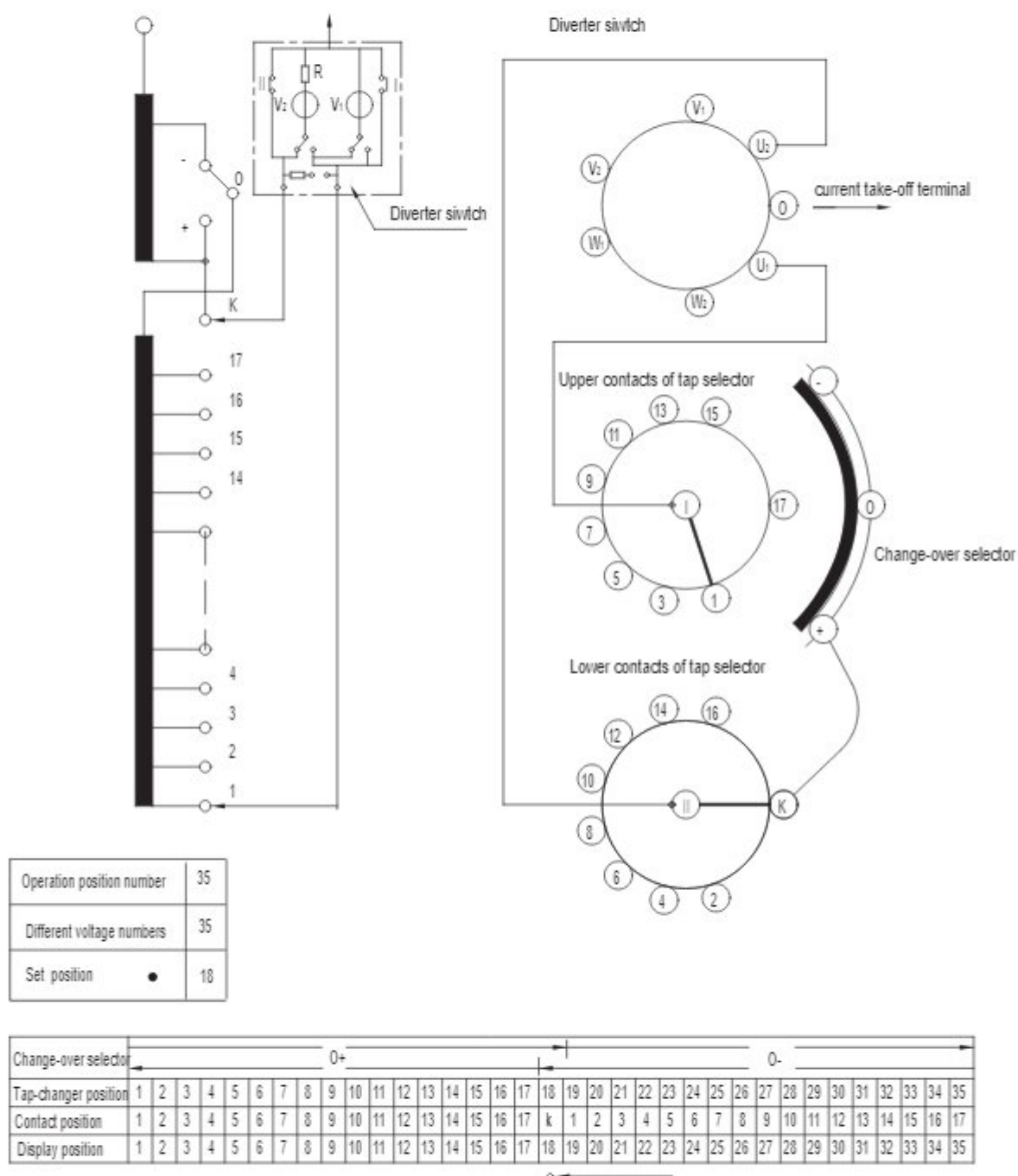
Operation position number	35
Different voltage number	35
Set position ●	18

Change-over selector	</																																		
----------------------	---	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Note: It is suitable for cage tap selector structure only

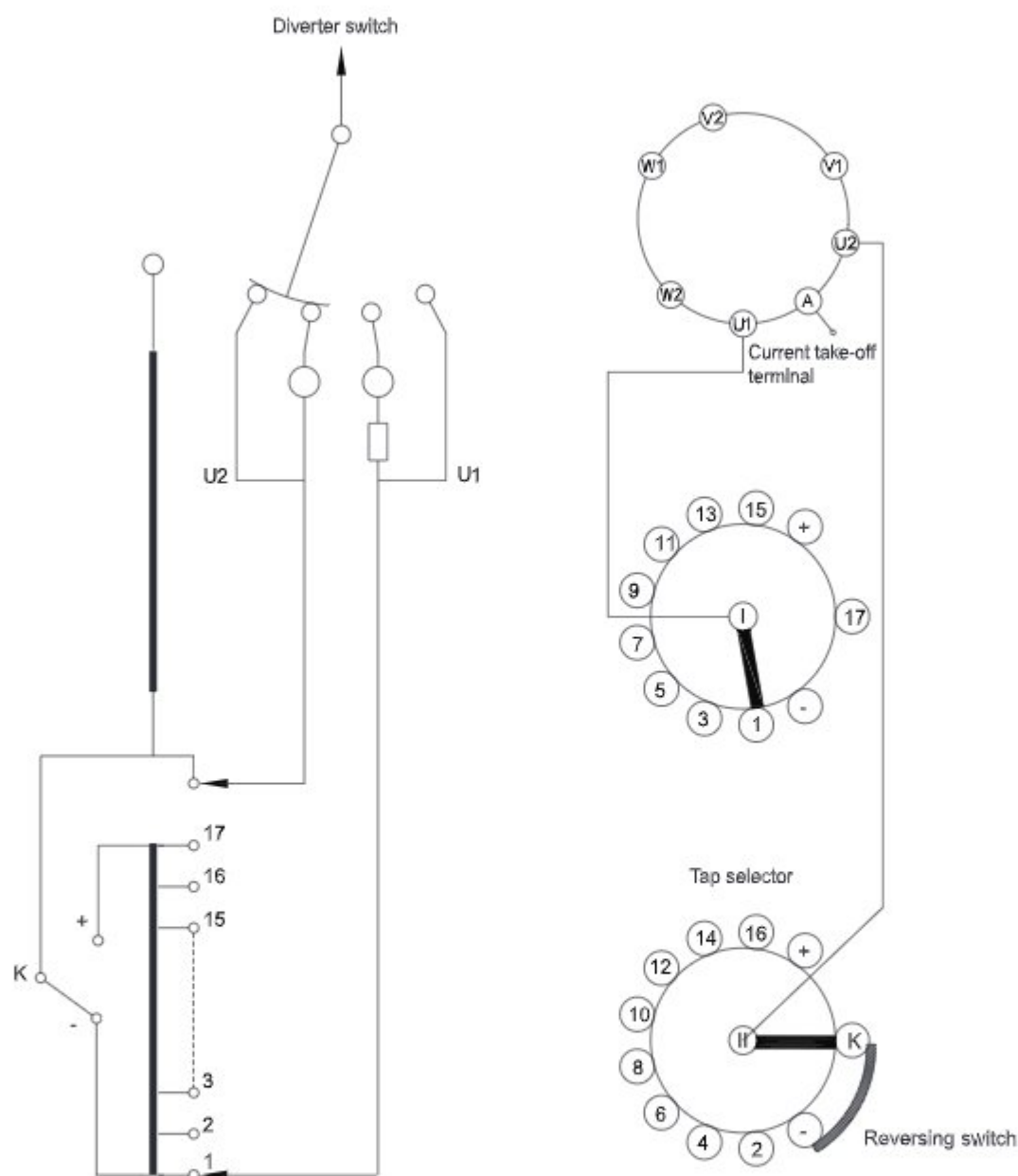
● Drawing is shown at the set position

Appendix 64. SHZV (18351G) operating position table and connection diagram



• Drawing is shown at the set position

Appendix 65. SHZV (18353W) operating position table and connection diagram



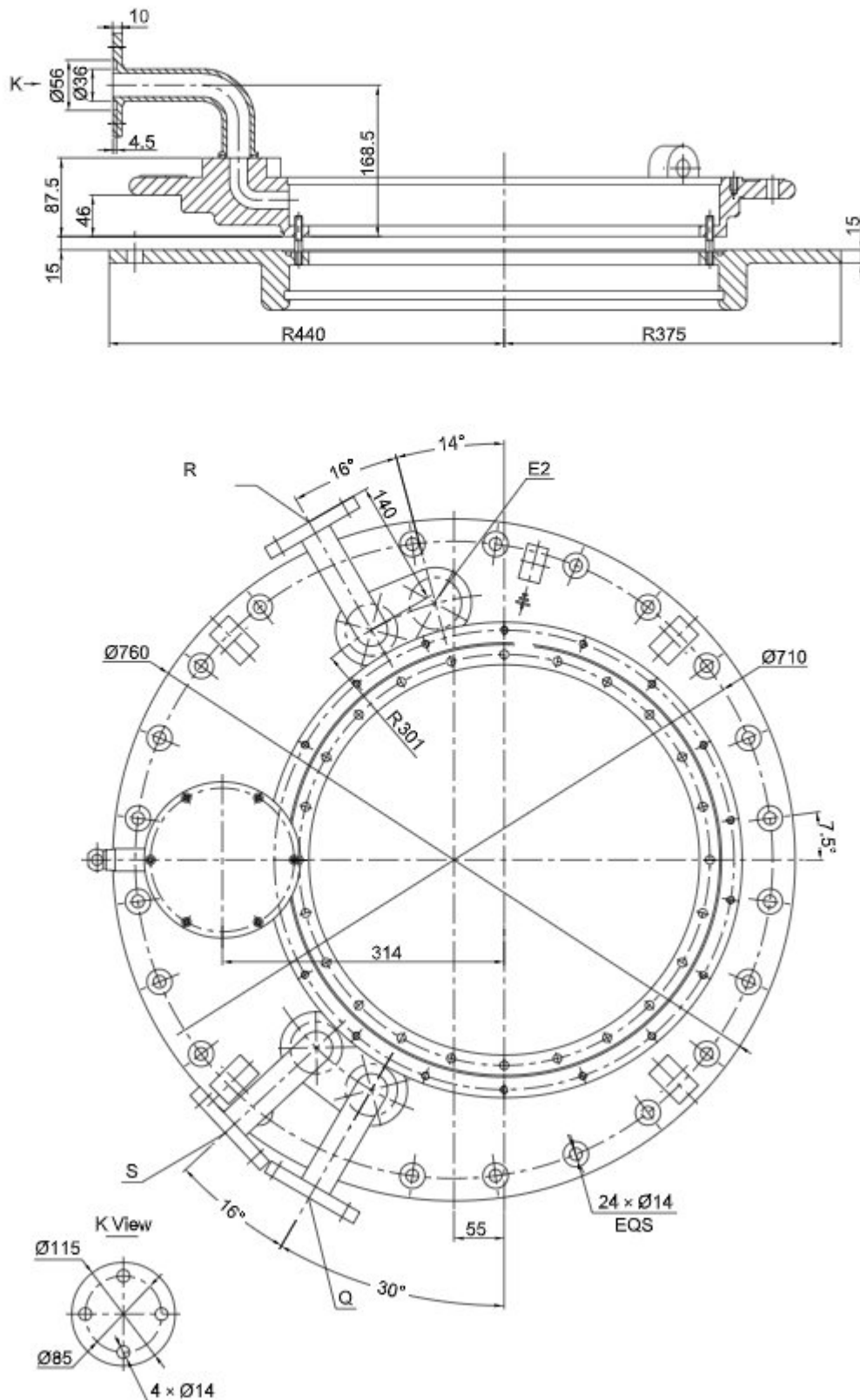
Operation position number	35
Different voltage number	33
Set position ●	17b

Change-over selector	← K+ → ← K- →																																		
Change-over selector location	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17a	17b	17c	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33

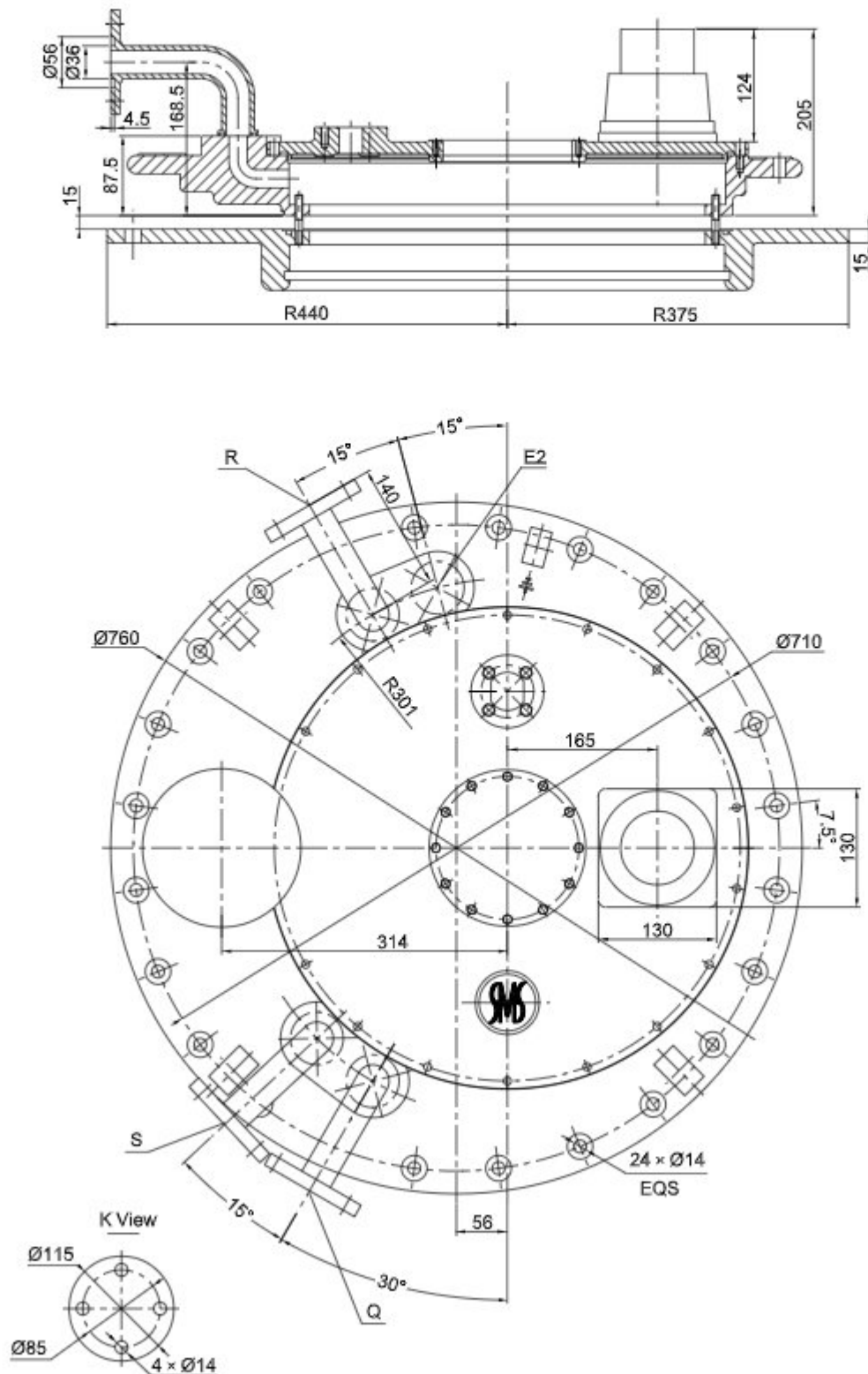
Note: It is suitable for cage tap selector structure only

● Drawing is shown at the set position

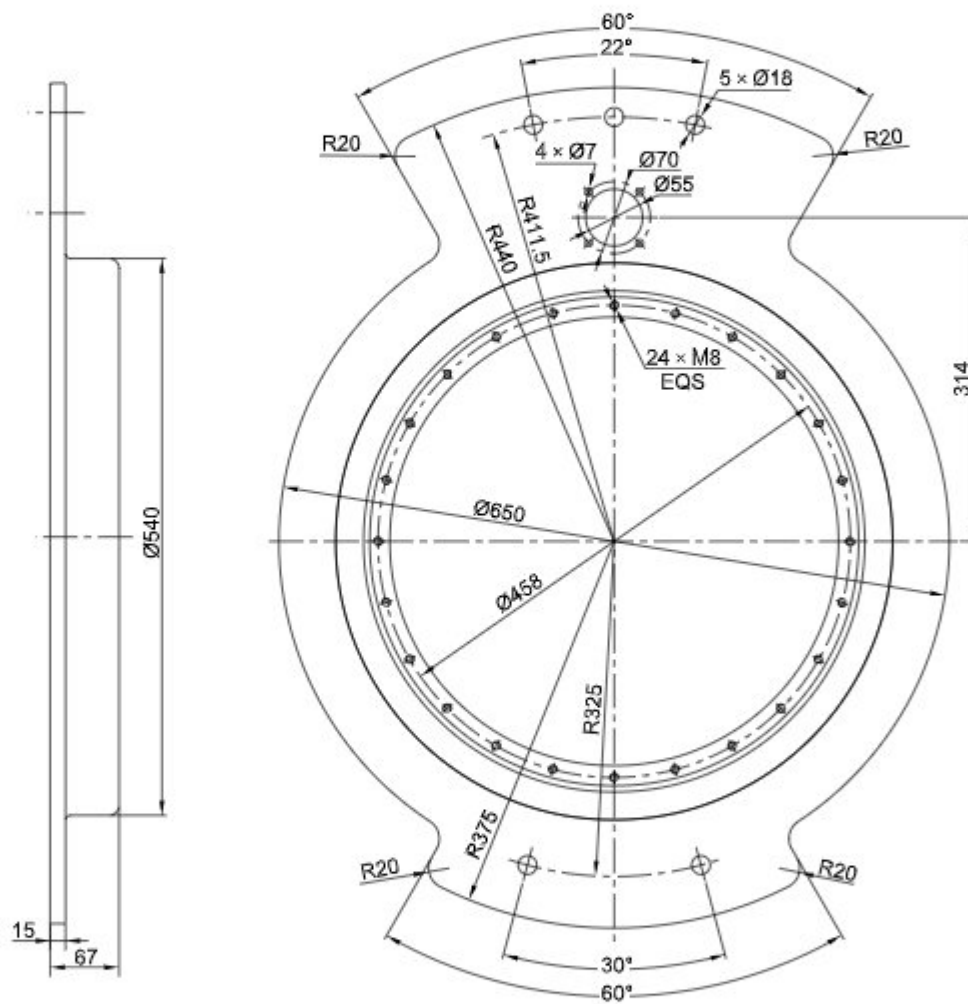
**Appendix 66 SHZV bell-type head flange,
overall dimension, cylinder tap selector**



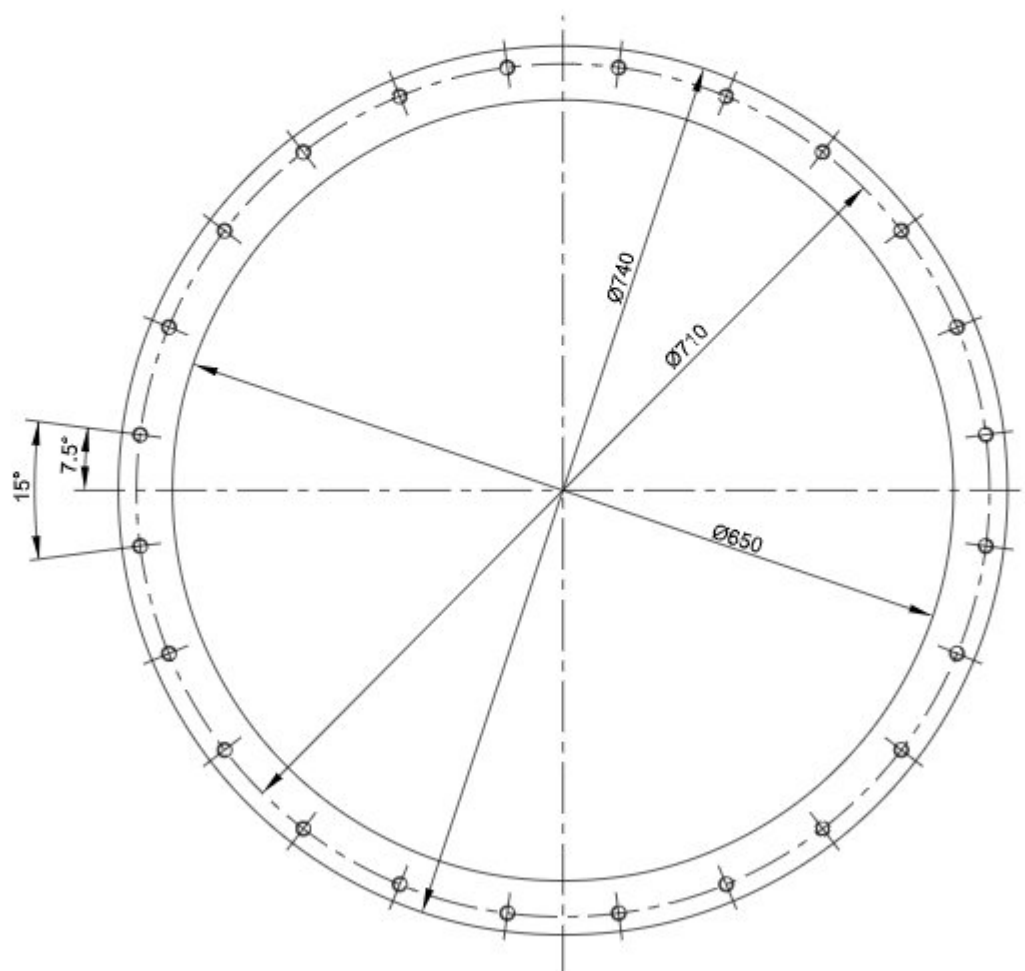
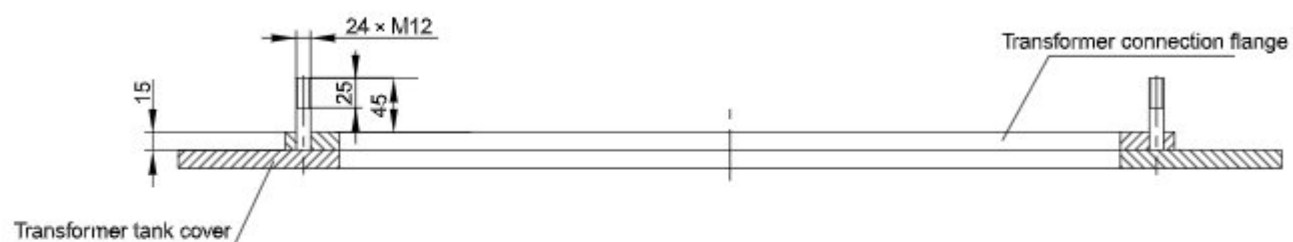
**Appendix 67. SHZV bell-type head flange with pressure relief valve,
overall dimension, cylinder tap selector**



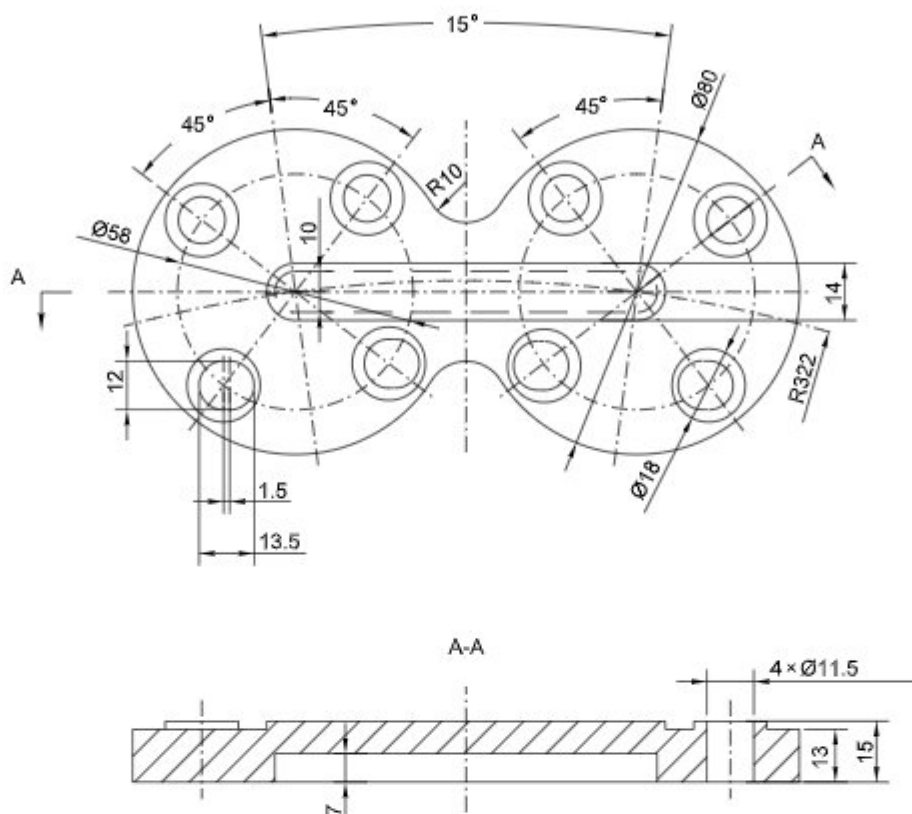
Appendix 68. SHZV supporting flange, overall dimension, cylinder tap selector



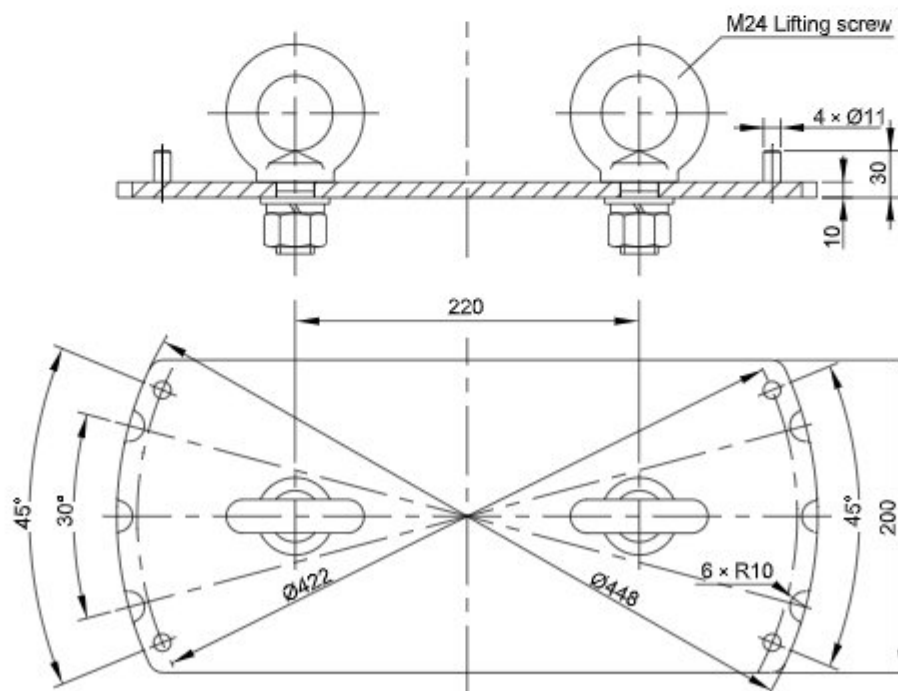
Appendix 69. Transformer mounting flange for SHZV overall dimension



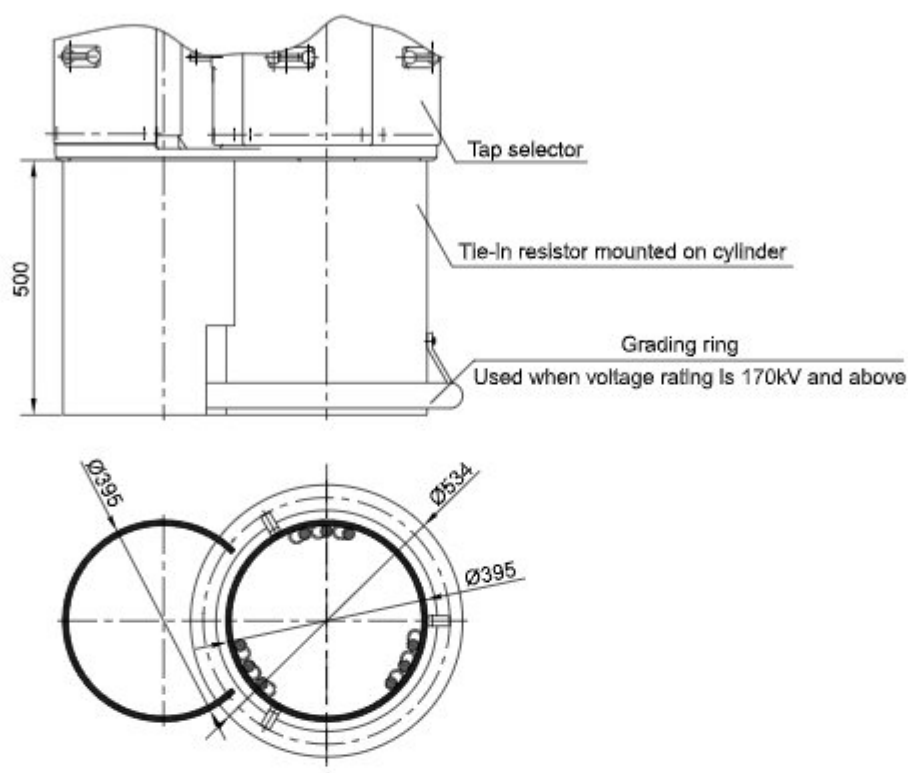
Appendix 70. By-pass pipe overall dimension



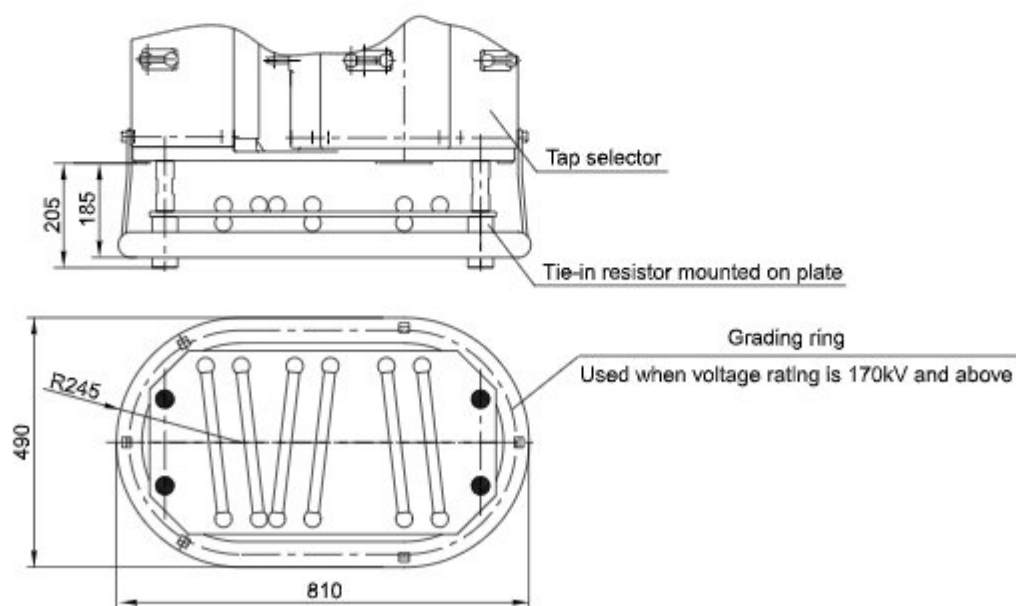
Appendix 71. Bell-type lifting plate ,overall dimension



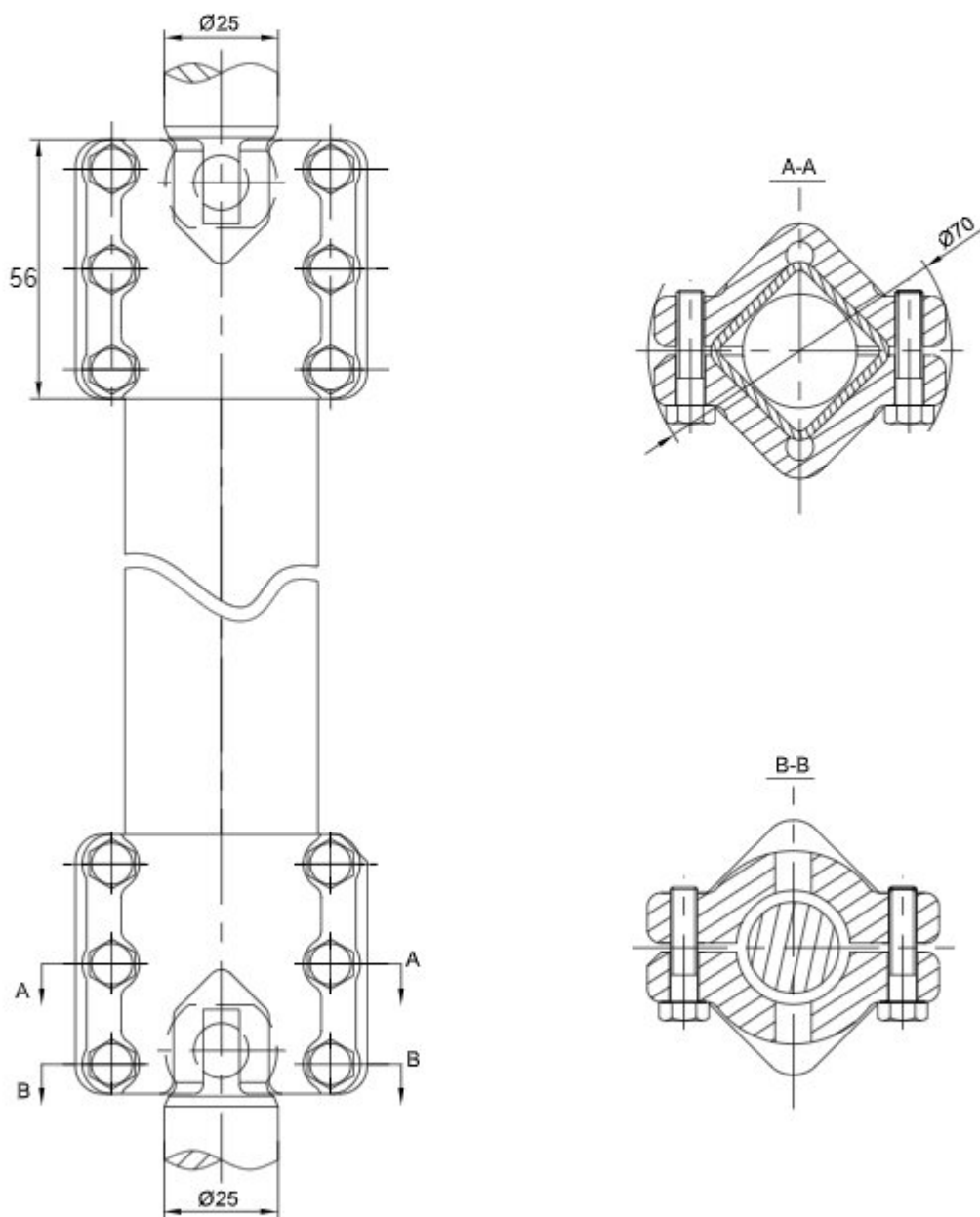
Appendix 72-1. SHZV OLTC mounted with tie-in-resistor on cylinder overall dimension



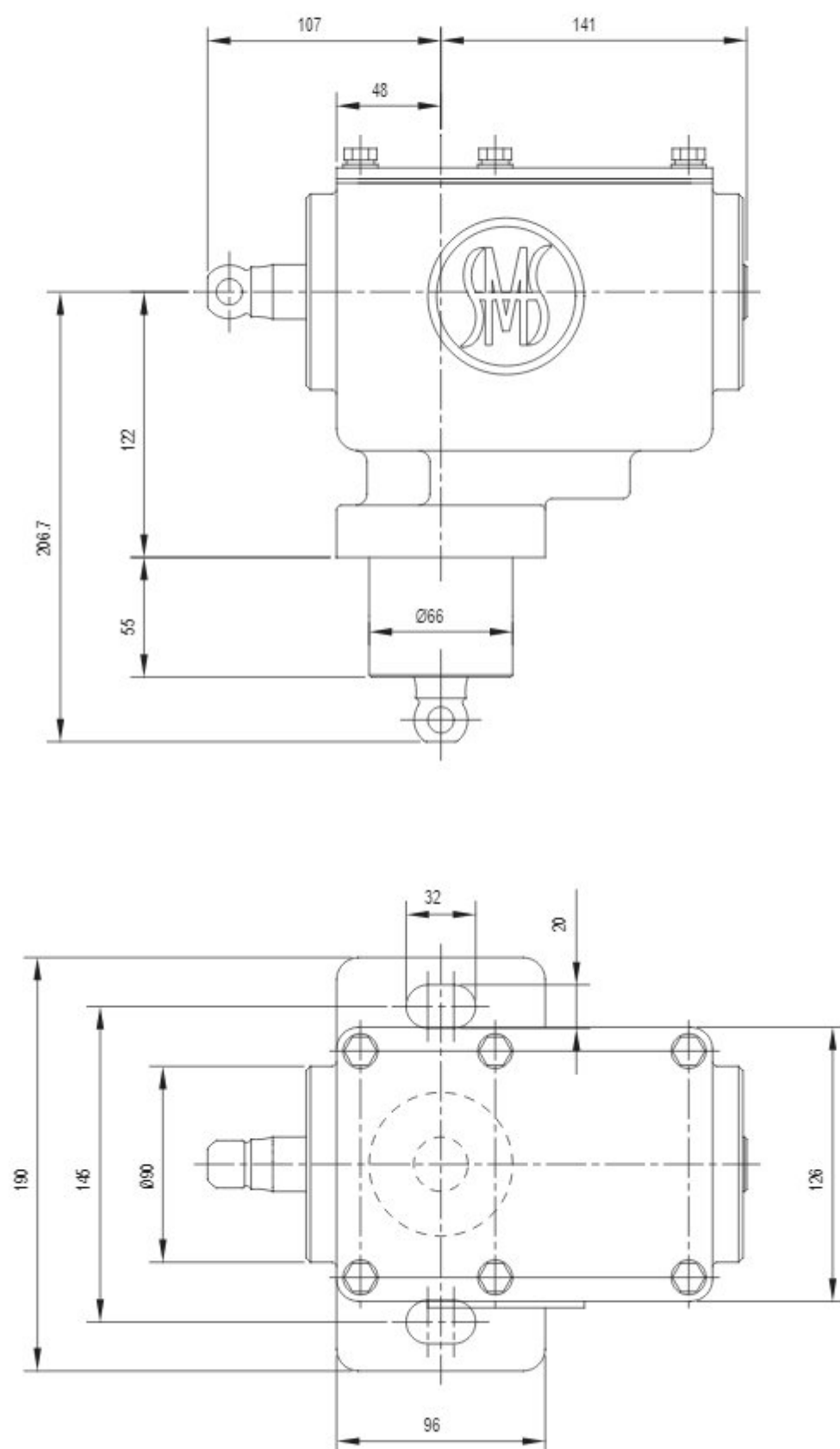
Appendix 72-2. SHZV OLTC mounted with tie-in-resistor on plate overall dimension (cage tap selector)



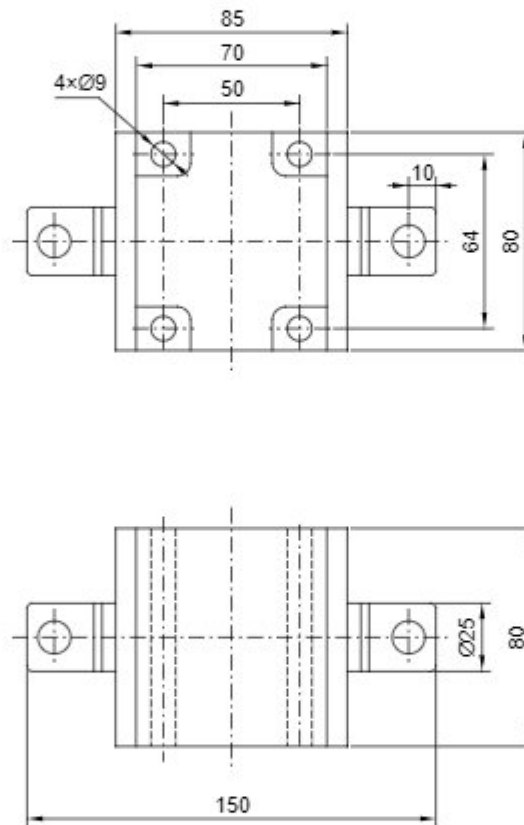
Appendix 73. Schematic drawing for connecting of horizontal shaft and vertical shaft



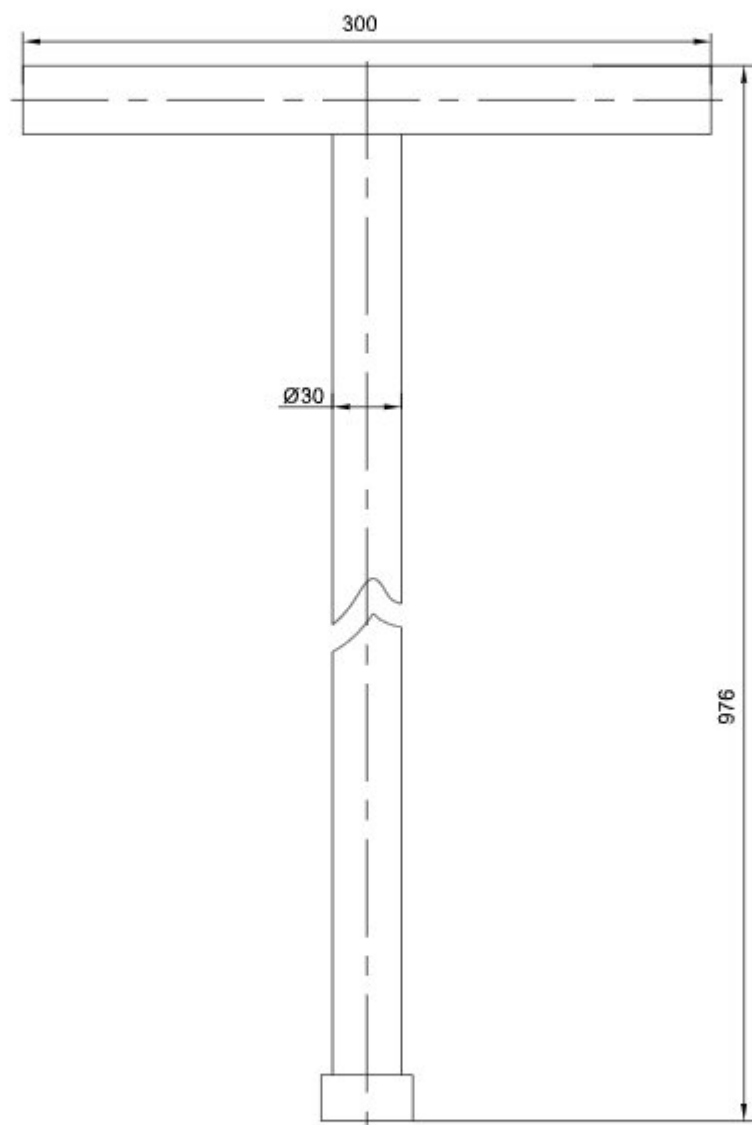
Appendix 74. Bevel gearbox overall dimension



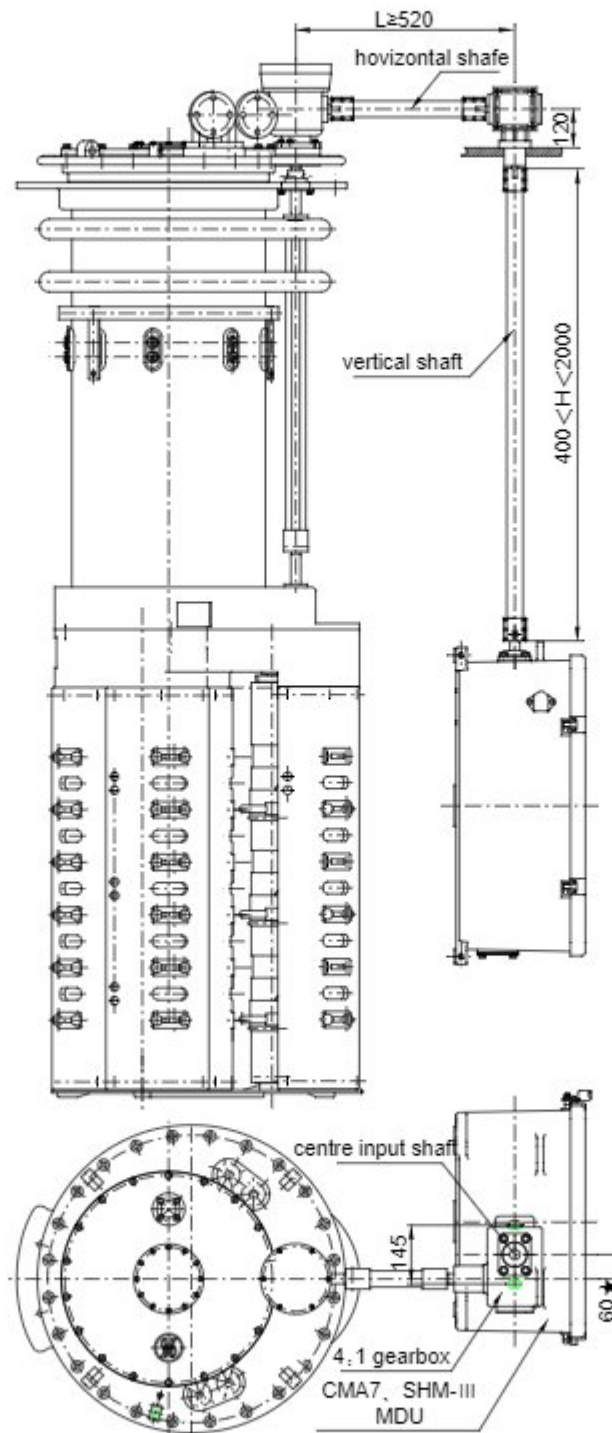
Appendix 75. Middle supporting box overall dimension



Appendix 76 . Operation key for oil discharge inside tapchange oil compartment

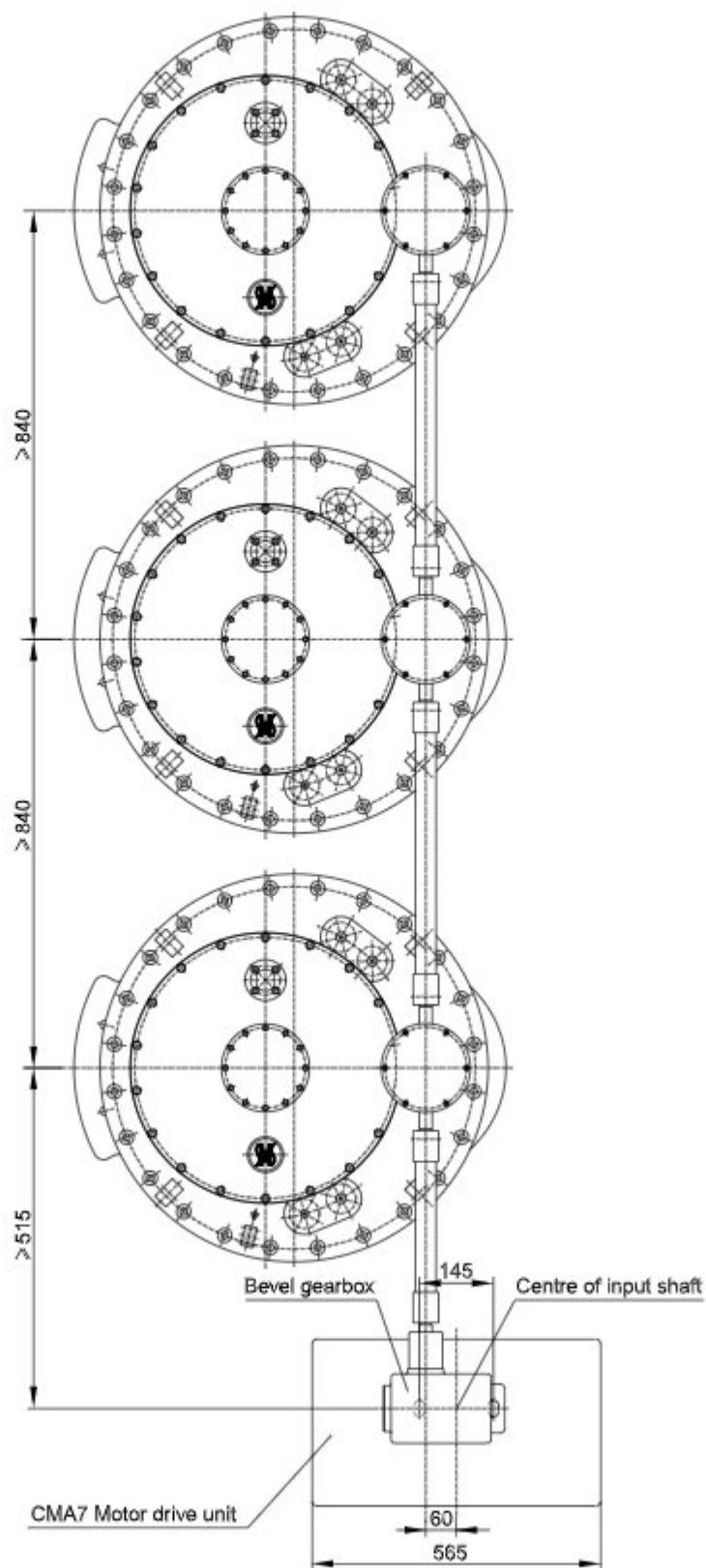


Appendix 77. Schematic drawing for connecting of SHZV OLTC and MDU



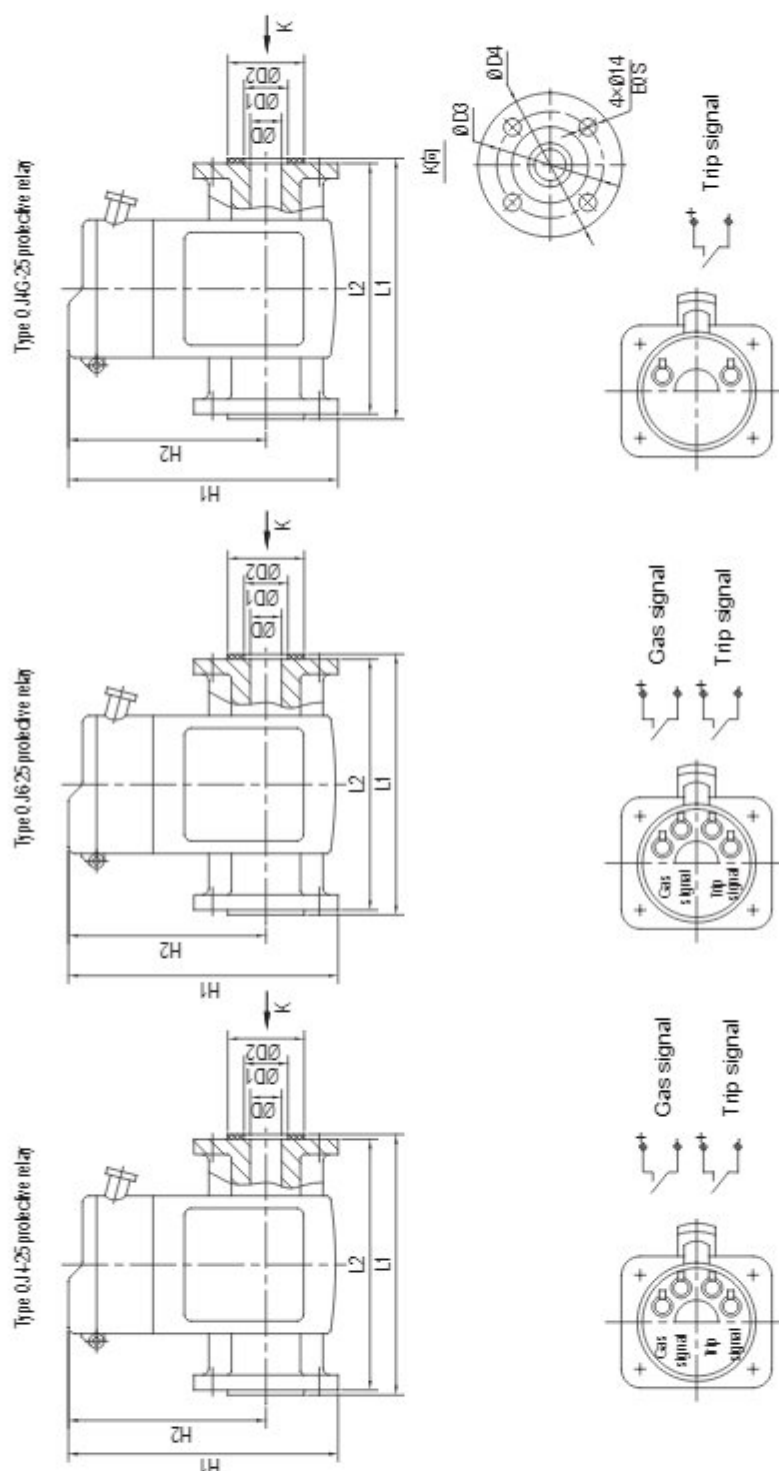
Note: When vertical shaft $H \geq 2000\text{mm}$, Middle supporting box need to be installed (middle supporting box see appendix64)

Appendix 78. Schematic drawing for 3 units of single-phase SHZV connection arrangement



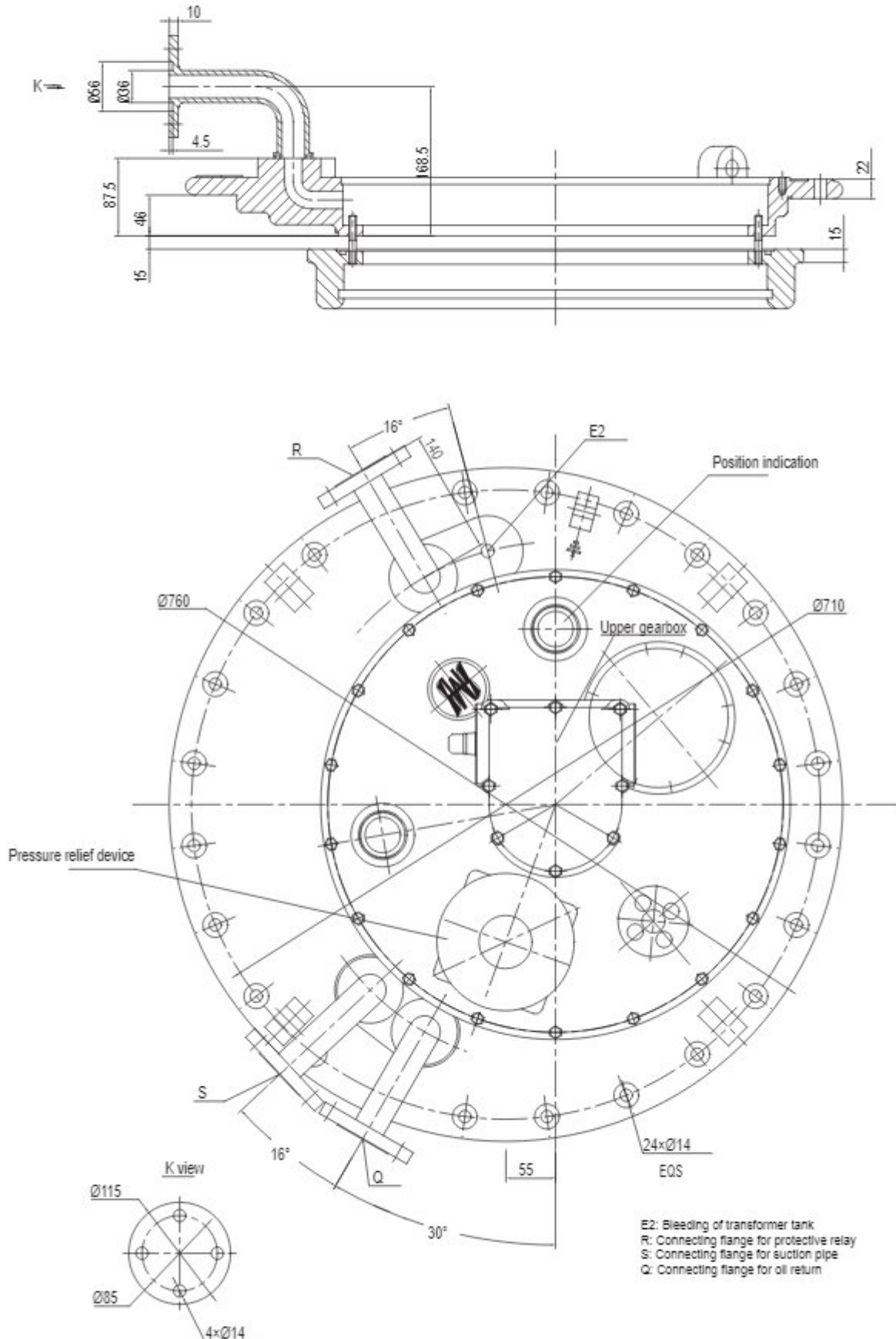
Unit: mm

Appendix 79. Protective relay overall dimension

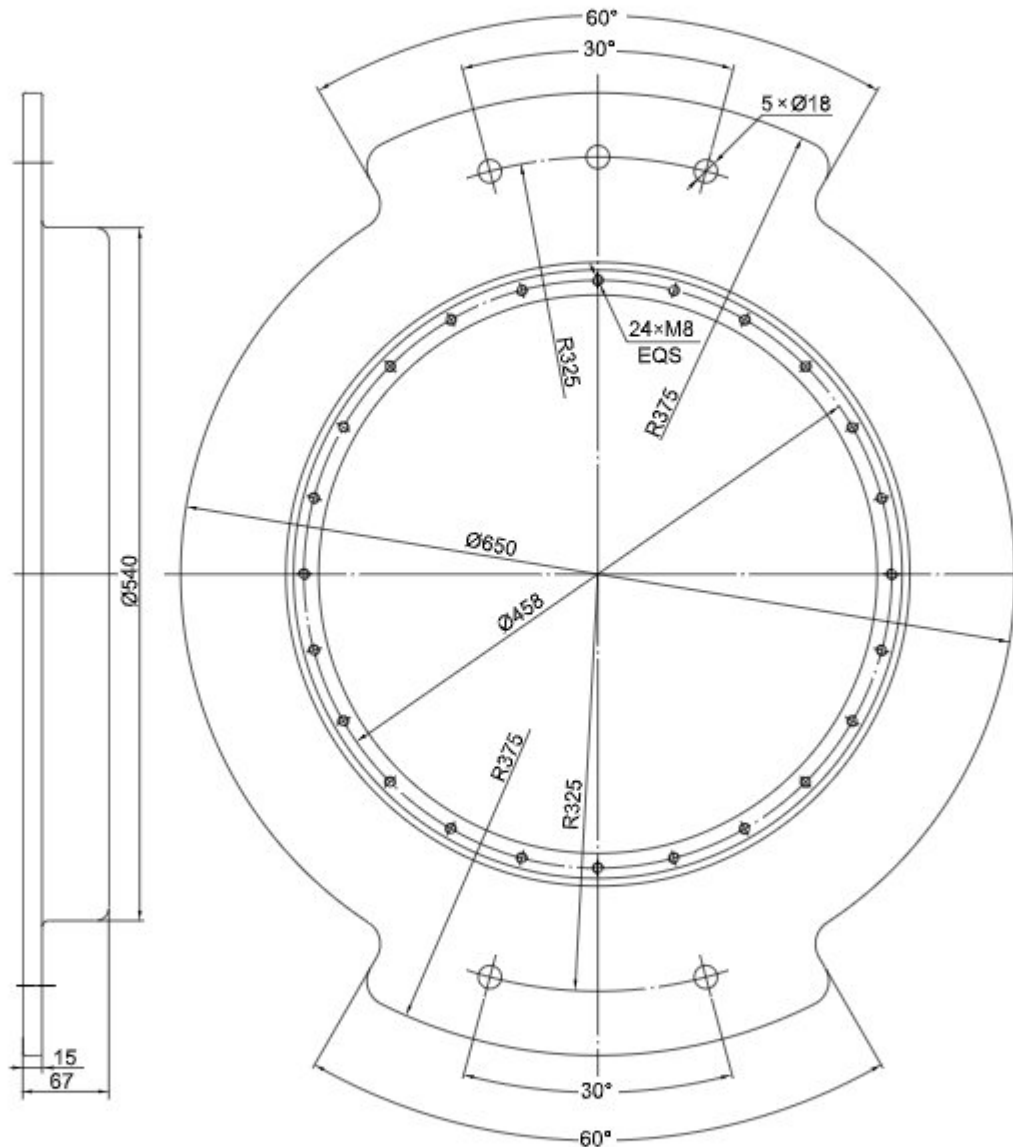


Type	D	D1	D2	D3	D4	d1	H1	H2	L1	L2	Remarks
Q.4-25	25	35	65	85	115	14	215	153	208	200	1 pair of gas signal and 1 pair of trip signal
Q.4G-25	25	35	65	85	115	14	195	133	208	200	1 pair of trip signal
Q.46-25	25	35	65	85	115	14	215	153	208	200	1 pair of gas signal and 1 pair of trip signal

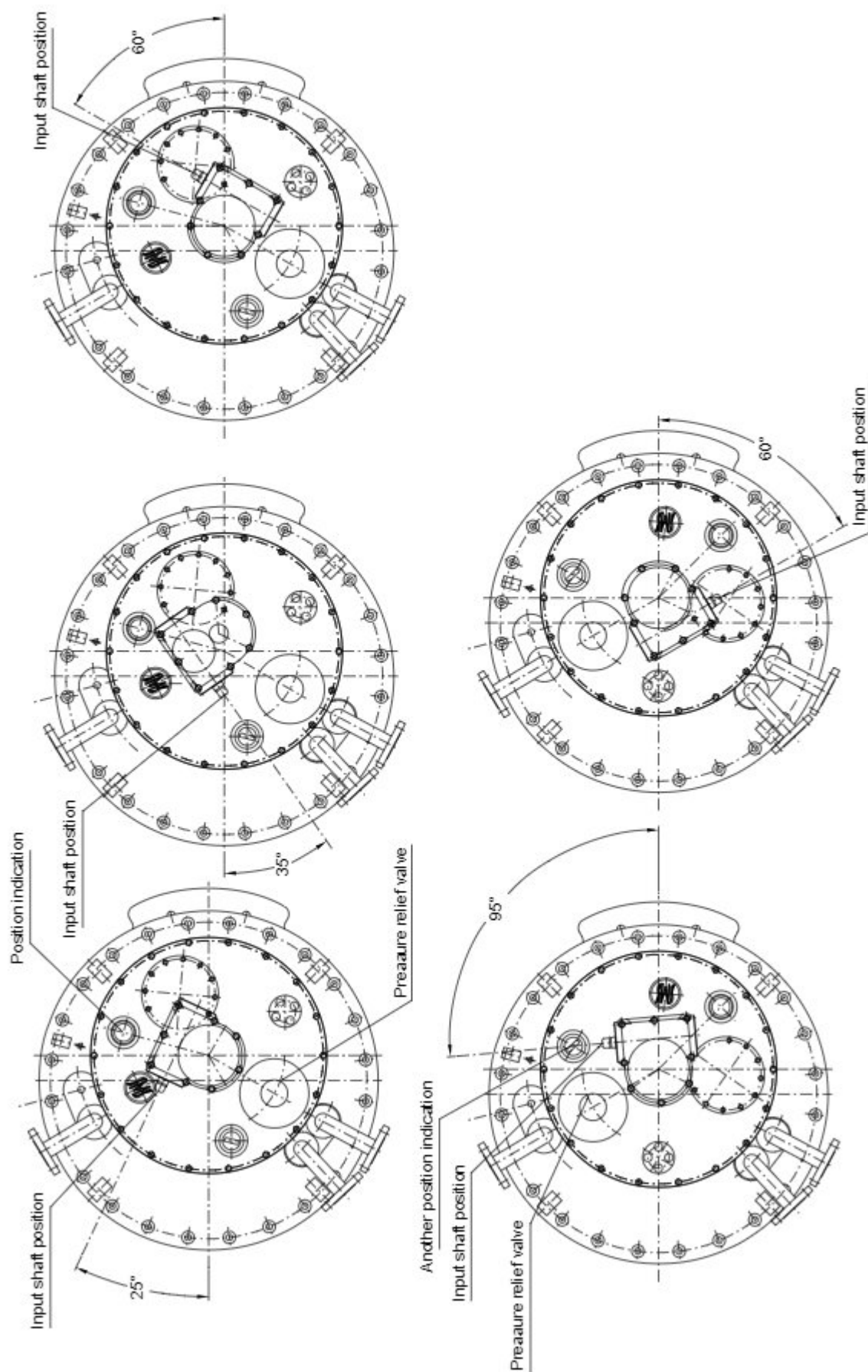
Appendix 80. SHZV bell-type head flange overall dimension, cage tap selector



**Appendix 81. SHZV bell-type supporting flange
overall dimension, cage tap selector**

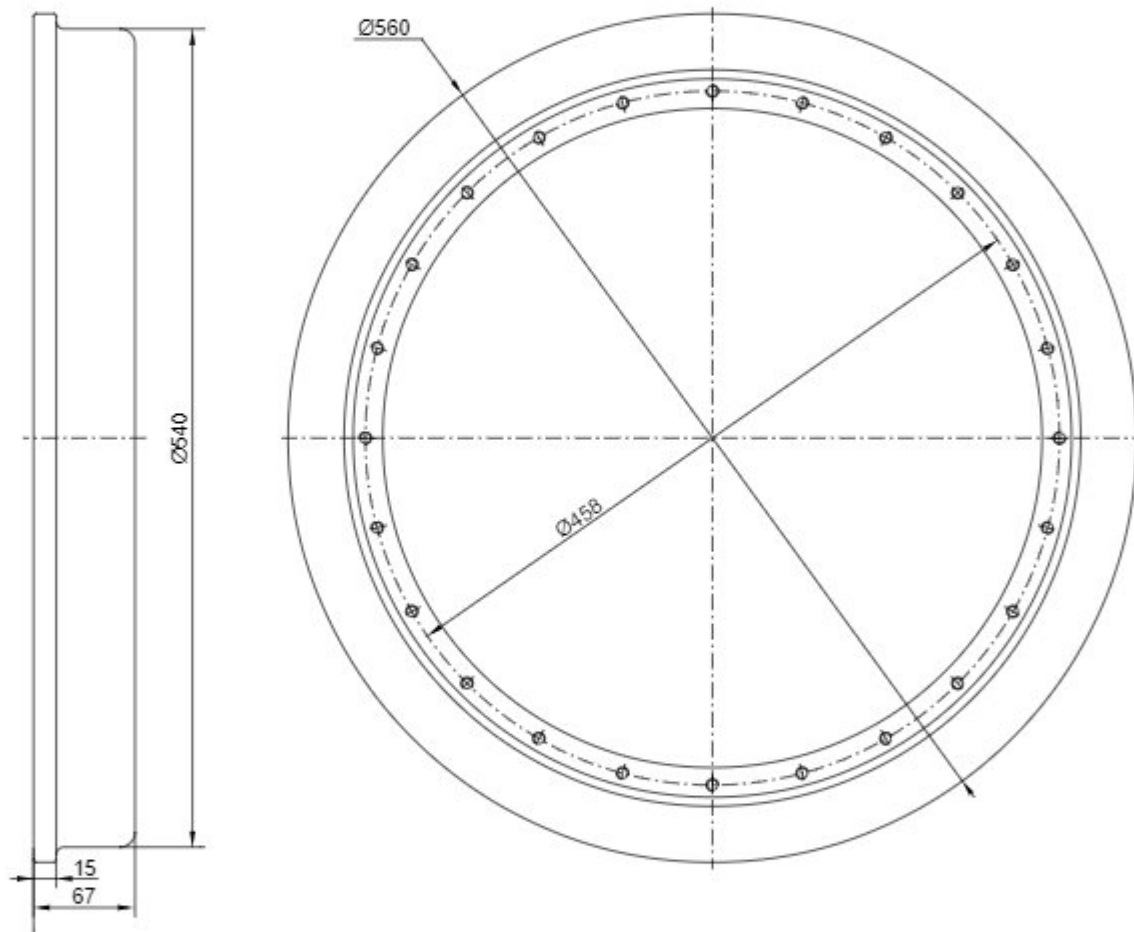


Appendix 82. SHZV gear mechanism drive shaft position, overall dimension, cage tap selector



Unit: mm

Appendix 83. SHZV standard tank type supporting lange, overall dimension, cage tap selector





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